

# **Pols 320 Political Demography: The Politics of Population Change**

*Numbered Transcript*

POLS 320 • Dr. Eric Kaufmann

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## Lecture 1: Demography, Migration and Power

**At a glance** — Kaufmann introduces political demography — how population change, migration, and power interact.

- 1.1 All right. Welcome, everybody. Today, I'm going to be speaking to you about political demography, or the politics of population change. And that's a new discipline that brings together two traditional fields, one being demography, the other being political science, and also sociology. So the politics of population change, which really asks, "What happens to politics when population shifts occur?"
- 1.2 So the first thing I want to talk about a little bit is demography, which is a well-established field. It's the study of population. And then we'll go on to see how that affects politics and, to some degree, politics and culture affect population, because we're in a very important time in human history right now, particularly between countries, where population shifts in different parts of the world are having major political effects.
- 1.3 So what is demography? Demography has an internal component, which is about, essentially, natural increase, on the one hand, including fertility, the number of children a woman will bear across her lifetime. Mortality rate, which is a death rate, so if you like a birth rate and a death rate, that gives us the natural increase. That's all happening endogenously, or internally, in a population.
- 1.4 If you imagine a country, and there were no other countries in the world, just France, every change would have to occur, every demographic change would happen within France. Death rate, birth rate. But of course, we have many countries in the world, so it's not just about death rate and birth rate or the internal aspects of demography. But there is also an external or exogenous dimension, which is net migration, and that is immigration minus emigration equals net migration.
- 1.5 So even, actually, if you have a declining population, if you have positive net migration from the rest of the world, then your population may not actually go down. So when we're talking about demography, we have to think both about natural increase and about net migration. So we now have these two components of population within a country.
- 1.6 But then we also have, in demography, mathematical formulas that link aspects such as fertility and birth rate with age structure, or migration with age structure. If you have a lot of young people coming into your country, then your average age is going to go down. One of the things that you notice about these formulas in demography is that there's a relationship between, for example, what's happening to your population in terms of increase or decrease, and the age structure.
- 1.7 Typically, if your population undergoes a decrease, again, not always, but sometimes, especially if it undergoes a decrease through declining birth rates, then you will tend to

get population aging. Or if the birth rates increase rapidly, then you will tend to get a younger population. And both population aging and a very young population have been connected to particular social and political phenomena, which we're going to look at.

- 1.8 There is a term called total fertility rate, TFR, and you'll see that used. The phrase replacement, total fertility rate, is 2.1. Total fertility rate is essentially the number of children a woman will bear in her lifetime, so when fertility starts in the teens till the mid-forties, let's say. How many children on average will a woman bear in her lifetime?
- 1.9 To make that a bit more complicated, there's something called a period total fertility rate versus completed total fertility rate. Completed total fertility rate can only be answered when you have women who are 45 or more, because you know how many kids they'll have borne in their lifetime. The period TFR, however, is kind of a prediction. You're looking at women of ages from, say, whatever, 15, 16, all the way to 45. And you've got the actual number of kids each of those little slices has, and you squash them together to create a kind of synthetic prediction of what the TFR is.
- 1.10 But of course, that period, total fertility rate, especially if you get a period where women are delaying having kids, and then they suddenly have them in their 30s, when they used to have them in their 20s, it's going to lead to a depressed period total fertility rate, but when they actually complete, into their 30s, the total fertility rate will rise.
- 1.11 And so, to some degree, when the average age of women having children is rising, that will underestimate the TFR. So, sometimes the TFR, say, in developed countries, might be a little higher than the period TFR. That's just a little technical. When you're looking at people posting online, you'll have a sense that often they're talking about the period TFR, rather than the completed total fertility rate.
- 1.12 The other thing to note is that demography, I mentioned birth rates, death rates, migration, but demography is the mechanism behind the evolutionary process of natural selection. Certain adaptations, if they're more successful, lead to more births and fewer deaths, expansion in populations carrying a particular gene. This is another thing to bear in mind.
- 1.13 The final thing I'll say is, demography is the most predictive of the social sciences, the most predictive of the social sciences. If we make an economic forecast, we have certain inputs that we can make a forecast. What's going to happen to the British economy in the next 10 years, the next 20 years? There's a lot of error in those kinds of forecasts. What's going to happen culturally to religion? There's a lot of error in that.
- 1.14 Demography, there is error, but there's also a lot that's fixed. So we know the characteristics of the 40-somethings 30, 40 years from now because they've already been born, in terms of ethnicity, for example, or religion. Now, those things can change to some degree, but we know population size, so we can project ahead. Based on the characteristics of the under-5 population, we can look at the under-45 population in 40

years, and there's a lot of predictive power there, much more predictive power than in other social sciences. So, prediction and forecasting is part of political demography.

- 1.15 Still sticking with demography, a lot of literature revolves around this idea of the demographic transition. Through most of human history, the human population has been flat or growing very slowly, or you've got feast and famine, but basically, the average is pretty flat because you've got high birth rates and high death rates. That is an equilibrium. High birth rate, high death rate.
- 1.16 That is the human condition since our species began 100,000 years ago or 50,000 years ago, all the way up until very recently, a sort of blink of an eye in historical time. Beginning in Europe in the late 18th century, we begin to see the death rates start to come down. Infant mortality, better sewage systems, better nutrition, eventually better medicine starts to bring the death rate, particularly the infant mortality rate, down.
- 1.17 And that leads to a population explosion because the birth rates are still quite high. The death rate starts to drop, and then birth rates are above death rates. And so you get natural increase, and you get a boom in population. During the 20th century, that process, which began gradually in Europe, gradual improvement in sanitation, in medicine, in wealth, in nutrition, leading to a gradual increase in population. Yes, a population explosion, but a moderate one happened very quickly in the rest of the world in the 20th century.
- 1.18 We had much more of a population explosion because all of this medicine and technology hit the post-colonial world. You had a rapid drop in death rate, infant mortality rate, with those high birth rates, and you had a very big expansion of population. Now what's happened is, we've reached another equilibrium of very low birth rates and very low death rates. The death rate falls first.
- 1.19 After the death rate falls and you get the population explosion, you then get a birth rate decline, due to a whole bunch of factors, including the fact, "Oh, half the kids aren't dying before age 10. Maybe we don't have to have so many," or "We're not as agricultural. We don't need all those hands to work the land," or "We don't need an old-age pension based on having lots of kids," or "It costs a lot more to raise kids in cities."
- 1.20 There are all these factors that are implicated in that decline in the birth rate through the 19th century, 20th century, beginning in Europe, then spreading to the rest of the world, so that now we see the total fertility rate below the replacement level of 2.1, that is two kids per couple, replaces them, 0.1 for accidents and death before they're able to reproduce. So 2.1 is that TFR replacement rate. The West, Western Europe, goes below it in the 1970s, thereabouts. The US, maybe not until the 2000s, although it kind of goes up and down.
- 1.21 Now we're seeing the total fertility rate drop below replacement in most of the world, actually—Asia, most of Latin America. The only major regions that still have quite high

fertility are sub-Saharan Africa and parts of west Asia. The global total fertility rate now is somewhere between 2.2 and 2.3 children per woman. The replacement rate, it's 2.1 in a developed country, and it's 2.3 or thereabouts in a country that's developing.

- 1.22 Roughly speaking, the world as a whole is at replacement level now. But that's only a temporary phase. We're going below replacement very soon. Now, what happens, of course, when you go below replacement is, there's a delay, which we call demographic momentum, between when the total fertility rate drops below—well, there's momentum in any direction, but let's just say, sticking with the example of total fertility dropping below replacement, but in the past, that total fertility rate had been higher. And so there are still a lot of, there's still a bulge of mothers coming up.
- 1.23 If you imagine a pig in a python, and there's still that bulge coming through of mothers from when the total fertility rate was higher; and even if they only have one and a half children or two children per woman, that's still a lot of births and a lot of mothers. So there's still growth in the total population. And especially when you're getting people living longer, not dying off as quickly, the other end of the python, the total population, can still rise for decades after the total fertility rate has fallen below replacement, maybe on average four decades, let's say it takes.
- 1.24 So if we think about Western Europe, the TFR going below replacement in the 1970s, it's not until the 2000s, even the 2010s, depending on which country, that, if we screen out immigration, if there were no immigration, then we would start to see populations declining, total populations dropping. That's not generally happened in Western Europe because of immigration, but it has happened in Eastern Europe. Places where there isn't really a lot of immigration, you are seeing those declines, like Russia, for example, Bulgaria.
- 1.25 So that's a bit of background on the demographic transition from one equilibrium, high birth rates and death rates, to another equilibrium of low birth rates and low death rates. We're now perhaps in a new period where, in fact, the birth rate is below the death rate, so you're going to get population decline. We can talk about that so-called lowest low fertility scenario.
- 1.26 The point about population change, though, is, and the reason is political, like if population change happened at exactly the same rate, in all parts of the world, within every ethnic group, within every religion, within every ideology, there would be a lot less politically going on. There'd still be some political things happening because the age differences would create certain dynamics, but it would be a lot less political.
- 1.27 The issue, however, is, it's not just that population rises and goes down; it's that it rises and goes down at different rates in different populations. And that's why it becomes political, is that means certain groups are growing faster than other groups and making up an increasing or decreasing share of the population.

- 1.28** We can even look at something as small as the Jewish population of England. So let's take even a small unit, like the Jewish population of England, and I've got two charts here showing the age structure of two Jewish populations within England. Salford, which is outside of Manchester, has what we might think of as a classic developing world population pyramid with, if you look at, you can read the age ranges in five-year segments, from age zero to 4, up to age 90-plus. You can see a bulge in the under-20s. A lot of young people in the Salford Jewish population and very few old people. It's got that pyramid shape to it.
- 1.29** Whereas in Leeds, the Jewish population looks like a tower. There's a lot of old people, not that many young. There are quite a few young people, but given that people tend to die off in the older ages, it's surprising to see how many old people there are in the Jewish population of Leeds. That really is referring to the fact that, in Salford, the Jewish population is ultra-Orthodox, where the average woman has 6.5, on average, 6, 6.5 children per woman. And there aren't so many older ultra-Orthodox people.
- 1.30** Whereas the Leeds Jewish population is like most of the Jewish population of England. It's made up of more liberal or non-Orthodox, or certainly non-ultra-Orthodox Jews, and so it has a very low birth rate and a lot of population aging. What that's going to mean is that, by 2050, the fast-reproducing ultra-Orthodox, much younger population, will become a majority of observant Jews in Britain, and that's going to radically change the British Jewish population, its character. And these are almost effectively separate groups within the Jewish population.
- 1.31** It's an example of the kind of change demography can bring that's coming out of that very big difference in the total fertility rate between these two subpopulations. So let's bring in the politics of this. We've talked about demography, which is about birth rates, death rates, net migration. How does that map onto human populations? Now, the first point to say is that, particularly when we're talking about ethnic groups, religions, nations to some degree, there is a degree in human society to which groups are socially constructed. The boundaries are not 100% fixed; certainly with ethnicity, that's the case.
- 1.32** And so, you could argue, well, even if there are different birth rates, say, between different populations, if you imagine, let's say, in the case of the ultra-Orthodox Jews and the mainstream Jews, if there was high intermarriage between those two groups, or if, let's say, ultra-Orthodoxy as an idea was changing very rapidly and conceptions of who is a Jew were changing very rapidly, those demographic dynamics wouldn't matter so very much. Whereas if the groups, in fact, as they are, don't intermarry, are very rigidly defined, then in fact demography matters a lot.
- 1.33** Now, in terms of theories of nationalism, one of the arguments in nationalism theory is, you've got one group of people who say group boundaries are easy to move around, they're flexible, they're invented, they're constructed. Another set of theories say they're relatively rooted, not that easy to shift around, don't change very much over

time. However, what we know is that, depending on which world region we're talking about, in some cases, those ethnic boundaries are rigid, and in other cases, they're more flexible.

- 1.34 In India, the boundary between Hindus and Muslims, for example, two ethno-religious groups, is very rigid. There is almost no intermarriage between these groups. They're very sharply defined by religion. On the other hand, if we go to the Caribbean, say Trinidad is one example, the boundary between ethnic groups is more fluid and there's more intermarriage. What we might say is, any birth rate differentials between groups is going to matter a lot more politically in India than it will in the Caribbean.
- 1.35 Why, in some cases, are those boundaries rigid and why are they looser is a sociological question. Andreas Wimmer argues that particularly patrilineal societies, where descent is passed through the father, tend to have a more rigid boundary between groups than matrilineal descent communities. That's one argument. Another is, perhaps if there's memories of conflict, so Northern Ireland is a place with relatively rigid ethnic boundaries within the UK, compared to England. There's a history of violent conflict there. So that might be another reason, especially where you have a demarcation by religion.
- 1.36 So political demography, that is the politics of population change studies, not so much the increase and decrease of population. It does study that as well, but the unevenness in the growth rates of, and the growth and decline of populations, maybe within a country, maybe between countries. So it's that unevenness, which is really the focus of political demography, unevenness over territory in particular.
- 1.37 The unit of analysis, we might be studying the nation state, but we could equally study civilizations. Is Christianity growing more than Islam, or is Islam growing more than Christianity, through birth and death rates and migration rate? We can take the civilization as the unit of analysis. We could take, I looked here at the British Jewish community. Something as small as that, we can look within that at dynamics of demography, which subpopulations are rising and falling within the British Jewish community.
- 1.38 Now, unlike animal species such as birds and toads and butterflies, we don't have perfectly neatly separated groups. When we're talking about social and cultural groups, it's not the same as different species that are totally separate. If you've got gray squirrels expanding and red squirrels declining, that's not what we're talking about. Because there's a certain degree to which boundaries can be reconstructed, there can be assimilation and conversion if we're talking about religion.
- 1.39 There is a certain amount of fluidity, and one of the questions is always going to be, how much fluidity versus how much rigidity socially, and then what's going on demographically? So there is the possibility of ethnicity being somewhat fluid. Of

course, different theoretical perspectives say it's really fluid, and others say it's not very fluid. This is one of the things that people debate.

**1.40** For example, people can change their identity. If I use the example of the United States, at one time, there was very little intermarriage between Protestants, Catholics, and Jews. Then as we enter the 1970s, intermarriage rates between those groups go up. Then there's this new group created called white, whereas in the past, there had been this Anglo-Protestant WASP group, Catholic groups, and Jews. There was an assimilation into this larger entity called white.

**1.41** That shows that, even though the Catholics probably had a higher birth rate, but once you get intermarriage, that sort of doesn't matter anymore. In fact, the political implications were not as great as we thought. There are different units of analysis. We can look at dynamics at the city level, the regional level, the ethnic group level, the national level, religion, civilizations. We can go really big. We can go global, what's happening in the globe.

**1.42** We can also look at differences by age and sex. Some civilizations, especially, there was a period where there was a lot of, and this still does go on, where there's sex-selective abortion of girls. You can get a country like China, which evolved a unbalanced sex ratio of more men than women in certain age groups. There's one question, what does that do politically to a society? Does that lead to, for example, greater potential for violence in the society or not? That could be a question.

**1.43** Another question is the relationship between age groups. If you get a lot of young people in a society, again, does that lead to more violence? If you have a lot of old people, does that decrease the amount of violence? But then there are perhaps other problems. A young society might be more economically dynamic than an older society. That's also something that political demography looks at—which groups are we talking about as our focus?

**1.44** Then we look at the different relationships between population and politics. Political demography, I'd say, is mostly focused on how population shifts affect politics. But there is also a degree to which there's a traditional interest in demography, that is, birth rates, death rates. How is that being affected by politics as well? For example, is a region such as Israel-Palestine, which has an ethnic conflict, is that going to have higher birth rates? Because there's an ethnic conflict, and both groups perhaps see having children as patriotic.

**1.45** Monica Duffy Toft, at Tufts University, uses the term "wombfare", quoting, for example, the PLO, Palestinian leader Yasser Arafat used the term that "the womb of the Arab woman was the ultimate weapon of the Palestinians." Of course, now the Jewish birth rate is higher than the Arab birth rate in the region, but at the time, the Arab birth rate was much higher.

- 1.46 This was very important in thinking about the future of politics. And this is similar in a lot of conflict regions where you have very tight ethnic boundaries. Northern Ireland, what's going on with the Catholic birth rate, which was higher than the Protestant birth rate? Is that going to mean Catholics will become the majority and take over politically?
- 1.47 It's worth saying that demography matters more in certain situations. So in a democracy, it's one person, one vote. If you have more people, you have more votes. So a democracy, you could argue in a democracy, demography matters more than in, for example, a dynastic kingdom where the soldiers are recruited as mercenaries from elsewhere. The population doesn't have a chance to vote. They're just subjects. Maybe in that situation, demography doesn't matter as much. So there are certain conditions.
- 1.48 Maybe if there is, if robots get good enough to do the fighting, then maybe demography doesn't matter as much for warfare, for example. So under certain conditions, demography matters more; under certain conditions, it matters less. Also, we're saying then that demography works in tandem with other factors to produce political effects. In general, this is what political demographers find. It's not usually the case that you get some demographic change and a one-to-one cause into politics.
- 1.49 That can happen, but often demography works in the background. It's like background noise that's conditioning what is happening. It works in tandem with other factors, such as economic problems, political instability, cultural memories, state fragility. Sometimes it's the interaction of demography with those factors that can lead to, for example, the outbreak of war.
- 1.50 Let's look at some examples of the way demography works in terms of causing political effects. There's an article, actually, that talks a lot about this, this idea of first order, that is, demography causes social or political change. Second order, demography causes something which causes social and political change. Third order, demography causes something which causes something, and that causes the change. So let's look at an example of first order change.
- 1.51 I mentioned the Jewish community in Britain. To the extent there's a political and cultural conflict between the ultra-Orthodox, with their interpretation of the Torah and of politics and the mainstream Jewish community, which is not ultra-Orthodox, which is either Orthodox or reform, more liberal, call them denominations. The high birth rate of the ultra-Orthodox means that they will be the majority by mid-century, and that will shape the Jewish community in important ways, and it'll shape Judaism in important ways. So that's a direct effect. Higher birth rates leading to a political effect in terms of political weight.
- 1.52 We've seen that in other ways. The rise of the evangelical Protestants in the United States in the 20th century was, some would argue, a predominantly demographic event, because evangelicals had about a one-child birth rate advantage through much of the 20th century, so they became a much larger share of the Protestant population.

And that in turn had other effects on politics, such as the rise of the moral majority in the 1980s.

- 1.53 We then get an example of second-order change, which is that demographic changes affect something which then affects politics. In this case, we use the example of birth rates or death rates affecting age structure. That's the second thing. And then the age structure has an effect on politics.
- 1.54 So, for example, if you imagine a fall in the birth rate, so that creates more old people and fewer younger people, which means it's an older population, and that increases the pension burden on workers. So that's a conversation you'll have heard, undoubtedly, and we'll hear about more and more. So there are demographic factors in terms of, falling death rate, falling birth rate means a more elderly population, means a bigger pension burden.
- 1.55 Another example could include a boom in young people through a high birth rate. If you had a high birth rate, a large increase in young people, they reach age 18 to 25, they reach an economy designed for a smaller number of people. There aren't the jobs to employ that bulge of 18, well, of young people, there are going to be a lot of young people who don't have work. So that's another example of a two-stage process, starting with a demography in which you have a lot of young people, and then you don't have employment, which leads to mass unemployment of young people.
- 1.56 The third order effect would be where demographic change affects, let's say, the age composition, that's the first order, second order, and then that indirectly affects outcome. For example, you have high fertility, for example, which leads to more young adults needing work. That leads to unemployment because the economy is designed for a smaller population, which in turn leads to insurrection.
- 1.57 Why? Because the people have no other options, and it's therefore a low-cost option for them to be recruited into an insurgent movement. And so it may be the case, therefore, that a very young population is connected to more violence, more civil war, more rebellion. That's the third order effect then, going from high birth rates to a very young population to large-scale unemployment to insurrection.
- 1.58 We can even talk about fourth order phenomena. This is showing you how demography works in the background. I use this quote from Prime Minister of Northern Ireland, Terence O'Neill, back when Northern Ireland had its own government, its Stormont. It does now, too, but doesn't have a Prime Minister. But Terence O'Neill used this phrase. I don't know, it's been attributed to him, the quote, "The basic fear of Protestants in Northern Ireland is that they will be outbred by the Roman Catholics. It is as simple as that."
- 1.59 So we'll take that statement from Terence O'Neill in 1969. 1969 was the beginning of the civil war in Northern Ireland, known as the Troubles. The Good Friday agreement of 1998 put a definitive end to that, but it was a very prominent civil war. When I first

came to Britain, that was still talked about, and there were still armed guards going around. What was going on in the Northern Ireland case?

- 1.60 You could argue that high Catholic fertility compared to Protestants was leading to an increase in the Catholic population share in Northern Ireland, which had been designed with a two-thirds Protestant majority when Northern Ireland was created in 1921. The Catholic fertility rate then meant that Catholics were gaining on the Protestants, which meant, potentially, on a one-person, one-vote basis, the balance of power would start to shift towards the Catholics, and they would eventually perhaps bring Northern Ireland into a united Ireland.
- 1.61 That was the Protestant fear. Catholic fertility and growth, therefore, leading to Protestant fear and anxiety over the growth of the Catholic population. It's that anxiety which led the Protestant-dominated government to say, actually, we don't want to reform the voting system, because they gerrymandered the voting system to ensure that they actually could dominate across all parts of Northern Ireland, including Catholic-dominated areas. So they didn't want to reform.
- 1.62 They didn't want to make things too good for Catholics, particularly who were the working class and poor in many cases. And so they didn't want to reform, and that led to frustration from Catholics, which was very important for the IRA to actually recruit people into the movement, the Northern Ireland Civil Rights Association. That's the beginning. Frustration over the lack of democratic reform is the beginning of the success of the IRA.
- 1.63 And then, of course, you get the civil war, as you get Protestant retaliation against Catholic violence. So this is a multi-step process in which fertility matters a significant amount. That is, the higher growth rate of the Catholic population through higher fertility was a factor in the Northern Ireland civil war. Now, you wouldn't say it's the only factor, but it's playing a role.
- 1.64 All right, I want to come now to discuss the more global picture and why we're having this conversation now. Why now? Why are we talking about the politics of population in the year 2025? Why does it seem like it's more important? I edited a book on political demography in the early 2010s. This discipline, if you want to call it that, was really just started. No one had really written about this in academia, in any serious way. There had been a few books, isolated, little book here, little book there. Now, it seems like you can hardly go a week without people talking about population shifts.
- 1.65 Why is that happening now? If we look at this line of world population, you can see that it's really flat, gentle rise, and then we come into the 19th century, and then we come into the 20th century, and then just explodes, the population. We reached 2 billion people, maybe around the Second World War. Two billion around the Second World War. We're now at 8 billion. To get from 1 to 2 billion, that's basically from about 1825 to 1940, got us from 1 to 2 billion. 1825 to 1940, it's over 100 years. Going from 2 to 3

billion looks like it's maybe 20 years, and then from 3 to 4, five years, 4 to 5, suddenly we're at 8, this takeoff of population.

- 1.66** Now, what this graph doesn't show, because it only goes to 2050, is that, in fact, there are projections now that the peak of world population will be at around 10 billion. So we're at 8 billion. We're going to go to 10 billion. But what's likely to happen is, we're going to plateau at around 10 billion. And then it's—now, it's not going to collapse the way it exploded, but the curve could still be logarithmic.
- 1.67** Because the reason you get this population explosion, as I mentioned, is the falling birth rates and death rates, but the death rate falls first, the birth rate comes later. So the death rate fell first, and that meant more kids survived, which meant more mothers, which meant more kids, more mothers, and that's the exponential growth rate.
- 1.68** We now hit this peak. We're at only a replacement-level birth rate in the world. We'll peak at around maybe 10 billion people. The total fertility rate will be low, but there will still be quite a few mothers in the system because of previous population increase. So we'll gradually go down, but if nothing changes with that total fertility rate, what will happen is, the curve will start to look like this. And so we'll get an acceleration of population decline through this kind of a curve.
- 1.69** That's the discussion now. Now, of course, what's important is, this boom and decline is happening at different pace in different parts of the world, within countries to some degree. It's happening at different rates between groups, and that's upsetting the mix within countries, globally as well; and especially where you have democracy, one person, one vote, that becomes important for power.
- 1.70** So the question of why now? Well, we're going through a demographic revolution. Now, that revolution picked up pace in the 20th century, and so it's been going on for a while. But there are some other factors, too, that might make our period of history even more important in terms of the difference between different parts of the world, maximal difference in terms of the rate of demographic transition. And it's that difference in rate of transition between different parts of the world across territory that really brings the politics in.
- 1.71** That unevenness of change, of what is very much a rapid change, that boom and then that decline, and the unevenness of that, is bringing about political consequences. We also have what's new, you could argue what's new in our time is, there is a more advanced globalization. So more people around the world are aware of what's happening in the West. People in the West are aware of what's happening in other parts of the world. Not always, but yes. So you have global communication technologies, satellite television, the Internet, social media, cheap flights, compared to, let's say, a hundred years ago.

- 1.72** So globalization is an added ingredient in the mix, which might make—so one of the things that we see, for example, with migration is, you always had—the number of people who live in a country other than the one they were born in hasn't changed very much. But long-distance migration between the Global South and the Global North has increased a lot. So the number of people who were born in the Global South who live in the Global North, if you like, that's mainly the Western countries, has risen steadily from the 1980s. That is actually different from the past.
- 1.73** Democratization. Now, there was a phase in the 1970s, really up through the 1990s, where more countries were democratizing. Again, democracy means one person, one vote, means that demography or population change is more important. When Iraq moves from a small clique around Saddam Hussein controlling things autocratically, when Syria moves away from Assad's regime, which is an autocracy and people aren't voting, it matters.
- 1.74** I'm not saying—clearly, the balance of the population does matter, particularly militarily, but it's not as immediate in terms of its effect on power. Whereas once you get democratization, if Syria actually has democratization, then minorities such as the Alawites or the Druze or the Christians, to the extent they are identified with one particular political party, then they're going to be out of power every election. If the Sunni Arabs vote a certain way, and the non-Sunni Arabs vote another way, the non-Sunni Arabs will be out of power.
- 1.75** Similarly, in Iraq, the Shia are a demographic majority. They were disenfranchised under Saddam, but under one person, one vote, unless there are other principles like power-sharing, the Shia are going to be in power in that country. So, democratization is another factor which makes demography important.
- 1.76** If we look at a map of the world, we can see that total fertility rates are lowest in the developed world, and they're higher in that tropical belt. If we start in the West, it's sort of the Andes going into Central America, which has the highest birth rates in Latin America, although even their now birth rates are, the total fertility rate's going at or below replacement. If you go over to Africa, it's sub-Saharan Africa that really stands out as having considerably higher total fertility rates than either North Africa, or South Africa actually, not to mention the rest of the world. Asia, North America, this is all below replacement.
- 1.77** In fact, it's the case that almost all of the world's population growth is taking place in that tropical belt we call the global south, including sub-Saharan Africa, Southeast Asia, West Asian countries like Afghanistan and Pakistan. That is really where all of the, essentially 97% or something like that, of the world's population growth is taking place. That's going to mean that, if we have immigration from the growing parts of the world to the parts of the world that have population stagnation or decline, almost all of the sources of immigration are going to come out of that tropical belt, which is where all of the population growth is in the world.

- 1.78** That belt will become increasingly African as we move into the middle and late 20th century. So sub-Saharan Africa will dominate increasingly in terms of population growth. But even now, the question is—of course, sub-Saharan Africa is also going through the demographic transition. So the black population of South Africa has more or less a replacement birth rate. So we are also going to see transition in sub-Saharan Africa, but it's slower because this is a more agricultural, less urbanized.
- 1.79** Women's education, which is the primary correlate of fertility rates, is still quite low, particularly secondary female education. When women are in their reproductive ages, if they're still in school, that reduces the birth rate. Secondary education is not yet, is not yet extensive throughout sub-Saharan Africa. It will be a while before those total fertility rates get to the replacement level. This is one of the reasons why the share of the world's population made up of sub-Saharan Africa is set to grow dramatically between now and 2100.
- 1.80** I finish with a set of what Jack Goldstone, who I've worked with and who is a key figure in the literature on political demography, calls the megatrends. Now, these megatrends, I remember being on panels where Jack rolled out the megatrends. And one of the ones that was actually the most prescient—he was talking in the late 2000s, so 2008, '09, '10, this period—so he went through a number of the megatrends.
- 1.81** And one of the ones that really strikes me now is this idea that humanity becoming a common disease pool. So you don't have isolated parts of the world that have no contact with the global economy, with global transportation. And therefore, all of humanity becoming a common disease pool, and that's just very interesting because we saw that demonstrated powerfully during COVID, that something can happen in China and it can suddenly go global. So this idea that we are now united into a common disease pool, I thought was an interesting demographic megatrend.
- 1.82** What are the other ones? The rise of the non-West and the decline of the West in population terms. That's one megatrend, clearly. The West has been dominant in one way or another since the 1500s. What does this mean for the political and cultural dominance and economic dominance of the West in the world, when its population is shrinking as a proportion of the total? The aging of the developed world of the West, East Asia, that's a second megatrend.
- 1.83** The third megatrend, which may be becoming a little bit less of a megatrend but, is the youth bulge. The youth bulge really refers to a large number of young people. That's really taking place in developing countries that are still relatively early in their demographic transition. That is, they've just relatively recently seen the decline in death rates, typically infant mortality rates.
- 1.84** And because of the decline of the death rate, the death rate falls below the birth rate. The birth rate has yet to fall, so they're in that population explosion phase, which happened in Europe in the late 18th, 19th centuries, creating a lot of young people,

who therefore may be recruited into insurgent movements. So youth bulge is another megatrend.

- 1.85** Mass urbanization, particularly in the developing world, you are getting the growth of cities of tens of millions. And we will see more and more of that in places like Ethiopia, Addis Ababa, Lagos, but in Southeast Asia as well. Mass urbanization. As in the developing world, in particular, you tend to get large-scale growth of cities, initially through the development of slums, which may eventually become integrated into the larger city region. And so, mass urbanization, mass movement from the countryside into the cities, is another megatrend. That's a demographic megatrend as well.
- 1.86** We mentioned the common disease pool, the creation of a common disease pool on the planet. Then there is this question then of migration. Because of the aging of the developing world, the youth bulge, that combination means that the difference between the Global South, that tropical belt, and the developed world, which is more outside the tropics, the difference in terms of population growth will reach a maximum between now and 2050. That's going to create tremendous migration pressures.
- 1.87** That's somewhat mitigated by the fact that countries tend to send the most migrants when they are in the middle income levels, rather than when they are very poor. When they're very poor, people don't have the money to move. As they become slightly better off, people get the money to move, and that's when migration tends to increase. So the migration potential, for example, of sub-Saharan Africa, may actually be higher in the future than it is now because people will eventually get the means to migrate more. But mass migration clearly is the last megatrend, and that, of course, is having considerable political effects.
- 1.88** Thank you very much.
- 1.89** So you mentioned there's a decline in the total fertility rate in developing countries. What contributes towards that?
- 1.90** The most important correlate is women's education levels, especially secondary schools. So the more young women that go to secondary school, the more the birth rate is likely to fall as a result of that. So that's the main driver. Now, of course, also urbanization means that it's more expensive to raise children in cities than it is in the countryside, and they're less useful in terms of the household economy, and therefore it makes sense to have smaller families in cities.
- 1.91** In Africa, we take the case of Ethiopia. The capital, Addis Ababa, has a replacement level fertility rate, but the country as a whole has a much higher birth rate. That's an example of where urbanization plays a role. Women's education plays a big role as well. There's also this question of, if you know that medicine is good enough, if you know that if you have two, they will survive, you don't have to have four to make sure two survive. Then that also will have an impact. In a way, to some extent, the state of medical technology also matters.

- 1.92** You mentioned one of the megatrends is mass migration. So one question is, how do we integrate these people that are coming from such different cultures into Western society through assimilation? Can you think of ways that we can do that, and countries where that has happened successfully?
- 1.93** So I mentioned that human populations, unlike animal species, that group boundaries are to some extent socially constructed, to some extent fluid. It depends which group, how different they are, what the line of difference is. Now, I don't think there is evidence that state policies of assimilation are integrated, certainly state policies in Western countries. There are not many examples of successful integration policies led by the state, I would argue.
- 1.94** When I say integration, I mean reducing, let's say, residential segregation, increasing intermarriage, for example. There aren't a lot of examples of that. It seems like an organic process. Now, there are examples in the past of successful assimilation. I mentioned the US case where you had immigrants from Southern and Eastern Europe arriving starting around the 1890s. They essentially lived in separate areas, relatively separate ethnic areas. They went to separate churches. Catholics attended Catholic church, often separate schools.
- 1.95** It was only in the 1970s, the late '60s and '70s, you started to get more intermarriage across that religious boundary. It was an organic process, actually. We're talking 70 years, three generations. It takes a long time for, I think, what I would call deep assimilation to take place. Even within groups that are European, it took that long. Now, the Protestant migrant groups, the German Scandinavians, assimilated more quickly into the Anglo-Protestant group. The Catholic groups intermarried.
- 1.96** All of that happened a little earlier, but these processes take time. This is one of the things I'm not sure there are many policies that will make a difference in terms of the speed. But yes, it has happened successfully in the past, obviously. I used that example of the US case, where Irish Catholic or Catholics and Jews have melted in, become this ethnic majority white population, where those differences don't matter very much.
- 1.97** The next question, of course, will be, which other groups are going to melt in and how? So in this country, for example, the assimilation and intermarriage rate of, let's say, Chinese, British, and Afro-Caribbean British is much higher, let's say, than Pakistani or Hindu or Sikh British. One reason might be, there's a bit more of a religious difference. Maybe that means that assimilation is slower.
- 1.98** You pointed out that research suggests that one factor, very, very important factor, of declining birth rates is more accessible education for women, especially second education levels. How does it work then? Because we fix one problem, we have a side effect on another side. Are there any countries or any systems which found a solution to that? How can we deploy world-class education to women without shrinking the birth rate in long term?

- 1.99** Well, yeah. There's almost a different debate in the developing world, where it's about women's education to reduce a very high birth rate. Then the developed world where, in fact, women's education doesn't make that much difference to birth rates. In fact, the highest educated women have actually somewhat higher birth rates than moderately educated women. Then those with very low education, again, have slightly higher. But it doesn't matter that much.
- 1.100** In any case, everyone's more or less going to school and going to secondary schools. I would say that the education issue is not so much of an issue in the developed world when it comes to fertility rates. It's much more to do with, for example, religious attendance, to some degree ideology. These kind of things seem to matter more than education in the developed world.
- 1.101** Which has more of an impact on society, in your opinion? People choosing not to have children, like couples not having the money, or the rise of infertility?
- 1.102** Well, I think it's the choice. I don't think infertility as a biological thing is a major reason why we have below-replacement birth rates in the West. I think it is almost entirely about choices. Because, one of the things—if we think about human history, essentially before the age of the birth control pill and contraception, you had to have as many children as you could. There was high maternal mortality, high infant mortality. Now what's happened is, with the pill, we've cut the link between sex and procreation, so it's a choice. So now having kids is a choice. That brings culture in much more.
- 1.103** And you mentioned the cost of having kids. The cost of having kids is partly governed by culture. So for example, if you are determined to have a large family, even if you're poor, even if you can't give your kids all an iPad, then that's a cultural choice. And that's one of the reasons why some of these world-denying religious sects, like the Amish or the ultra-Orthodox Jews, have such high birth rates. They don't necessarily care about achieving the markers of material success of other parts of society.
- 1.104** Actually, fundamentally, this is much more cultural. Whereas if you were to go back to much of human history, if you didn't have kids, then you might starve eventually, because you have no one to work the land. And so, it was much more a material necessity for everybody. So it didn't really matter if you were a religious conservative or an atheist; everybody had to engage in the same behavior.
- 1.105** The so-called second demographic transition theory argues that, once we enter a new world where having kids is a choice, it's not necessarily so much of an economic decision; it's much more governed by norms and values. I would say norms and values are now more important than they were in much of human history for determining fertility rates.

## Lecture 2: Population and Environment

**At a glance** — Population and environment — the nexus of population, resources, and climate.

- 2.1 In this lecture, what I'm going to be addressing is the nexus between population, environment, and resources, in the context of climate change. So even though most of this course is going to be focusing on the unevenness of population growth between groups, between civilizations and countries, there is a part of political demography that isn't focused as much on differences between, but is focused on the relationship between pressure on resources and politics.
- 2.2 And this involves a debate between two sides, what we might call the Malthusians and the cornucopians, people who believe that population pressure from, let's say, well, that population pressure does lead to violence in a world of scarce resources, and another view that says human ingenuity will always get us out of these population resource pressures through innovation.
- 2.3 If we take Malthus, when he was writing in the early 19th century, he focused a lot on this idea that food production grows arithmetically. By the addition of new technology, for example, you can improve food production arithmetically. You can improve it on a line like this. But population grows geometrically. More mothers, more children, more mothers, more children. The line of improving food production is linear and arithmetic, and the line of population is geometric like this.
- 2.4 The theory, therefore, is that, in the long run, population increase will outstrip the food supply, resulting in starvation, which will be a natural rebalancing of the system. As people die off, the population will then be forced below the line of food production, and the cycle begins again. This is obviously quite a pessimistic view that, in a way, the only way you get stability is through some kind of a natural check. The environment is a natural check on human reproduction. This is part of the dismal history of humanity.
- 2.5 This tradition, this Malthusian tradition, has also been important in a number of more contemporary works. Most famously, Paul Ehrlich's *The Population Bomb*, from 1968. 1960s was the peak in terms of global population increase. Population of the world in the 1960s was, of course, much lower than it is now, but the rate of increase was very fast in the developing world. This led to predictions that the population would simply balloon and explode out of control in the developing world, leading to mass starvation, famine, violence, etc.
- 2.6 That theme is, to some degree, echoed in books like Stephen Emmott's *10 Billion*. This more in the context, not so much in the context simply of mass starvation, but the context of global climate change and pressure on a world with finite resources. So, there is a tradition then of Malthusian theorizing and writing when it comes to population and society.

- 2.7** Against this, we have the Julian Simon cornucopian school, which very much believes that no humans are—human ingenuity and technology has been able to solve many problems in the past. We went from being a hunter-gatherer society to an agricultural society, where we could support a much higher population density, to an industrial society, and can support an even higher population density, and we'll continue to make those innovations to support higher levels of population density.
- 2.8** Different groups from both the left and the right support different positions. So in terms of cornucopians, we have an unusual set of bedfellows. You have free market neoliberalism, which is very cornucopian, believes in technology and progress. Then you have Marxism, as well. Marx and Engels very much opposed Malthus' view, and they argued that, to quote Engels, "Scientific progress is as unlimited and at least as rapid as that of population." So both the Marxists and the free market economists are in bed together on this question of population pressure.
- 2.9** And indeed, we see in the 1960s and '70s, at the time Ehrlich was writing, the results of what's called the Green Revolution, which really revolutionized seed production and food output. In fact, if you look at the chart, the production of cereals and other food stuffs was rising fast enough to feed a growing population. In fact, GDP per capita continued to rise during this period in the developing world.
- 2.10** Why? Well, chemical inputs such as fertilizers and pesticides, new irrigation technologies, farm mechanization, and high-yielding varieties of staple crops such as rice, wheat, and maize. That is how the Green Revolution staved off mass starvation from that 1960s population explosion in the developing world. The question, of course, is, can this be sustained indefinitely?
- 2.11** The skeptics, the more Malthusian skeptics would say, "Actually, the Green Revolution and the excesses of the Green Revolution are already starting to show, for example, soil depletion and erosion, and that this kind of innovation is not going to be possible indefinitely." And so, in fact, there will be a moment of truth when the rubber hits the road and Malthusian logic kicks in.
- 2.12** There were a number of books written in the 1980s and '90s on the potential for growing conflict in the world. There was a whole genre of Malthusian writing about demography in society. Jared Diamond's book, *Collapse*, for example, talks about how environmental collapse historically has played a major role in the decline of civilization. You think of a place like Easter Island, whose civilization just suddenly declined and had to be abandoned, or perhaps Mayan Mexico.
- 2.13** Why did these civilizations decline? This is a question, and one argument is that ecological collapse, the depletion of forests and soils or other kinds of external ecological factors, are what led to the decline of civilization. And that in that context, we have to think about the global depletion of resources in the context of climate

change. Robert Kaplan's work on the coming anarchy, describing the civil war in Sierra Leone, for example, as being in some way related to pressure on scarce resources.

- 2.14** This, again, all tying in with the rhetoric on global climate change, and what the effects of global climate change are going to be, as we move towards a warmer world. We're going to see an increasing number of severe weather events, which are more intense, heat waves, droughts, pests, loss of agricultural productivity, possibly a nonlinear tipping point in which temperatures really rise very rapidly. Melting sea ice, rising sea levels. How is that going to alter the climate? And will that lead to climate migration, pandemics, famine?
- 2.15** This is this Malthusian tradition of pressure from the environment and from population on scarce resources. In terms of mapping the impacts, a number of writers and policymakers have sought to try and map the impact of climate change in the world. First of all, there is the temperature and sea level rises. Where in the world—clearly, if you're in the Maldives and you're low-lying, then you're going to be very threatened by rising sea levels, much more so than if you are a large country like Canada or Russia.
- 2.16** That's one question, is the issue of which parts of the world are going to be most affected. Bangladesh, low-lying, again more exposed than a country that is more, obviously, that is inland. But equally, you then have the question of which countries are best able to adapt. Even if you have a lot more rain or extreme weather events, if you're an advanced society, you can build flood defenses, you can have evacuation routes, you can have better fire protection for forest fires. You'll be able to adapt much better to these more extreme weather events than a developing country that simply doesn't have the resources to do that.
- 2.17** Actually, the areas of greatest vulnerability to climate change will tend to be in the developing world, the tropical belt, Central America through sub-Saharan Africa, through into west Asia. That is going to be more exposed, perhaps to the effects of climate change; even if climate change itself is not necessarily going to hit harder in those regions, their capacity to adapt is going to be weaker.
- 2.18** In this vein, writers like Thomas Homer-Dixon argued that, "We are on the threshold of an era in which armed conflicts and security concerns will come frequently and primarily from environmental change." That's a very Malthusian worldview, writing in the '90s. It's worth saying, 1991 was the peak of ethnic civil war. Since then, we've seen somewhat of a decline. 1991 had a lot of ethnic civil war, in a context where there was also discourse around climate change, and so this may have shaped Homer-Dixon's theorizing.
- 2.19** What are the mechanisms that Homer-Dixon puts forward? One is a decrease in the supply of renewable resources, such as food, due to climate change or exploitation. Not just food, but also perhaps forests, which are very important in developing countries as

fuel for heating homes, cooking, but also help actually to stabilize the soil and prevent erosion, which then can affect agricultural production.

- 2.20** So, one argument is that we're going to have a decrease in the supply of these renewable resources due to climate change, and heavier exploitation because of growing populations on finite resources. An increase in demand for renewables, therefore in a world of scarcity. That is also going to exacerbate inequalities in the distribution of resources across the world, but also perhaps within countries, and that is where politics is going to enter the picture. And so, we're going to have almost this growing population pressure on resources, leading to revolts and rioting.
- 2.21** The first place we'll likely see it, Homer-Dixon argues, is in terms of local environmental pressures—deforestation, overgrazing, the way in which pesticides poison the soil, more intensive fish farming, perhaps leading to problems with species being wiped out or getting disease, draining of a water table due to more intensive cultivation, erosion, silting up of rivers, bacterial blooms. The damming and diverting of water supply could lead to drought as rivers are depleted or don't have as much water flowing through them.
- 2.22** Then you have industrial processes, such as nuclear power, mining, and industrial production that can lead to more toxic waste and pollution of groundwater resources. These are some of the mechanisms that Homer-Dixon talked about. It's entering into an interplay between massively increased population growth in the developing world because of, again, that demographic process of death rates dropping, infant mortality rates dropping, birth rates remaining high. Birth rate minus death rate is positive, leading to a boom in population and more pressure, particularly in rural areas.
- 2.23** So sub-Saharan Africa will be the main zone where this will happen, but also places in south and southeast Asia that are poor and least able to cope with climate change and pressure on resources. We're going to see this degradation of fresh water supplies, pressure on agriculture, possibly rising sea levels and flooding, leading to environmentally-induced mass migration. Some claim that, Stern claims that 200 million or more may be permanently displaced by climate events, all of which will make reducing poverty much more difficult.
- 2.24** Now, against the Malthusians are a series of more what we might call cornucopian skeptics of the processes that Homer-Dixon lays out, and Kaplan and others. One way in which cornucopians rebut the Malthusian claim is to say, "Let's look at history." The historical record does not seem to show many cases in which environmental problems led to large-scale conflicts. Now, of course, we have books like Diamond's book, *Collapse*, that claimed that environmental change was the cause of the decline of civilization. This might have been the case in islands like Easter Island, but it's far from conclusively proven.

- 2.25** If we take current models of conflict, where we might look across the world, say which countries are involved in civil war, and say what are the characteristics of the countries that are involved in civil war, versus the ones that are not involved in civil war, we don't find that the countries that have more degradation, more population pressure, seem to have higher levels of civil war.
- 2.26** So the current models of violent conflict in the world, particularly civil war, do not find many strong correlations between variables like population growth and density, or intensity of resource use and civil war. You don't find a place that is very densely populated, for example, having more civil war than a place that's lightly populated.
- 2.27** The other point is that climate change induces, typically, gradual change. Let's say you have, for example, more rain or drought. That change may come in over time, but gradual change is difficult to tie to sudden events like the outbreak of war. So gradual processes are often not directly linked to civil war outbreaks, for example.
- 2.28** It's also the case that climate change can have different effects. For example, the Sahel region, which is just south of the Sahara, has actually been getting greener rather than drier, as was predicted, say, in the '80s and '90s. So it seems to have been getting greener. In some areas, climate change seems to be leading to more rain, in other places to less rain. So it may aid agriculture in some cases. It may harm it in other cases. So on balance, the effect may not actually be negative.
- 2.29** One example of this cornucopian skepticism towards the Malthusian case, that we're going to see mass conflict and migration as a result of climate change, is, there's a literature on so-called water wars, whether those who share a lake, like Lake Chad or a river like the Nile River or the Mekong River, different countries share these bodies of water. They use these bodies of water. Do we see countries fighting over water use? Does, for example, Sudan, which is on the Nile, has Sudan gone to war with Egypt, which is also on the Nile, or Ethiopia gone to war with Sudan?
- 2.30** One of the arguments that a scholar named Geoff Dabelko makes is that no nations have gone to war specifically over water resources for thousands of years; that, in fact, there isn't a record of this being a major cause of warfare. Others would say, well, De Belko has a point, but this is a bit of an overstatement. One of the questions we could ask is, well, actually, why is it the case, given that water is a scarce resource, particularly in dry environments, like Egypt, why wouldn't it be the case that Egypt and Sudan would fight over controlling water? If Egypt is using too much of the Nile, why wouldn't Sudan be angry or vice versa?
- 2.31** Egypt and Sudan, actually, if you look historically, they have a number of agreements in terms of water use. The first was reached in the colonial era between Egypt and Britain in 1929, and that was later renewed in 1959 in the postcolonial period. It seems as though Egypt and Sudan reached an agreement on water use on the Nile. And so, this also, this limits dam-building, it limits development on the Nile through agreement. So

that seems to have been resolved quite peacefully, despite the incentives, given how scarce a resource water is.

- 2.32 And we can see other examples of where this has taken place, often in regions where countries are at war with each other or on different sides. So Vietnam, Cambodia on the Mekong River, Israel and Jordan on the Jordan River, India and Pakistan on the Indus River—in all these cases, there were river commissions which have functioned even during times of war.
- 2.33 Jeff De Belko argues, in fact, sharing a lake or sharing a river actually can be a source of interdependency and trust-building, rather than a source of war between different countries. We don't have strong evidence then, that in this case, a particular resource, fighting over that resource has been common.
- 2.34 What about then the question of resource depletion, pollution, and degradation? Now, Jack Goldstone, who is a political demographer who's written a lot on this topic and other topics around political demography, argues that long-term environmental degradation, that slow change which climate change would exacerbate, long-term environmental degradation has little or no significant role in fomenting conflict.
- 2.35 Why might that be the case? Because if changes are happening gradually, it's easier to adapt to change. If it's happening over a period of decades, people make decisions over a much shorter time horizon than that. So you can switch from resource-intensive crops, for example, very thirsty crops like almonds, to less thirsty crops, so a less water-intensive crop can be planted. We're also in a situation of urbanization. The growth of cities now means people in rural areas have the option to migrate to the cities.
- 2.36 Now, that can lead to its own pressures, but it does offer a kind of safety valve. So rather than thinking necessarily about long-term climate migration, say, from the developed world to the developing world, it may be that urbanization relieves that kind of pressure, if we're getting desertification, droughts, or floods. So, changing crop use, urbanization, these are two means in which people can relieve the pressure of those long-term, slow-moving environmental changes.
- 2.37 It's also the case that rural areas may not actually be politicized or organized for rebellion against the center. If you have a particular, let's just say, politics runs on ethnic lines in a country like Myanmar. Well, in the main ethnic Burman region, if you have environmental pressure in that region that is politically dominant, it's difficult to see how that would necessarily play within a political conflict which is structured around ethnicity.
- 2.38 It may just be that these are rural areas. They have their local problems, but that doesn't necessarily map onto any obvious political cleavage, so it won't necessarily lead to political rebellion. It may lead to certain pressures. It may lead to certain democratic pressures, but it doesn't necessarily lead to violence. In order for violence to happen,

the discontent in the rural areas has to connect to a faction within political conflict, which often takes place in the center in the capital.

- 2.39** So it may be, if the environmental pressure is taking place in an ethnic periphery in Myanmar, that can add to the grievances that that periphery has vis-a-vis the Burmese center. And in that case, perhaps it can contribute to separatism and civil war. So population and environment, the Malthusians or others might say, well, actually, yes, it's the case that it doesn't have a direct effect. It is the case people can adapt to slow environmental changes in different ways, and states can address this also in different ways nowadays because modern states have technologies that can mitigate.
- 2.40** Even poor countries have some capacity to mitigate these environmental changes. Now, most of the war in the world takes place within states. Almost all of the war is happening within states. We want to look at intrastate violence, so civil wars, rather than wars between countries. It's also the case that, it seems from the literature, that if environment is going to matter for conflict, it'll matter more at a smaller scale, rather than at the larger scale, certainly of interstate war; even of ethnic civil war, it may be that these are small-scale conflicts.
- 2.41** What about sudden climate events? We talked about climate change through drought, for example. But equally, we might get, for example, a hurricane that is a much more sudden event. Could a sudden event like a hurricane not be connected with violent conflict? We don't seem to see that, actually. But it's not that these sudden climate events don't matter at all. It may be that they are conditioning factors in conflict, rather than direct causal factors.
- 2.42** It may be that they interact with other factors, and it may be that a whole series of events have to fall into place, including climate pressure, for there to be the impetus for conflict. Koubi argues in particular that climate change does matter, and it matters in this indirect interactive way, not in a direct way. So, drought more than flood, droughts can actually put pressure on the incomes of rural people, and that can be a conditioning factor, one element amongst others that could lead to, could sort of spark a conflict.
- 2.43** But this often affects low-level conflicts in the countryside, rather than a larger-scale civil war that would involve battles between factions, possibly in cities. That what this affects is not necessarily whether a conflict takes place, but how intense the conflict is, how long it lasts, how severe it is. So that it still matters, but even if we can't tie it specifically to one particular civil war, or as the ingredient that caused a conflict. For example, the Syrian conflict, some people have talked about the role of drought as a factor in that conflict, but it's been very difficult to prove.
- 2.44** Most civil wars take place in the countryside. That's where rebel movements can hide and organize. There is a history of land conflict, and some have said that, in fact, this can be tied to rising food prices as a result of growing population pressure on

environment. So there's a whole literature that looks at food prices as an indicator of rising population and food prices. So if we get rising populations, that leads to rising food prices, and that can be a precondition for revolutions.

- 2.45 Jack Goldstone's work on revolution shows that, prior to the French Revolution of 1789, prior to the English Civil War, prior to the Russian Revolution, the 1848 revolutions in Europe, and even the Chinese Revolution of 1949 that brought the communists to power, those were periods of rising food prices. And rising food prices, of course, does involve, can involve growing populations. So where you have growing population, as a result rising food prices.
- 2.46 Now, of course, food prices can rise for other reasons. It could be that the climate gets colder, you have a hard winter, so food production is lower. So it could be environment, it could be growth of population, as Goldstone argues in his book. But this combination of rising food prices as a result of either population or climate is a factor that contributes to the risk of revolution, argues Goldstone.
- 2.47 Henrik Urdal, who is a Norwegian researcher who has done a lot of work on the relationship between population, environment, and resources using quantitative data sets, argues, OK, it's not the case that population density or growth rate in a country is necessarily linked to a greater risk of civil war within the country. But if we go to the sub-country level—now, most data sets in the world operate at the country level.
- 2.48 We have data on Burkina Faso. What's its population density? What's its population growth rate? Is it at war or not? We have that country-level population data, and that very rarely shows a connection between a variable like population density or growth and violence, however—when you control for other things, such as GDP per capita. But if you go below the level of the nation state and you have data on regions within a country, Urdal's argument is, looking at that regional data, actually you do find some significant correlations now.
- 2.49 The regions that have, particularly rural regions that have fast population growth and high population density do seem to show a greater propensity for local-level violence. There does still seem to be some relationship between population pressure on, particularly, food resources and the likelihood of conflict. This can also happen because landlords, particularly where people don't own their own land, the land that they're farming, the landlords may be linked to a particular faction at the center.
- 2.50 If there's anti-landlord sentiment as a result of population pressure or environmental pressure on resources, then this can find an outlet against those landlords and against the political party they support. Insurgent groups can harness this rural discontent over pressure on resources to recruit for insurgencies against whoever is connected to those landlords, which may be the government. If the government has relationships to the landlords, then population pressure can fuel discontent with the existing government, and that can lead to rebellion.

- 2.51** Now, urbanization, moving from the country to the city, can relieve that pressure. This is one of the arguments that the cornucopians make, as well. Again, people can change crops, they can change occupation, they can move to the cities, and that will remove some of this pressure that may be leading to rural discontent. Again, that rural discontent has to be mapped onto one of the factions, the established factions within a conflict.
- 2.52** So let us say, an ethnic separatist movement vis-à-vis the center, that if the population pressure on food resources is in the countryside where the ethnic minorities live, and there's a separatist movement there, then that can help the separatist movement to recruit people. In that roundabout way, then we may get an effect from environment and population on the likelihood and the severity of civil war. That is one way in which this works.
- 2.53** What about sudden events? We've talked about long-term climate change, its impact on, for example, droughts and erosion or floods and what that will do to crops. But what about sudden events, like Hurricane Hazel in Haiti, 1954, or the Chernobyl nuclear disaster, or the 1988 Armenia earthquake, the Pakistan 2013 earthquake. Now, earthquakes, of course, are not due to climate change, but hurricanes are and floods are. So what happens with these sudden events? Does that then lead to revolution?
- 2.54** One of the arguments is, it very much depends on how the government responds. If the government responds well and is seen to have responded well to a catastrophic event, then that can boost the government's popularity. If they're seen to have not responded well or in a partial way, then that can actually exacerbate conflict. A lot then with these events, with a lot of environment and also with a lot of population dynamics, there are interaction effects between the population or environmental event and other factors. In this case, well, it depends on the way in which the government responds to the disaster. If it responds badly, then that can lead to discontent.
- 2.55** We also have a literature on non-renewable resources, again not linked to climate change, not necessarily linked to population pressure on environment. But of course, if populations are growing, even non-renewable resources are fixed, and that can lead to more pressure on those resources, such as oil or such as diamonds. Now, here we definitely do see in the literature a relationship between lootable resources like oil or diamonds and the likelihood of civil war. If there are lootable resources in a particular part of the world, Liberia, Angola, or Congo, for example, with diamond resources, then those parts of the world are more likely to experience violent civil war.
- 2.56** But again, it's not clear exactly how this is related to global environmental change. And even population change, particularly rural populations that are poor, are not necessarily the ones extracting the oil or the diamonds for themselves. And so, it's not obvious that population growth would exacerbate conflicts over oil or diamonds and other non-renewable mineral resources. Where these resources are valuable is to

either the state or to insurgencies that are trying to raise money for their conflict with the state.

- 2.57** Now, there are crops such as opium or other drugs which can be grown, coca leaves and so on, which perhaps might be affected somewhat by climate change. But I think it would be quite a stretch to argue that the way climate affects the growth of these narcotics would necessarily be a contributor to the likelihood of civil war. Where there may be an interaction is between climate change, population growth, and ethnic conflict. Almost all of the wars in the world are wars within countries, or civil wars; almost all civil wars involve ethnicity to some degree, religion.
- 2.58** Whereas, maybe during the Cold War, they might have involved ideologies like Marxism more, they're now involving ethnicity and religion more. So it could be the case, for example, that population pressure may lead, within a country, one group of people in a densely populated area to move to less densely populated areas. But in doing so, they may cross from what is seen as one tribe or one ethnic group's territory into another ethnic group's territory, within the same country.
- 2.59** So moving, for example, from more densely populated coastal regions of Uganda or Kenya. In Kenya, the Kikuyu, who are more dominant in the higher-density coastal regions, have migrated into the Rift Valley, for example, areas in the interior of Kenya. Same thing's happened in Uganda, where there's been migration into southwestern Uganda from the coast. These are people of different ethnic background, migrating within a country to another area that is seen as the homeland of a different ethnic group.
- 2.60** This has led to so-called sons of the soil wars, as James Fearon and David Laitin describe it. And so, they have tracked the rise of these so-called sons of the soil conflicts. To what extent then does climate change or population pressure then lead to internal migration within a country, but across ethnic boundaries? Sometimes you have a state that gets involved in settling people. In Sri Lanka, the Sinhalese-dominated government actually had a policy of settling Sinhalese people into the Tamil regions in the northeast, for example, as a deliberate policy; or the Chinese government settling Han Chinese, ethnic Chinese, in Xingiang, where the Uyghurs live, or in Tibet, for example.
- 2.61** That changes the composition of the population in these ethnic regions. That can be an irritant and lead to civil war. Donald Horowitz, for example, in his book, *Ethnic Groups in Conflict*, talks about in-migration of ethnic others into a territory can be a precipitant for civil violence. Now, the question is whether that in-migration has been caused by population growth and density in, for example, more developed or coastal regions. Has it been caused by climate, or has it, in fact, not been caused by those factors and would have happened anyway? It's hard to say.
- 2.62** I think it's arguable that some kind of population growth and density is a contributing factor to this internal migration. Another example is Javanese in Indonesia being settled

into West Papua, where there's a lot of tension between the settler groups and the indigenous groups. Now, in addition to this, we have migration as a result of civil wars—Palestinians being settled into Lebanon, for example, or perhaps the Zimbabwe conflict, many Zimbabweans moving to South Africa. Some of this is economic migration. Other could be civil war-related migration. That has been linked to civil conflict.

- 2.63** The question, however, is, how much of this is related to population increase and climate change? It's very difficult to say in many of the most high-profile cases, such as the civil wars which led Palestinians to move to Lebanon or Zimbabweans to South Africa, it's arguable that this population pressure was not necessarily a major factor. Again, it's very difficult to identify population and climate as being behind some of these movements.
- 2.64** Very often, in fact, it seems as though it is more advanced groups that have more advanced education, more advanced technology, that are maybe closely linked to the government, that are moving into these areas, perhaps to exploit resources, to move in as landlords into more sparsely populated areas. That doesn't suggest groups that are driven out by high population density. It may be a deliberate strategy of the government to settle members of more, quote, unquote, "advanced" groups into less developed regions in a country, such as the interior of Uganda or West Papua in Indonesia.
- 2.65** There can sometimes be a role, again, of deliberate government action working through ecological factors. So, building dams which floods out different groups and forces them out of their historic territories—that can be an ingredient in discontent. But it's very difficult to piece together whether this migration is voluntary or not. Is it encouraged by the government? Is it organized by the government? You have this interaction then of what are seen as aliens coming in to areas dominated by autochthonous groups, who claim that these people are foreigners and interlopers in their regions.
- 2.66** Now, sometimes you have voluntary migration into a region within a country like the Ivory Coast. Ivory Coast had a lot of migration from Burkina Faso, and also from the northern part of Ivory Coast, which is more Muslim, into the southern coffee- and cocoa-producing regions, which are more Christian. That changed the composition of the population in the south and of Ivory Coast as a whole. It's argued that that was a major ingredient in the civil war that broke out in Ivory Coast. But again, that had to, that had to map onto existing cleavages within the Ivory Coast political system.
- 2.67** It's also not clear whether that migration for work in the south was due to population and environment pressures. Although Burkina Faso is a part of the Sahel region that has been identified as a place where these population and environment pressures have been more intense, so it could have been a factor inducing people to migrate to the south of Ivory Coast. Another example is Assam in northeast India, which has seen

Bengali Muslim labor migration and has led to a certain amount of violence, including killing of migrants in that region.

- 2.68** There's always a question in terms of this migration within countries. How much is voluntary? How much is driven by population? Or is this simply people looking for opportunities, which would have happened anyway? Thank you very much. Just to say that the conclusion here is, we've been looking at the question of climate change, population pressure on resources, and the extent to which that pressure then leads to rebellion against the government or feeds into civil war.
- 2.69** To what extent might we see large-scale climate migration has been mooted by some? And that will then take us away from these conflicts within countries, possibly to tension across countries, if we have large-scale international migration as a result of climate pressures. Now, I think there are reasons perhaps to be skeptical of whether that large-scale climate migration is going to occur, if we look at certainly the cornucopian case, which would argue that, in fact, these migrations are often not as a result of population and environment pressures, and they only very rarely seem to lead to violence. I leave it there.
- 2.70** Are there any regions where the opposite might happen, where there is climate change or population growth and there is less conflict, actually?
- 2.71** Yeah, that's an interesting question. One good place to look at might be the Sahel region, just south of the Sahara, which has received more rainfall than in the past, arguably due to climate change. Has that dampened down, possibly, or reduced some of the kindling, let's just say, for a conflict to burst into flame? I guess, if you're a cornucopian who is skeptical of the climate change hypothesis, you would say, well, probably the drought conditions don't have a big impact on violent conflict; and equally, good weather and better harvest likely aren't going to have much impact on dampening down violent conflict.
- 2.72** And of course, we have seen quite a bit of violent conflict in that Sahel region, even during this period of greater rainfall. So in theory, if you are a Malthusian, you might say more rain is going to sort of lead to better times. It's going to lead to fewer grievances and less conflict. If you're a cornucopian, you'd say it doesn't really matter either way.
- 2.73** The Malthusians, like Paul Ehrlich, have made predictions about famine increasing and other kinds of food-related issues due to population growth, and these predictions have not come true. So do you think that, therefore, the cornucopian case is stronger than the Malthusian case?
- 2.74** My view is not as binary. I think it's true that there hasn't been mass starvation, and the share of people in absolute poverty has been dropping. Probably at the time of Paul Ehrlich, it was very difficult to predict the declining fertility rates in the developing world, the speed at which that's happened, which has reduced Malthusian pressure.

But I think it would be equally naive to assume that population pressure has not contributed to, for example, famine risk in places like Sudan and Ethiopia and the Sahel region.

**2.75** If the fertility rates had been lower, probably, I think there's a good case to be made that the degree of immiserization and the number of people dying in these famines might have been lower as well. But if the case is, well, we're going to have this runaway mass starvation, because we're going to have large-scale populations which will have nothing to eat and will then start more or less killing each other, no, that hasn't come to pass.

**2.76** I think that's fair. It's fair to say that probably the cornucopians have got the better of the argument, if we take the global scale and if we take particularly the more catastrophic claims that were made by various writers, perhaps Homer-Dixon, perhaps Malthus. Then no, we haven't seen that.

## Lecture 3: Young Populations

**At a glance** — Young populations and the political effects of youth bulges.

- 3.1** All right, welcome back. If we go back to the first lecture, I was talking about demography or population, which involved two components, one of which is endogenous, or has to do with natural increase based on births and deaths, and the other, which is to do with migration, which is interacting with other populations. If we just take the natural increase part, I focused on aging populations, but also very young populations.
- 3.2** That's what I'm going to be talking about today, is this question of young populations, mainly in the developing world. What is the connection between very young populations and phenomena such as violent conflict in particular? There's a well-developed literature in international relations and comparative politics on this question of how important the youth or elderliness of a population is in being related to the likelihood of a place breaking out in war.
- 3.3** Now, Samuel Huntington, who's one of the more important international relations scholars, who wrote a book called Clash of Civilizations, the book was written in the mid-1990s, and this was just coming just prior to a wave of Islamist terrorism and Islamist civil conflict within the Middle East. Huntington had written that, in fact, there were more violent conflicts, either within Muslim-majority countries or on the borders between Muslim countries and non-Muslim countries.
- 3.4** He was explaining this just after the 9/11 attacks, arguing, well, I don't think Islam is more violent, but the key is demography; that, generally speaking, the people who go out and kill each other are men between the ages of 16 and 30. And during the '60s, '70s, and '80s, there were high birth rates in the Muslim world, which has given rise to a huge youth bulge. And so, he's arguing that this demographic surge in the Muslim world is the reason why a lot of violent conflict was breaking out in Muslim-majority countries and regions.
- 3.5** And one of the questions we want to ask is, how important is the age structure of a population in determining how violent or peaceful that society is? I want to begin by looking at crime, and particularly violent crime, and scholars find a pattern, which obviously men commit far more violent crime than women. That's not exactly news. But if you look within convicted male criminal suspects, you can see a very clear age pattern. There's a spike of suspects aged 16 to 24, falls somewhat, those 25 to 34, 35 to 44, 45, and then 55, it's pretty much nonexistent or low.
- 3.6** So there's this pattern of young men, particularly in that 16-to-24 age group, being the most violent, accounting for a lot of the violence, relatively high violence among the 25-to-34s, and then it really drops off. There's that age pattern of younger men committing a lot more violent crime. Now, it turns out, you can go back in time and look at British

crime records. You can look at crime records in the United States, and they both have this same shape. There's a lot more murders and a lot more crime in Chicago than there is in England and Wales. But the age pattern of murderers in Chicago and England is identical.

- 3.7** You can lay one on top of the other, and this chart just shows the age of the murderer along the bottom from age zero up to age 80-plus. Then it's got two indices of homicides per million per year. The Chicago index, which goes from zero to 100 to 200 and so on, up to 900, and the England and Wales numbers go from zero to five to 10, so it's almost like one-thirtieth as large on the English scale as the Chicago scale. Chicago is a far more violent city than England as a country. But the age pattern of murderers in both societies in this data is identical. So age does seem to be connected.
- 3.8** This tells us two things. One is that age is very much connected to violence, but it also tells us that age can't be the only thing, or even possibly the main thing, because that age doesn't explain why England is a much less violent place than Chicago. So there are clearly going to be other factors determining rates of violence, but equally, age does seem to play a part in this. And so, we might expect a younger population, on average, to have more murders per capita, to be more violent than an older population.
- 3.9** Michael Ignatieff's book, *Blood and Belonging*, was written during the war in the former Yugoslavia between Croats, Serbs, and Bosnian Muslims. As a journalist, he was writing about what he saw, which was young men, quote, "Young men who loved the ruins, loved the destruction, loved the power that came from the barrel of their guns." And that's a kind of argument around young, testosterone-fueled men who are attracted to violence and therefore are more likely to commit violence.
- 3.10** And so, there's something then in that age group of young men that means they're going to be more violent or more open to violence. And so, where we have demographic shifts that might, for example, if there is a decline in infant mortality, and a country is early in its demographic transition, remember, where that death rate falls, but the birth rate is still high, and you get a boom in population, you're going to get a lot of young people surviving, and you're going to get a large young population.
- 3.11** And so, most of the population in many poor sub-Saharan African countries, the average age in the population might be in the teens. So we have a very young population; there may be more potential for violence. Although, as we'll see, we have to be very specific about which age groups we're talking about. It's not just anyone under the age of 30, it's that 15, 16 to 30 group that is the most important.
- 3.12** In terms of looking at the correlation then between age and violence, we see that, amongst countries that are rated as very young in their age structure, so typically, an average age under 20, in the teens, their risk of being involved in a civil war in the 1970s was in the mid-22%, in the 1970s and '80s. By the time we get to the 1990s, it's above 25%. It's just worth saying that the pattern of civil wars in the world, after 1945,

there was a rise in the number of civil wars in the world, peaking about 1990, 1991, and then declining till about 2007, 2008. And then we did see a rise, a little bit of a rise again, but not much.

- 3.13 If you compare the very young countries, their rate of civil violence is much higher than the youthful or the transitioning in the mature countries. It looks as though young countries certainly have a lot more civil war than older countries. But you might say, well, is that really because they're young? Yes, we see a correlation. A country that is younger is more likely to be involved in civil war. Is that because it is young, or is it because the country might be poor or uneducated or relatively less well educated?
- 3.14 It's worth just saying what the numbers show, in terms of the risk of conflict. Countries in which more than 60% of the population are under age 30 are at least four times more likely than countries with more mature population age structures to experience new outbreaks of civil conflict. There's definitely a correlation, but is that, again, because of age structure, because there's a lot of young men who have a lot of testosterone? Is that why they're more violent?
- 3.15 We also know, for example, that poverty is heavily correlated with young age structure, and we know that poor countries are more likely to have civil war. Maybe it's the poverty doing the work here, not the demography. Clearly, we can see in this chart that very young countries are poorer. Even if they might have faster economic growth, they tend to be poorer than countries that have an older age structure. There are a series of skeptics of the youth bulge hypothesis. The youth bulge hypothesis is really the idea that younger countries are more violent, more likely to be involved in war.
- 3.16 Paul Collier and Anke Hoeffler in 2004, James Fearon and David Laitin, these are all quite important scholars in this literature on civil war. They argue that, when you control for GDP per capita in particular, also for democratization and various other factors, perhaps the roughness of the terrain in a country, which is also correlated with greater risk of civil war—when you control for these factors, actually, you don't see age structure mattering and predicting a greater risk of civil war.
- 3.17 However, the youth bulge hypothesis proponents would argue that this is because scholars like Collier and Hoeffler haven't measured youth bulge properly. I'll just talk about this question of how you measure the youth bulge. What they say is, these other authors have miscalculated because their calculation of the youth bulge simply says, what is the share of young adults, like 18- to 25-year-olds, as a share of the total population? That's how they're measuring the youth bulge, and that shows no significant relationship.
- 3.18 But if instead you were to divide the 18 to 25s by the adult population, then you see that, in fact, the youth bulge share, the 18-to-25 share, does significantly predict the risk of violent conflict. What is the difference then between these two hypotheses, the skeptics and the youth bulge hypotheses? I'll try and illustrate that with the population

pyramid. Again, that's showing the age groups along the bottom, starting with the zero to fours, five to tens, going all the way up to the senior citizens, as we go from the bottom to the top.

- 3.19** And then we have the two sexes, the men and the women, men in brown, women in blue. If we take the 15 to 30s, that's a few bars up from the bottom, and then you divide by the total pyramid, which is the total population, you get one number. If you take those bars and you divide only by what's above those bars and including those bars, then that is dividing the 15 to 30s by the adult population over 15.
- 3.20** If you have a very poor developing country like Mali or Ethiopia, they have that classic pyramid shape. So if they have a lot of people that are aged 0 to 15 in the population, they also have a lot of people aged 15 to 30. They don't have many people above the age of 30. And so, actually, you can have a society where they've already started reducing the birth rate, so they don't have many people aged 0 to 15, but they still have a lot of people aged 15 to 30.
- 3.21** A country like Iran, for example, in 2005, during the unrest of that time, it still had a large youth bulge, even if its population, age 0 to 15, had begun shrinking. Which is why you actually, according to Urdal, you should classify Iran in 2005 as having a youth bulge, whereas using the other system, in fact, it wouldn't be classified as such. There are a lot of countries that just have really young populations, but there are also countries that have a smaller number of 0 to 15s and a larger number of 15 to 30s.
- 3.22** Actually, what really matters is the 15 to 30s divided by the adult population. That's the measure you need to understand what is the potential for violence in that society. For example, if we take the case of Iran, which had a violent revolution in 1979, when it had a really young population, it did have a youth bulge with a young population. But in 2005, it also had a youth bulge. But if you were to measure the 15 to 30s and divide it through by the entire population, because the 0 to 15s have shrunk, it would provide a different measure than if you take 15 to 30 divided by the adult population only.
- 3.23** Looking at these two population pyramids, in Iran, in 1980, it has the classic pyramid shape, very wide base, a lot of young people coming up into the peak. But in 2005, actually, the base is a bit, is narrowing. You have a bulge in the middle, 15 to 30, and then it narrows again. They are both youth bulge populations, and so they both should be considered the same way, which they are under the Urdal methodology, but not under the Collier methodology. That's the difference, argue the youth bulge hypotheses.
- 3.24** Iran, therefore, had moved down its curve of demographic transition in terms of the percentage of the population that is under 30. That had declined, but the 18-to-25 share, or the 18-to-30 share, the critical age range for the youth bulge, had not changed. So the countries had become somewhat more mature in its population shape, but the youth bulge share was powerful in 2005.

- 3.25** Now, right now, in Iran, the youth bulge share is much smaller than it would have been in 2005. So youth bulge theorists like Urdal would say, "OK, Iran in 2025 has not got the same chance of engaging in a violent rebellion or an uprising as it did in 2005 or in 1980." That would be the argument from the youth bulge hypothesis. But in 2005, they would say, "Actually, youth bulge looks the same, so they do have a significant chance of rebelling."
- 3.26** And of course, we did see the uprising, 2005 and 1979. We don't quite see it now. And is that because the population of Iran is not as young and therefore is not likely to be as risk-taking perhaps, as it was in the past, and therefore, a violent revolution is less likely in Iran? Now, in other situations where Iran already had a below-replacement fertility rate by the 1990s, so it's long had below-replacement fertility, although it's interesting, as a country, where the fertility rate went below replacement, but it stayed close to replacement for quite a number of decades.
- 3.27** But then you have other situations, other countries where there is still quite a young population, like Pakistan. This chart here shows two variants of a youth bulge, where you might have a society like Iran that started to mature, but the 15 to 30s are still showing that youth bulge. Then you have a society like Pakistan where there's a youth bulge, but also with a lot of people under the age of 15.
- 3.28** We talked a lot about how population shifts work with politics, and it's often not a direct effect. It's often working in the background. It's working indirectly. It's working several steps up the causal chain through second-order, third-order, even fourth-order effects. Similarly, in the case of the youth bulge, it's not a direct effect. Lots of young people, young people hate the old people and they fight against them. That's not what's going on.
- 3.29** I'm not saying you can't have generational conflict, but in this case, in the case of civil wars, this is not what's going on. There's no direct effect of conflict between generations. That's not the main effect. What is going on is a second-order effect in which the youth bulge—which, don't forget, is caused by declining infant mortality, allowing for more young people to be born— that youth bulge reaches adulthood, and there are no roles for that bulge of young people in this economy that's designed for a smaller number of people.
- 3.30** You've got unemployment and you have a lack of a role, and therefore, you have people who can be recruited into insurgent movements. It increases the risk of insurrection. There is also a more positive story in which you might say, well, a youth bulge, a bulge of young people, reaches maturity in a society which can employ them. And so there's an economic boom, and especially if you have a declining share of people under the age of 15 because of falling birth rates, then you can actually get a boom in the number of people who are in the working age of 15 to 65.

- 3.31** So you might get a demographic dividend, what's called a demographic dividend, from young people who are put into work. Young people are more energetic, creative; they then generate more economic activity. We'll talk about which societies are in the Goldilocks zone, if you like, for having a very favorable age structure.
- 3.32** There's a bit of a difference in the literature on civil war, as to the motivations of those who engage in civil war. One view says, well, greed is the dominant factor. People look at what they can get out of joining a rebel movement, maybe pay, status, power; maybe they get their hands on some wealth. The benefits and then the costs, in particular the so-called opportunity costs. If I join an insurgent movement, what do I have to give up?
- 3.33** Now, obviously, there's a risk of being killed or injured, which has to be traded off. But if, for example, there were good employment opportunities, steady jobs, then if you were just about the money, factoring in the risk, you'd say, "Well, I can get a job that pays me a moderate amount. Maybe I don't get to loot. Maybe I can't realize quite the upside that I might be if I was in a rebel movement, but on the other hand, it's very risky, so why would I take that chance?"
- 3.34** On the other hand, if there are no jobs and there is no other alternative source of income, then the so-called opportunity cost, what you're giving up to join the movement, is less, and that might change your calculation to join the rebel movement. In particular, when you have a youth bulge that leads to large-scale youth unemployment, then that can lower the opportunity cost for young men.
- 3.35** And it makes the decision to join the rebel movement more sensible from that rational choice perspective, when it's all about, "OK, costs and benefits, factoring in risk, yeah, I'm going to join the rebel movement because I don't really have good alternatives." That's what's known as the greed perspective, which takes the view that people are making rational choices as to whether to join insurgencies or not.
- 3.36** There's another view, however, which emphasizes grievances, which is more psychological and not necessarily as calculating and material. So the grievance perspective argues, in fact, that what happens is, young people are unemployed. One problem there is, and so they may be resentful of society. They may have a lot of built-up resentments. It may be that, without a job, particularly in traditional societies, in the Muslim world perhaps, or in religious societies, that means you can't get married, if you don't have the money as a man, and that therefore can lead to other kinds of frustrations.
- 3.37** And so, that is the grievance view. Or for example, yeah, you don't have upward mobility. There's no jobs for you. There's no career advancement, for example. That leads to resentment. So once we start talking about resentment, we're talking about grievances, and that's more psychological. That then is something that can lead to

people to be motivated emotionally, to be involved in violent conflict, not just in terms of their calculation of what their opportunity cost of joining a rebel movement is.

- 3.38** Both of these explanations, the greed view and the grievance view, lead to a greater likelihood of joining an insurgency. So they both lead to the same outcome, but they are different theories of how demography might work. This blocked upward mobility point is interesting. So if you think of something like the military, the civil service, where the number of jobs is relatively fixed and it depends on the government, and you suddenly get a surge of young people, they may resent the people who are sitting on those positions within these organizations.
- 3.39** They don't have upward mobility; that's being blocked by the system, so they become anti-system. Youth bulge theorists would argue that there are, as with many demographic, political demography phenomena, this works in a very indirect way. So in order to have violence, it's not enough to have the young population; you need to have some other factors involved.
- 3.40** Sometimes urbanization can help to concentrate these young unemployed men. If you have cities where they can congregate, cities tend to be where political mobilization happens more easily, and people can, rebel movements can communicate. Maybe a more urbanized society that has a lot of young men hanging around on street corners, that could be a factor that can magnify the impact of the youth bulge on conflict.
- 3.41** Poverty, obviously, the poorer a society is, if it has a youth bulge and it's poor, that can increase the likelihood of violence. If the society is one in which men have to amass a certain amount of wealth before they can get married, and they have to get married before they can have sex, then that's going to create certain sexual tensions in the society. This was one argument that was sometimes made, that there are these "excess" men who don't have a place in the society. They've got nothing to lose. And so they're more likely to use violence to get status and wealth in that kind of a society.
- 3.42** So there are all of these other factors that can interact with a very young population to lead to an increased risk of violence. It's worth saying that there are other theories of ethnic violence, including theories that would argue that there are genuine cultural, or particularly since most civil wars are ethnic or religious, genuine grievances against other groups, if you imagine a situation where one group actually tends to dominate in the civil service and the military.
- 3.43** And this is often the case in postcolonial societies. Certain ethnic groups are overrepresented in certain positions, maybe because their coalition is in power. If the southerners are in power in Nigeria, if the Kikuyu are in power in Kenya, and they're controlling the government, and politics runs on ethnic lines, then actually, that can feed into the grievances of youth bulge populations.
- 3.44** But it's not always going to be a negative story with young populations. It's possible that young populations can be also associated with economic growth and with boom

times. Let's look at the more positive story of the so-called demographic dividend. What is a demographic dividend? Demographic dividend, the argument here is that the population of workers compared to the total population, when the population of working age is at its maximum, then the economy is benefiting the most from the age structure.

- 3.45 What you want is, you want to have the maximum number of people in that working age bracket of 20 to 64, or 18 to 65, let's say. You want to have as few young and old dependents as possible. Let's look at the case of Ireland, for example, which in Ireland, the fertility rate comes down. And so the population of young dependents, 0 to 19, starts to fall around 1980, and it goes into decline.
- 3.46 So the population of young dependent people, that is, people who are not working because they're too young, they're in school, they're dependent on their parents, that starts to decline as a share of Ireland's total population. The population age 65-plus is flat in these years, from the 1980s to 2005, so they're not increasing their population of elderly who will also be dependents. And as a result, the working-age population, age 20 to 64, increases from 50% of the population in 1980, up towards 65% by 2005.
- 3.47 So you're getting an increase in the share of people who are working and paying tax, and a decrease in the people who require public services like either schooling or daycare for the young population, or elderly care for the older population, pensions and elderly care. Dependency goes down, contributors go up, that's going to lead to an economic boom, which Ireland enjoyed.
- 3.48 Part of what one of the arguments would be, well, the Celtic Tiger phenomenon of Ireland's spectacular economic growth owes something to the demographic dividend of having a large number of people in working age and very small numbers of people that are either young or old dependents. Of course, this is often a temporary phenomenon as a country transitions from, again, the high birth rate, high death rate, through the population explosion, down through to low birth rate, low death rate.
- 3.49 In between, you get that population boom of people in working age. If you have an economy that can—if you can educate those people and employ those people, you can realize that demographic or the economic boom from that youth bulge. If you don't have an adaptive economy and you can't find jobs for the young people, they could become a problem for your society, and you'll see crime, perhaps even a risk of insurgency, during that youth bulge. So the trick is, how do you actually create enough opportunity for those young people?
- 3.50 That's where a lot of countries are, and of course, when you have a vibrant global economy and you have reasonably good government and economic development, then you can absorb that labor and actually realize this demographic dividend. Maybe that's what's happening in places such as Vietnam, perhaps parts of Latin America, where

maybe they are harnessing the— China, no longer so much, but earlier in its development.

- 3.51** This theory of the youth bulge, then—this is not, again, it's not often a direct effect of the youth bulge leading to particular political outcomes. It is usually an indirect effect. We could either have the youth bulge reaching the population in a poor, not well-governed society, with perhaps not very good education, leading to a lot of unemployment, which then leads to insurrection. That's path A.
- 3.52** Or perhaps we have youth bulge reaching adulthood in a more dynamic, better-governed society with better education and is able to attract investment, perhaps to do low-end manufacturing, and then move up the value chain with a lot of energetic workers of productive age. Maybe Vietnam is an example of that today. You then get this demographic dividend where the number of under-15s in Vietnam, due to its falling birth rate, is dropping. There aren't yet large numbers of senior citizens that require care, and therefore, most people are working and paying tax and producing, and so there's high economic growth.
- 3.53** Now, one of the questions is, how exactly does the youth bulge link to the level of violence? It may be, there's two questions. One, does a young population lead to a greater chance of that society being involved in civil war? Then there's also the question of, well, if the country does go into civil war, is the level of violence higher in a younger society than in an older society? It's worth distinguishing here between different types of civil conflict.
- 3.54** We have two main types of violent conflict in the world as a whole, but especially in the developing world. One would be civil wars, largely fought in the countryside. The second are urban riots, largely in cities. So urbanization tends to be linked to rioting. And the question is, how does the youth bulge affect both types of conflict? It may be that a young population is more linked to, for example, urban rioting, than it would be to civil wars, because civil wars do require a structure of grievances.
- 3.55** For example, I use the case of Myanmar. It could be that the Shan and the Mon and these peripheral groups in the countryside, they have established grievances with the center, and so you have the structure there for violent conflict. And it may be, in that case, the young population does lead to more violence. Whereas maybe in another case where you have, like Botswana or a country, society which is more homogenous, Vietnam, for example, you might not have the same conditions, and therefore you might not have this mapping onto civil war.
- 3.56** I mentioned a lot about interaction where demographic factors are present in interaction with other kinds of factors, like poverty, for example, or weak state capacity. These things might be present together. When they're present together and you have multiple factors, then you might get violence. It's a bit like disasters. With a lot of disasters, you don't need just one thing to go wrong; you need a whole series of

things to have gone wrong and to have failed. That's maybe sometimes the way demography works.

- 3.57** Another example that we could use for how youth interacts with violent conflict is age and ethnicity interacting. If one ethnic group has an older age structure than another ethnic group, you can then get resentments either from the older ethnic group, the older members of the established ethnic group, against the younger members of the growing ethnic group, or vice versa. That could even happen in the developed world.
- 3.58** I'll use the example here of the United States, where we're looking here at—and this comes from 2010, William Frey's work, where we're looking at the difference in the ethnic composition of the old population and the young population in different American metropolitan areas. We can see, for example, in the city of Phoenix, where I know Peterson Academy is based, as of this data from 2010, the minority share, mainly Hispanic share, of the senior citizen population, was only 15%, but of the under-18 population was 56%.
- 3.59** So we don't have just a difference by age, but we have an age-ethnicity interaction difference. One factor in the politics of Arizona, arguably, is that relatively white seniors, or relatively white taxpayers in older age brackets who are more prosperous, may not be willing to pay tax as much for a predominantly minority younger population. So this age-ethnicity interaction can therefore affect the politics in interesting ways and lead to different kinds of conflicts within a society.
- 3.60** Another example of a similar phenomenon is in Northern Ireland, for example, where the Catholic population, growing faster with a higher fertility, is therefore younger than the Protestant population, which is older. And it leads to, so for example, with a lot of young people, the IRA might find it easier to recruit young men, especially in a place with high unemployment, partly as a result of the young population, partly as a result of the Protestant-run state perhaps discriminating against the Catholics in housing and employment. You have a wider availability, a wider pool of potential insurgents that could be recruited to the IRA.
- 3.61** Equally, the Protestant population being older, seeing a younger population, that dynamic of an older population looking at a younger population can be fearful, can be more fearful of that population. Its anxiety levels go up as a result of the younger composition of the rival ethnic group. And that stokes anxiety and a desire not to reform, whether it be one person, one vote, whether it be the welfare state and so on.
- 3.62** What we then saw was a decline in the Catholic birth rate in Northern Ireland. That then starts to shift the calculus. OK, so for the Sinn Féin IRA, it might become harder for them to recruit. They also may worry that the Catholic share is not inevitably going to carry Northern Ireland into a united Ireland. So it might increase the—the strategy might be, well, we've actually got to make a push now while we can, while we have the

numbers and the growth, because we're no longer going to have the numbers and the growth in the future.

- 3.63** But on the flip side of that, maybe the Protestants are saying, well, their birth rate is going down, so in fact, the threat level is going down. Actually, we can be a bit more relaxed. And that might make it more likely for the unionists to be willing to make a deal that would involve power-sharing with Catholics, for example. So there are all kinds of ways in which age structure is interacting with other variables, whether those be poverty, whether that be ethnicity. That shows how demography, again, it's not that single cause; it's working in combination with other factors to produce change.
- 3.64** Just to conclude, what have we really been talking about? Well, we've been talking about age structure, which is one aspect of demography. Age structure is one aspect of demography. Young populations, as opposed to older populations. So a youth bulge, where you have a large number of people aged 15 to 30 in the population, how does that affect politics? One of the arguments of the youth bulge theorists is that the higher share of 15 to 30s in a population makes that population more susceptible to violence.
- 3.65** Now, that could be criminality, but equally, it could be insurgency, especially in a situation where you have some other factors at work, an established conflict frame between a number of different groups, where you perhaps have unevenness in the age structure between one group and another, leading to anxieties, leading to perhaps easier recruitment into an insurgency. We looked at this question of greed versus grievance in civil wars.
- 3.66** The youth bulge can shape both people's rational calculations, as to whether to join an insurgency. So weighing costs and benefits. What are the costs of me joining the insurgency? Those costs are lower when the population is younger, because there tends to be fewer employment opportunities in that young population. So the youth bulge lowers the cost of getting engaged in insurgency, and therefore, it may make it more likely for young people to get involved in rebel movements.
- 3.67** But equally, there's a grievance aspect, which means that, if jobs are monopolized by older generations and the young people don't have upward mobility, they can be resentful. Especially if government positions are monopolized by one ethnic group, and they are members of another ethnic group, that can add to this sense of grievance. We can have a grievance element, as well as a greed element. And the youth bulge can be involved in both mechanisms that lead to an increased at risk of civil war.
- 3.68** We also talked about different types of violence, urban rioting versus more rural civil wars, and how they tend to be affected by population dynamics. Easier to see the effects in terms of urban riots. Civil wars tend to require more elements to fall into place. So it may be that the youth bulge affects urban rioting more than civil war. We've also seen an example. When we look at the youth bulge, it's an example. It's the

same as if we're looking at climate change and environmental pressures, that these sorts of factors tend to work in indirect ways through interaction.

- 3.69** A number of other factors have to be in place for very young populations to lead to conflict or more intense conflict. Poverty, ethnic differences, perhaps rough terrain, a whole bunch of other things have to be there, too. Perhaps poorly managed economies and weak states have to be there. But if you have these other factors, if you have a youth bulge, it's more likely that the country tips into conflict than if it doesn't.
- 3.70** So, interactions in second-order, third-order effects, tends to be the way in which demography operates; and that's certainly the case when we're looking at the phenomenon of the youth bulge in developing countries. Now, it wasn't all doom and gloom. I should say, by the way, as one of the predictions of this model—of course, a lot of this literature came out in the early 1990s, which was the peak of civil wars since 1945. That's gone down to a great extent. There's been a big decrease in the amount of civil war.
- 3.71** One argument we might make is that, as societies become more mature and the youth bulk shrinks as a percentage of the total, we're going to see less civil war. This is one of the themes we're going to talk about with population aging. Does population aging mean less warfare, less war in the world in general? Perhaps the happier part of the youth bulge hypothesis is to say that, with the decline of the youth bulge, we're likely to see less civil war.
- 3.72** It's worth reminding you, however, that just, if you take a very young country, just because that very young country gets its birth rate down, there's going to be a 15, 20-year delay or even more, before that actually starts to shrink the youth bulge, because people who were born 15, 20 years ago, as they become aged 15, 20, are then going to be in that youth bulge zone. So it takes probably 40 years of a country's fertility going down before the youth bulge really passes through the system, and those people actually become 30 to 40, instead of 15 to 30. Then they start to pass out of the youth bulge zone. Then the risk of conflict would be predicted to fall.
- 3.73** The other thing we mentioned was the demographic dividend. Just looking at youth bulge as a problem is also the wrong way to look at it. It may be an opportunity, too. If a country can get those other factors, can be a little bit better off, a bit better organized, can attract investment, then actually it can harness the youth bulge. It's like a fire. Instead of burning yourself, you actually use it to heat something and eat some nice food, maybe.
- 3.74** But you can harness that population of young working-age, who are very creative and energetic, and that can give you an economic boom. We're also seeing that as well. So this whole question of the youth bulge, typically studied in terms of its risk to the chance of violent conflict, but also studied by economists as something that can fuel an economic boom. With that, I look forward to your questions.

- 3.75** What are the main risks associated with the other end of the spectrum issue, like a youth gap? If you have a very severely low birth rate that contributes to a super low 15 to 30 age population, or you've just got a lot of emigration out of the country, what are the main risks associated with that sort of dynamic?
- 3.76** Yeah. So the other end of the spectrum, in a way, with population aging, and you have very few people of working age, and we are going to talk about that more next time. But of course, what happens is a number of things. One is, there are some who would argue there's less dynamism in the economy without a significant young working-age population. You may not be able to fill certain physically demanding jobs. It could be the case you don't have enough young workers to pay the pension costs, the medical costs, particularly if you have a large old population.
- 3.77** Now, it may be that you, like in Russia, they had a lot of old people dying as a result of alcoholism, so they actually didn't have the rise in the elderly population that was therefore an increase in dependency. But basically, if you have fewer workers, then it's fewer workers to support older age dependents, and therefore, that leads to various kinds of economic problems, arguably. So there isn't as much capital to invest and to grow the economy.
- 3.78** Consumer demand might be weaker. There's a whole host. Then, of course, and that's without getting into what happens when you try to then substitute by bringing in immigrants to make up for your lack of young people, which then introduces political conflicts into the society. This is part of a much wider conversation about population aging and what that means for society and politics.
- 3.79** You mentioned that youth bulge is likely to affect and bring about urban rioting. And compare this to maybe the youth bulge in a civil war or in a conflict where the populace is, where you find that 15- to 16-year-olds are mainly the ones involved in a civil war. And then the senior citizens having control of what is happening in the war, for example, in sub-Saharan Africa, in some of the countries. Why is it the case that the youth bulge is likely to affect more and bring urban rioting, rather than the civil war? Maybe in the case of sub-Saharan Africa, where you find younger populace of 14- to 15-year olds in the war?
- 3.80** The reason for that is that, often civil war, which is a much more violent, typically, involves a lot more battle deaths and requires you to mobilize people into armies, to hold territories, to have bases and headquarters. Usually with civil wars, there is a larger national-scale grievance. Most of the time, that'll be an ethnic or religious grievance nowadays. Not if we go back to the Cold War, but certainly nowadays. You have to have more factors fall into place before a civil war will happen.
- 3.81** Whereas maybe with an urban riot, there's less at stake in terms of violence, and it's not as necessarily linked into the politics of a country, in terms of the main issues that are being debated, which is often over power between ethnic groups. You will get more

of these urban riots, which may be more small-scale and may not have the national significance, but they could. Of course, if it's about overthrowing the government, then yes, then it's much more important.

- 3.82** So this is just where, I guess, when you have a young population, and migrants are also younger; I should say that people migrate when they're younger. So cities will often have a younger population because people have migrated in. Older people won't migrate as much. So that concentrates things in the urban center, and that may be another reason why you may have more urban rioting.
- 3.83** It's worth saying, however, that civil wars can also be organized from cities, but they typically happen in the countryside. So they require a different set of organizing principles, but yeah, the youth bulge can affect both, but it's just more difficult for it to ignite civil war, arguably than urban riots.
- 3.84** We talk a lot about integration, when we have a lot of people from different countries coming into any other country, integration between races and religions. But I think it's very interesting just to speak about the integration between the older generations and the younger generations, because the gap is so massive in regards to how far we've come and just, for example, in the last 100 years. Do you think there is enough being done in societies to integrate the older generations and the younger generations?
- 3.85** That's an interesting one, the generational question. There's a lot going on there because you have this issue in Western countries of the young generation; it's harder for them to, say, get on the housing ladder, for example, or to accumulate savings, perhaps because the cost of property has gone up a lot. The question is, is that going to lead to resentment against older generations? That is one question, but I guess political demography would tell us, it's going to be about how that interacts with other factors.
- 3.86** It may be that people from a well-off background, the parents will help their kids onto the housing ladder. And so what might happen is, you then have a class-age interaction. So it might be the working class, or people from a poorer background, who are young are the ones who might be manifesting the resentment. But then the question is, who are they resenting? Are they resenting the old people from a well-off background? So there's a class dimension. There's an age dimension. It's complicated as to how that will manifest.
- 3.87** I think with the case of ethnicity and how that plays into that, in the case of the US, where the people with the wealth tend to be older, but they also tend to be whiter, the politics of that might be to say, "Well, we're going to resist tax increases." It might shift. If older people are more likely to be voters, and white voters tend to be more common, so if whites vote at higher rates, which is not really true, but to some degree it's true, especially vis-à-vis Hispanics and Asians, you might get a coalition in power who would, say, be very resistant to cutting benefits to senior citizens, like Social Security and Medicare in the US case.

- 3.88** In the case of Donald Trump, for example, very protective of Social Security and Medicare, which tends to benefit relatively well-off older white voters. But perhaps there may be resistance to spending money on social programs or public health care that might benefit younger members of particularly, which would tend to benefit more diverse populations. You can tend to see politics break along these lines as well, as age is interacting with ethnicity, or age is interacting with class.
- 3.89** It's going to be interesting to see whether this idea of young people resenting the old actually does become politically important. I'm somewhat skeptical. I mean, we see different patterns of—you'd expect young voters then to vote in particular ways in particular countries. And it does vary a lot. In some countries like Britain, the young people vote for the left in very large numbers, but that's less, it's not as evident in US-Canada, for example, or in Europe. It is in some countries, but not in others. It's not often, it seems to me, it's not often the case.
- 3.90** But sometimes, yeah, you can get a young generation with its own generational consciousness. The baby boomers were one such generation. But of course, they were a big generation because they were products of a boom. Generation Z, for example, are a product of a baby bust. They're going to be a smaller force. There's not as much of a generational consciousness there. But it's a good question you ask.

## Lecture 4: Population Shifts

At a glance — Population shifts within and between countries.

- 4.1 We've talked about the impact of population shifts and pressures within countries domestically. What I want to do in this next lecture is to look at the impact of population change on international politics or international relations, that is, politics across borders globally, and to consider this question about global military power and the global balance of power in the world. How will the coming population shifts affect the balance of power between different regions of the world, between different countries in the world?
- 4.2 I especially want to look at population aging. In the past few lectures, we've looked at, in particular, population pressure in developing countries. We've looked at the youth bulge, very young populations in developing countries, and perhaps we'll shift now to consider more developed parts of the world, but in relation to developing countries. The first thing to note, if we look at a table of countries' populations, 2005, 2025, now this table was done in the late 2000s. The predictions are reasonably good.
- 4.3 A few changes, but the general dynamic, as it was predicted in the late 2000s, has held up reasonably well. We see Pakistan's population going from 157 million in 2005 to 229 in 2025. In fact, Pakistan's population has overshot that projection by a certain number. I think it's around 240 million. Russia's population, 143 million in 2005. Actually, that's remained pretty stable because they got their old-age mortality problem under control that had emerged after communism, and they've had some immigration from Central Asia and other parts of the region.
- 4.4 Their population has not yet declined that much. It's declined a little bit. But what's particularly noteworthy is some of the expansions. The Democratic Republic of the Congo, 57 million, about the size of Britain's population in 2005, is now over 110 million. If you were to look at Britain and Congo in 2005, similar-sized countries, 2025, Congo is now about, pushing towards twice, well, it's about twice Britain in just 20 years.
- 4.5 There are a number of other African population powers that are emerging. Ethiopia, which, of course, has had a lot of civil violence just recently, going from 77 million in 2005, here it says 118 million, as predicted. It's actually overshot that by, I think about 5-6 million. It's around 125 million. 125 million is a serious population power. Nigeria, 131 million in 2005, and it's now over 200 million. 200 million is kind of really where the US was when I was growing up, a significant population power.
- 4.6 We are getting the rise—even Uganda, 60 million. It's overshot that as well. That's about where France is. We're getting a number of new powers rising, and what we're going to see is a number of other countries declining in population share. To what

extent is that then going to reconfigure the balance of power in the world? How important is population size to be a world power in the world?

- 4.7 In terms of population growth, that's almost all happening in the Global South. The population of the EU-27 is basically flat and poised to decline without immigration, for example. Population growth of the US has slowed considerably. Why has this happened? Well, it's happened for the reasons we talked about in the first lecture, around the demographic transition from high birth rates and high death rates to low birth rates and low death rates.
- 4.8 And in fact, to this new phase known as lowest low fertility, where, well, the second demographic transition, which is really where the death rate begins to exceed the birth rate, and you start to get population decline. And so this chart looks at the total fertility rate, the number of children ever born to women, dropping below two in most European countries, around the Second World War. That is mothers born around the Second World War, who come into motherhood in the 1960s. From then onwards, broadly speaking across Europe, we have a below-replacement fertility rate.
- 4.9 And so, that's now been in place from the '70s till now. That's over 40 years. So were it not for immigration, we would be seeing absolute population declines across Western Europe. Again, demography, the most predictive of the social sciences. We know a lot about the characteristics of the 40-somethings of the 2060s because they've already been born. We know how large that population is. We know its ethnic composition. We can project that forward once we assume, we make certain assumptions, around the number of children that those people will have.
- 4.10 Now, of course, a total fertility rate is economically and culturally determined. So total fertility rate is less predictable. It's possible that it will go up. There are certain social forces that have led it to go down, so it's not as predictable. But once we know, we can make assumptions based on the age structure of the under-five population about the population in the future. Of course, there will be migration in and out. There could be catastrophes and possibly wars and famines. That will affect things at the margins, but still, in the long run, it is total fertility rates that matter the most for these projections.
- 4.11 In the short run, migration matters more. Short-run projections, migration is more important; long-run projections, it's fertility and mortality, but especially fertility, since mortality isn't changing a great deal any longer. If we look at, again, going back to Jack Goldstone's work in the late 2000s, you can see these projections of populations. The broad trend is correct, but in some cases, such as in Germany's case, because of immigration, there has not been the same projected decline.
- 4.12 But broadly speaking, across Europe, Southern and Eastern regions projected to decline first, Northern and Western regions later. Germany, Russia, and Eastern European countries like Czech Republic, Italy and Spain in Southern Europe, would be places that would have a lower birth rate. Just in terms of the mathematics, at a total fertility rate

of 1.3, a country is only a quarter of its size in 100 years. So a country like Italy would only be a quarter of its size.

- 4.13 At a total fertility rate of 1.0, that's starting to push towards a country being only one-tenth as large. So places like Korea, Hong Kong, coastal China, with a total fertility rate around one, in 100 years, that's going to be only a fraction. Those countries are only going to be a fraction of their size, perhaps only 10%. Korea would possibly only have 10% of its population in 100 years, 1% of its population in 200 years.
- 4.14 That is sort of the effect of compounding. Again, fewer mothers, fewer children, fewer mothers, fewer children, and it's like compounding effects. Just as the way we had the population explosion and the population of the world, the time it went to go from 1 to 2 billion was 100 and some odd years, 2 to 3, 3 to 4, a shortening length of time to get up, and the peak at 10 billion. And then on the way down, it's going to start off gradually, and then it's going to pick up speed, if the total fertility rate does not rebound to above 2.1.
- 4.15 At a total fertility rate of 1.5, the decline is less. It's country will be—so 1.5, Britain is at around 1.4, so we would say in a hundred years in Britain, without immigration, Britain will be somewhere between a quarter and a third of its present-day population. It's worth saying that a lot of the world's countries are in this lowest low zone, under 1.5 now. And so, in fact, being above 1.5 is relatively unusual in the developed world.
- 4.16 The US is probably just above that, but much of Southern and Eastern Europe, Canada is below that, and some countries, even in Northern and Western Europe, are below that. So, a lot of the world is headed towards this quite substantial population collapse, even. We can talk about the role of immigration in a minute.
- 4.17 But let's just look at some projections put out by the United Nations in 2004, based on the low fertility scenario, which actually seems to be the one that has come to pass in the ensuing 20 years. We have eastern Europe going from a population of 300 million in 2000 to a population of 60 million in 2300. And arguably, it could be faster than that, given the numbers that we've seen. Western Europe going from 180 million 2002, about 80 million in 2300.
- 4.18 What does all this mean? This substantial decline of populations as a result of total fertility rates being low, therefore, fewer mothers, fewer kids, fewer mothers, going down that curve, logarithmically. It's like a reverse Malthusianism. What is that going to mean, politically? And what I said at the beginning was that political demography is very much about the unevenness in population change over territory. So unevenness between the developed world and the developing world between, say, Europe, Asia, Africa.
- 4.19 That's going to express itself in the relative share of the global population that is going to be accounted for by different continents, for example. This is a chart from the United Nations, 2019, projected population by world region. Now, it's worth saying, the UN's

projections have consistently been wrong in the direction of overestimating fertility rates. So fertility rates, the UN, and many demographers, long has had this unquestioned assumption that fertility rates would just, if they had were gone below the 2.1, they would just somehow automatically rise in the long run towards 2.0.

- 4.20** It was just an assumption that, well, in the long run, fertility rates will be at replacement. But in fact, sub-replacement fertility has been extremely enduring for many decades now. If we start to use those population scenarios and fertility scenarios, it's the case that, for example, the continent of Africa, which only made up a small fraction of the world's population in, say, 1950, maybe one in 10, by 2100, is likely to make up close to half global population.
- 4.21** Europe's share will remain reasonably constant, depending on immigration levels. North America, reasonably constant. Asia will peak in influence in terms of population share, maybe around 2040, and then begin to decline. Certainly, the balance of population in different world regions is going to change dramatically, since currently about 40% of births are in Africa. That's going to rise to 50% by 2050 in the world. And so, with 50% of births, it won't be that much longer— well, with a number of generations—before Africa is then half of global population.
- 4.22** What is the relationship between population then and power? One factor that matters, of course, is GDP per capita. How wealthy is a country? Because power is related to having the money to buy advanced weapon systems. And if we compare different world regions between 1950 and, say, 2001, we look at their share of world population, Western Europe drops from about 12% in 1950 to about 6% in 2001. The former USSR sees a drop. Africa sees a rise from around 8% to around 14%, so there's a growth there.
- 4.23** In terms of GDP, however, despite the drop in Western Europe's population by half, the GDP, their share of world GDP doesn't drop nearly as much. It goes from 25 to 20%, and Africa remains relatively flat. So its population is rising, but its share of world GDP is pretty flat. So this is just to say that there are other factors involved. It's not just population growth. You've also got to have some kind of productivity increase in order to realize a rising GDP.
- 4.24** Asia, showing, well, a country such as Japan is also dropping, and China is pretty flat. But China's share, even though China's share of the world population is pretty flat, its share of GDP has gone up dramatically. Clearly, it shows that much of the story of rising GDP is not about rising population, but it's much more about rising productivity.
- 4.25** In terms of aging, different age structures around the world, the developing world in the tropical belt has got the youngest population. The developed world, the Global North, if you like, which includes the most southernly part of the globe, tends to have below-replacement fertility. We're going to see a real shift in terms of a population power in the world. If we want to think about which parts of the world are going to

have the population, I mentioned already, Africa is really the one region that's going to be growing dramatically.

- 4.26** Some of those countries, like Ethiopia, Congo, Nigeria, are going to be population superpowers in 2050; whereas a number of Western countries, such as Italy, for example, may fall out of the rankings considerably. It's worth saying that the share of old age dependence in the developed world will also rise substantially. By 2050, we're likely to see 40%, upper 30% of the population of many of the countries in east Asia, Southern and Eastern Europe, being 60 or above.
- 4.27** So, one of the questions this raises is, who is going to support all of these senior citizens? We have this ratio between the working-age population, age 15 to 64, and the population, age 65 and above, the old age support ratio. In the 1950s, that might have been 7 to 1. Workers to elderly dependents by the year 2000, that was around 4 to 1, in many European countries. By 2050, that's going to be more like 2 to 1.
- 4.28** We're seeing a smaller number of people of working age supporting a larger number of people, of senior citizens, who are less likely to contribute economically and more likely to require medical care, for example, pensions. This raises this question of the old age support ratios. When you have very few workers per senior citizen, that's going to mean less money for economic growth, more money having to go into elderly care.
- 4.29** Now, before I go further, it's not the case that, just because you have a low birth rate, the population just keeps getting older and older and older until everybody's 80 years old. It's worth just correcting that perception, which some people have. Yes, there is a relationship between low fertility and aging, but it is also the case that there is an equilibrium that's reached. So, one of the reasons that we have a really big problem in western and in eastern Asian countries with population aging is that the birth rates used to be higher in the past, and they've come down, and lifespans have increased.
- 4.30** And so that's one of the reasons we've got a lot of older people in the population being supported by a lot of younger people, is that there is that previous bulge, the baby boom bulge. It's like a pig in a python. The snake has eaten the pig, and it's working its way through. And if you imagine that right before, let's say, the boomers die off, you have that big bulge. Then you've got a small snake where you don't have as large generations. That's going to be the maximum for old age dependency.
- 4.31** But let's just imagine we've had a birth rate of 1.0 for 100 years. Well, then the people born 50, 60, 70, 80 years ago will not be large generations. And so, the snake will actually be relatively symmetrical. We're not necessarily going to have a worse aging problem. It will actually stabilize at some point. What percentage over 65 is the maximal point, I can't say, but maybe in the long run, it'll be maybe 35, 40%. It won't be like 50, 60. So there's a limit to this population aging, but we are just at a moment where that's going to get worse and worse over our lifetimes, because of previous baby booms and subsequent decline in fertility.

- 4.32 What is the relationship then with immigration? Is immigration a solution to the aging problem, the decline of the working age as a percentage of Western populations? Haas' argument is, and in fact, that of demography as a whole, demographers would agree, I think. There is some debate, but basically, what has to happen when you—if you want to maintain a fixed share of your population, age 15 to 64, as a percentage of your total population, then immigration has to be high and rising almost at an exponential rate.
- 4.33 A study was done in the European Union. The amount of immigration coming into Europe, typically from outside the EU, is maybe between 500,000 and a million. That would have to be in the tens of millions, consistently, year in, year out, in order to have a consistent and maintain the existing working-age population share. The reason this is the case is, it depends obviously on who the immigrants are.
- 4.34 But on average, if you assume that immigrant populations coming in will tend to converge with the host population's birth rates, then they're going to age. They're going to need elderly care. They're going to need pensions, and therefore, you're going to have to bring in new immigrants to pay for the older immigrants who have now aged, and you have to keep bringing in newer populations.
- 4.35 Now, if you were bringing in, let's say, populations that had the fertility rate of the ultra-Orthodox Jews, and so you were then bringing in endogenous growth, then that would be a different situation. But as we see globally, globally fertility levels are pretty much dropping everywhere. We know that immigrants and their descendants tend to converge with the host population.
- 4.36 What this means is that, in fact, immigration can only be a temporary solution, and the levels of immigration that would be required are so high as to be politically infeasible. So it's unlikely that immigration is actually going to—well, in fact, immigration is not likely to be a solution to the aging problem in the long term. It could be a stop-gap solution for one or five years, but not long-term.
- 4.37 Now we might say, well, fertility rates are likely to bounce back to the replacement level, but we haven't seen that. What we've seen is the demographic transition through the 20th century; since the baby boom, we've seen only decline, pretty much. Very little difference from that. We've seen it in country after country around the world. There seems to be something pretty structural and pretty enduring in this. And so, that means there's going to be fewer workers per old age dependent.
- 4.38 The question then, politically, is, well, older people will form a larger share of the electorate. If we're talking about the world today, going forward to 2050, more older people as a share of the electorate. We know that older people vote more than younger people in percentage terms. That's actually changed even in the last few decades. The gap between old and young people's propensity to vote has also widened.
- 4.39 What does this mean? Does this mean then that old people are going to vote for spending on old people and not for spending on things that matter to young people,

like education? As we get that drop in working-age population, are we going to be into this feedback loop of more older people voting to benefit themselves with generous pensions and retirement policies, sucking money away from spending on young people, inflating asset prices, leading to even further drop in fertility, leading to a population doom loop? Is that going to occur?

- 4.40 It's worth saying that, looking at the major powers in the world, Russia is likely to have a big drop, a significant drop in population. China, in particular, one of the rising powers of our age, the rise of China is a major theme in international relations, but with a total fertility rate pushing towards 1.0, does this mean that China will age very quickly, very soon become older than America, if it isn't already older than America, and as old as Western Europe, and therefore, no longer be in an optimal geopolitical position?
- 4.41 One of the debates that comes out of this is the question of, where are public funds going to be spent? Guns versus canes. If we're thinking about global power projection, especially now, we have this debate in the West about increasing military spending, vis-à-vis the Russian threat. European countries, Canada, NATO countries have to find funds to increase military spending as a share of GDP, to whether it be 3% or 5%.
- 4.42 At the same time, you have aging populations with increasing pressure on public health care systems. So guns versus canes—canes would suggest money needs to go into hospitals, social care, all of the things where demand is going to be mushrooming due to the aging population, taking money away from budgets that include, for example, military spending. So, is there a trade-off between military spending and health care spending? As countries get older, the shift arguably will be away from military spending towards health care spending.
- 4.43 And what does that mean then for the balance of power in the world? If you get fewer workers, you've got less tax revenue. Less tax revenue to spend on the military. Meanwhile, the elderly are demanding a greater and greater portion of your government budget. There may be a decrease in savings because there is a pattern, whereby people tend to save during working years and spend during their retirement years. Therefore, a country that's got more retirees is going to have more spending and less savings.
- 4.44 This is the theory, anyway. Now, there are some wrinkles in that in actuality. But still the idea is that, if a person is meant to acquire savings during their working years and spend it in retirement, then their savings will drop over their lifetime. Countries that are older will have less savings, which means less money for investment in economic growth, and certainly less money for military expansion or expenditures.
- 4.45 Just looking at the elderly care profile, so in Japan, it's argued that the increase in elderly spend will be 15 times more than the military spend. Germany, seven times, France, five times. And in some cases, cuts to military spending is already being justified to some degree, in terms of the demands of the elderly care system.

- 4.46 Now, in countries that have no real tradition of governments supporting the elderly, whether through pensions, whether through health care, maybe China is an example of this, then in fact, it'll be a less severe problem to have an old population. If provision for the elderly is not coming from the government, the government doesn't have to worry as much about a rising share of senior citizens.
- 4.47 Of course, the people who do shoulder the care burden are going to be the children of the elderly, and that could actually still damage, even if it doesn't damage the government budget, it could damage the economy and lead to discontent. So, this question of what an aging population does to military power could be a matter of government budgets being strained by health care spending, and therefore diverting money away from the military to health care.
- 4.48 But even within military budgets, it's been observed by Haas and others that, in fact, these military budgets are shifting into personnel costs and obligations to military retirees, military pensions, and health care, for example. Veterans' health care in the US would be an example of that. Even within the military budget, more of that is going to personnel and retirement costs and not towards weapons.
- 4.49 In Germany, four times the money within military budgets is going to personnel, as distinct from weapons. In the US, it's a better balance, but this idea of spending a lot of money on weapons and on active militaries, that may be just more and more difficult in an aging society. This leads to Haas' argument about the geriatric piece, that an older world is a more peaceful world. Why? Because in older countries, there's less military spending, less appetite for risk-taking.
- 4.50 Also, when people have fewer children, they're going to be less willing to risk the lives of their children in combat, as compared with very young countries where there are a lot of surplus children. That's really the argument of Gunnar Heinsohn's book, [Sons and World Power], that, throughout history, surplus sons have been a major source of global conquest and violence. So looking back through history, the Vikings, the conquistadors, these were often the younger sons who weren't going to inherit land, and they therefore had to find a role for themselves, particularly of the nobility.
- 4.51 But these surplus children have played a big part in global military conquests and movements. And so that, when you have a country like Afghanistan that has a very high birth rate, there are a lot of extra children, extra sons, and therefore, there will be more of a population that can be recruited for war. And also, as we talked about last time, when we talked about the youth bulge, men in that 15-to-30 age bracket are more risk-taking, and therefore, that also will contribute to war.
- 4.52 But just this fact that you're going to have a lot of children around, Heinsohn compares Lebanon in 1975, when it had a very young population and it had quite a long protracted civil war, with 2006, where it had a much more, a much briefer civil war, and argues one of the reasons for the briefness of 2006 in Lebanon, compared to '75 to '90,

was that Lebanon in 2006 had already had a more mature population age structure, and therefore, the civil war didn't get out of control. It didn't become as big as it did in the 1970s.

- 4.53** Taking that logic further, you take a country like China, which is rapidly aging. What is the appetite going to be for sacrificing what is very often one's only child? The nickname "little emperors" for the only children—are they going to want to go to war? Are their families going to accept them being drafted into battle? The Russian-Afghan war, Afghanistan and Russia, you're looking at one society that's got very young, high fertility, lots of surplus sons. Russia already, well, Soviet Union already had relatively low birth rates, and there was a lot of pressure from Russian mothers not to have their sons go into war.
- 4.54** Now, there's an interesting test now with the Russia-Ukraine war, is to know exactly whether these pressures, in a country like Russia that has relatively low birth rate, relatively aging population profile, is that going to make it harder for Russia to continue its war? Will it make it more likely for Russia to want to come to the table, if it's the case that there is a constraint on the number of young men willing to go to war, and when the electorate is very mindful of casualties? Whereas in Afghanistan, maybe people were less mindful of casualties.
- 4.55** So this was the Heinsohn hypothesis, which Heinsohn raised as well, looking at the American and coalition involvement in Afghanistan, that in Afghanistan, there was simply much more of a willingness to get involved in military conflict. It was much more risk-taking. We have to think about then the aging of the global population, and whether the aging developed world will mean a loss of military power in the developed world, compared to the developing world.
- 4.56** The first world will become more and more reluctant to intervene in conflicts around the world; whereas if you went back to the '70s, Vietnam, Iraq, the Vietnams and Iraqs and Afghanistans, that kind of willingness to project military power will become more unusual. If countries like the United States or regions such as the EU are unwilling to risk the lives of their citizens in war, that'll mean they're going to be less able to project power around the world.
- 4.57** Maybe there will be new, hungrier societies that are more willing to risk lives that will become more important in global affairs, when it comes to hard power. In terms of where populations are young, we can see on the map here, it's sub-Saharan Africa, it's Afghanistan and Pakistan, perhaps parts of Southeast Asia. Where might we look for the next rising global military power? Might we see a shift of military power and influence to the current developing world?
- 4.58** With Europe, Japan, Russia in decline, China not far behind, the US, which had had a reasonably high total fertility rate, close to replacement as late as 2010, that's now clearly heading to a much lower level. Where is it going to be? Middle East, South Asia,

Africa—where is the new power going to come from? But of course, there are a number of prerequisites for rising military power, including having a high GDP, which means having quite a productive economy. It's no good having a large population if, in fact, you're a poor country. Having a significantly complex economy is probably at least somewhat necessary in order to be a rising power.

- 4.59 If we turn to international relations theory, which has looked at the rise of military powers across history, going back to the ancient Romans, Hedley Bull makes the argument, today 100 million is necessary to be a great power, but even that is not sufficient. You have to have other elements. You have to have GDP, for example. In a world—there's a theory in international relations called power transition theory, where rising powers can cause instability in the world system.
- 4.60 China was one example of this. A power that is rising that was formerly not a power but is rising, is going to want to assert itself in the world. What will that power be? Let's look historically at the relationship between population and power. The British Empire is rising in the 18th and 19th century on the back of a major population expansion. So, to give you an example of how much of a population expansion, in the year 1700, France and England—France had about three times the population of England.
- 4.61 By the year 1900, England had pulled even with France, while exporting 25 million people to Australia, Canada, USA, South Africa. Britain had a real population explosion. France had a relatively muted population rise, because Britain industrialized, which allowed early marriage, which allowed more of a population increase. Britain's rise very much worried the French. The French were under Napoleon, and later were employing all kinds of strategies to raise the birth rates, without a great deal of success.
- 4.62 But they really worried about their population, and that not having a growing population would see them lose in the military race with their rivals, Britain. You then had the rise of Germany in the late 19th, early 20th century, again experiencing a population explosion. That was worrying Britain and France, whose populations had started to slow down due to the demographic transition. The next major power to rise was Russia, who, again, they were rising on the back of population boom. Then came China.
- 4.63 There is this relationship between rising population, rising power in the world. But of course, it's not a hard and fast one. If we look at the rise of Nigeria, Congo, Ethiopia today, would we say they are going to tread the path of Germany, Britain, Russia, et cetera? It's unclear, because not only do you have to have a certain amount of wealth, but if you're disunited and heavily riven with ethnic conflicts and divisions, like Nigeria, Ethiopia, Congo, then it's going to be harder for you to have a unitary foreign policy and act in that way in the world.
- 4.64 So yes, there are links between population growth and world power, but it's not one-to-one, and it's mediated by other things. As we've seen so often with demography, it

has to interact with other things to have an impact on politics. The same is true if we look at the developing world, where it's not obvious what the next major power. If you take the BRICS, Brazil, Indonesia, Turkey, and so on, problem in a way, with a lot of those countries, is their fertility rates have now peaked and are starting to go down, places like Turkey and Brazil.

- 4.65 So, you would then have to look, let's say, to the Nigerias and Ethiopias and Congos, which are not economically that sophisticated yet, and they are strongly divided by religion and ethnicity. So, again, maybe not likely to become world powers anytime soon, but would have the population basis to be world powers. Nigeria could be a world power, if it were—actually, Nigeria is one of the stronger economies in Africa.
- 4.66 Perhaps Nigeria has some of the ingredients, but it is heavily divided between the Muslim north and the Christian south. And you can imagine, for example, Nigeria, the Muslim north would be oriented politically more to the Muslim world into one set of politics, and the Christian south oriented more, perhaps to the west in a different set of politics. That's going to make it very difficult to have a singular view of foreign policy.
- 4.67 So, the question of population aging, how else might it affect power? One way is via culture. So Jackson and Howe's book, *The Graying of the Great Powers*, talks a lot about this question of social mood and how older societies have a different culture. They quote Alfred Sauvy, "A society of old people living in old houses, ruminating about old ideas." Is that what societies are going to be like in the aging West and East Asia, and increasingly now parts of South America and parts of Asia?
- 4.68 There's also the question of fluid intelligence and mental agility declining with age. Studies of Nobel Prize winners shows that their work peaks in terms of patents, and the age of Nobel Prize winners, it peaks in the 30s, declines sharply in the 40s. Now, that varies by subject area; in mathematics, it's much younger. Mathematics and physics is much younger than, say, history, which is older. It is true that the average age of a Nobel Prize winner has been going up.
- 4.69 There's a debate over to what extent this is genetic or to what extent it is something to do with culture. But still, there's no doubt that, on certain types of tasks, fluid intelligence peaks earlier. What this might mean then is a culture and age structure that is less conducive to developing new ideas, that is less creative. So one study of new entrepreneurs finds 40% under age 35, 70% under age 45. So very few entrepreneurs are over age 55.
- 4.70 People become more safety-conscious as they enter retirement. They might shift their stock portfolios to safer investments, less risky investments as they get older, because they want to be able to depend on their stock portfolios for their retirement. That might lead to a less risk-taking society, which is going to have slower economic growth as a result; whereas, rising populations create a climate of optimism, in which you get more new investment, new ideas, and new products.

- 4.71 Also, when a population is growing, you can depend on new consumers entering the system. That changes the nature of the economy. Therefore, economic growth is more vibrant in younger societies than in older societies. It may be that the stagnant economic growth rates in developed countries, particularly in the West, is partly a result of their aging populations.
- 4.72 It's worth saying that not everybody has this skeptical view that aging is inevitably going to lead to economic decline, cultural decline, political decline. So, one alternative view is Jennifer Sciubba's view that, in fact, old people, for example, are unlikely just to look out for themselves. They are likely to care about younger generations and about the nation as a whole.
- 4.73 So they can be persuaded to support policies that help young people and not just to look out for policies, such as not raising the retirement age or spending on old age care. That it may be possible for states to adapt to an older population by raising the retirement age and trying to bring new groups into the workforce, such as women or perhaps the elderly, but making better use of your population, particularly your older population, in order to combat.
- 4.74 This gets to a debate around this definition of the working-age population, 15 to 64, or 20 to 64. That very rigid definition of who is unproductive and who is old arguably doesn't suit an age where manual work has declined, and cognitive work where you don't necessarily need to be expending a lot of physical energy, and so maybe people can work longer. So one solution might be raising the retirement age, and more people, especially if they're not doing physical work, remaining in the workforce longer than in the past.
- 4.75 It also could be the case that you don't necessarily need a lot of young people. You can just use your young people better. As long as you're directing young people into tasks that require fluid intelligence, for example, and you're harnessing them correctly, then you can make up for the fact that there are fewer young people as a proportion of the total population.
- 4.76 And so, maybe Europe is not doomed, because maybe it can develop high human capital amongst its elderly population, and its elderly can be a lot more effective than the elderly population in younger countries. It can make up for the fact that it has fewer young people by having higher productivity amongst its older population.
- 4.77 In that sense, we have to rethink what we mean by the term dependency ratio, that if you have seniors who are economically productive, you can't think of them as dependents. Now, they may require some medical care, but then if the health of seniors is improving, then in fact, that will also lighten the dependency.
- 4.78 There's been some interesting work done by Vegard Skirbekk, a demographer, on this question of cognitive functioning in older populations. In terms of short-term memory, cognitive tasks as tested by psychologists, one thing researchers find across different

countries is that an old person in Scandinavia or in Western Europe is functioning at a much higher level than an old person in China, for example. So a senior is not the same thing in Sweden as in China or Mexico.

- 4.79** This is a chart just looking at short-term memory scores by age across different parts of the world. You could see that an 80-year-old in Northern Europe, scoring a 0.4 on short-term memory, is equivalent to a 65-year-old in Mexico and a 55-year-old or 53-year-old in China. What we're saying, this graph is suggesting, is that an 80-year-old in Scandinavia is scoring as well on short-term memory as a 50-year-old in China or a 60-year-old in Mexico.
- 4.80** In other words, if you have a population that is well educated and healthy, an elderly population that's well educated and healthy, they can be productive for very long. So maybe in China, aging really is a threat, if this is correct. Maybe in China, actually, there's going to be a big increase in old age dependency because the senior population is not so healthy, doesn't have as much cognitive development, which is related to education level.
- 4.81** The education of those older Chinese citizens might have been lower, and the nutrition might have been worse. Perhaps a solution for the West is in having a healthier and active senior population that can counteract population aging. Actually, that would suggest the problem may not quite be as bad as we expect, argues Skirbekk. Again, I mentioned short-term memory. That seems to be related to the population share with at least secondary education, which is much higher in Northern and Western Europe than in Mexico or China at the older age range.
- 4.82** So, that mental activity seems to correlate. Now, the question is, can you make use of those older voters and older workers in your economy? To what extent can you do that to counteract the problem of population aging? There are also other technologies that are emerging. If you don't have enough social care workers, can robots be part of the solution, as they are in Japan? To what extent can you use robot technology to substitute for the lack of young workers?
- 4.83** To what extent can you use innovation, labor-saving innovations, to get around not having as large a workforce? And artificial intelligence would be another tool that could be used to, in some way, get around a smaller workforce. So can these technological fixes compensate for having a smaller workforce? And this is going to be one of the big questions going forward. I would say, along with this debate is this question of whether we're going to see a golden age for young people in the West, because of population aging, because of the demand for young people.
- 4.84** Are they going to be able to demand much higher wages, and are they going to be in very high demand? We don't see that right now, but it's possible that we might be entering a period where there will be very high demand for young people because of their scarcity. Whereas, if they had come of age during a baby boom, then it's possible

that there would be higher unemployment. So maybe unemployment will become a thing of the past, possibly.

- 4.85** The last thing I want to say is just to briefly mention this question of whether there can be a rebound in developed-world fertility. And there was talk of a rebound in the early 2000s. One of the things we definitely saw was that highly educated women had— for a long time, the more education a woman had, the lower her birth rate, lower her total fertility rate. That was true in 1975. If we look at the blue line in this chart, it simply slopes downward. Countries with a higher human development index, so education, for example, and income, lower total fertility rate.
- 4.86** By the time we get to 2005, we start to see actually a different, more hockey-stick curve, where the countries with the highest human development index had a somewhat higher total fertility rate. So the Scandinavians of this world, somewhat higher total fertility rate than those with a moderately high human development index, say, Southern Europe, East Asia. Now, of course, the Afghanistans and Congos and the Malis, of course, have the highest total fertility rate and the lowest human development.
- 4.87** But this has just suggested that perhaps a different pattern was coming in, in which the highest human development index would be associated with higher total fertility rates. I think that since these charts were developed, the evidence is still reasonably strong for this. But I would say that the record of the Scandinavian countries and the wealthiest countries is that, in fact, total fertility rates haven't fallen, particularly since the pandemic.
- 4.88** It's far from clear whether this relationship will continue, but it's certainly something to think about. Even if there is a rebound, even if the most developed societies with the highest women's education do have somewhat higher fertility than countries with moderately high education, it's still the case that those countries, such as the Scandinavian countries, have below-replacement fertility. They have not rebounded to replacement fertility.
- 4.89** Even though, yes, it does appear like the relationship between education and fertility rates is no longer a negative one, it still doesn't mean we've seen a rebound to replacement fertility in developed countries. With that, I look forward to your questions.
- 4.90** We learned that world's population has been changing from 2 billion, now it's 8 billion. Projection is 10 billion, with all the pros and cons for states and things we talked about today. What is the real number we should be aiming at, for the whole world, for long-term, so we don't go from a maximum to a minimum all the time? What is the real sustainable number we should be focusing on?
- 4.91** Some people say we need to be at a million. Others say, who are more that cornucopian Julian Simon, they could say, "Well, no, we need to be expanding.

Population is a good thing. Each human being is more happiness in the world. We need to have more happiness." Others say, "Well, no, that's going to crowd the world and make people less happy." I think that's a philosophical question. I can't really answer it.

- 4.92 It's worth saying that, if you have above-replacement fertility forever, and you don't have settlement in an interplanetary sense, settlement of other planets, then at some point, you're going to get the Malthusian hypothesis of overcrowding, and the world will be just a bunch of flesh, right? No, not quite, but you will. There are these experiments where they grow bacteria in a jar with a certain amount of nutrition, and they just expand and expand, and then they hit some wall and then collapse.
- 4.93 You don't want to hit that wall. That's obviously human immiserizing. We ultimately want to have a replacement level fertility at whatever the population is that we want to maintain. But it just, that population size is extremely difficult to control. In a way, we have policies that can reduce it. Those policies seem to work better than the policies to increase it. It's just very hard to control.
- 4.94 You mentioned, the more a country has a younger population of men, they can maybe risk going to war. Given that this is the case, for example, China and their population, we've seen where they're getting to maybe to an older age dependency. Has this played a big factor in the way they may conduct warfare, in the case, let's say, invading Taiwan, which they have been wanting to maybe probably do that in the near future? Would you say that has been because of their population, having maybe older men and taking the risk of going to war? Has this played a big role in how they conduct warfare?
- 4.95 Yeah, I think one of the questions—China is not a democracy. If mothers didn't want to risk their precious sons in battle, in a democracy, they would vote against any politician who was proposing war. Now, of course, there are many factors that go into the decision to go to war, so I don't want to make it simplistic.
- 4.96 But the question is, in an autocracy like China, where the party and Xi make the decisions, to what extent do they have to take public opinion into account? There's an argument that, even in an autocracy, you have to take public opinion into account to some degree. If the public really doesn't want something, that might make it harder to do. Of course, the party members themselves will have only children, and so that might influence their thinking.
- 4.97 One of the questions is going to be whether technology might make—if you had robots, for example, that could fight wars, robot technology, or just drones and robots, it's possible that that might change the calculus to make it more feasible to launch an invasion, especially if you didn't think that you were going to then get bombed and attacked.
- 4.98 But the geriatric peace argument is that, if China had a higher fertility rate and a younger population, it might be more likely to be willing to risk going to war and losing lots of people. It's worth saying, however, that in any decision, the demographics is

only one part of the calculation. In the case of, say, Putin's invasion of Ukraine, yes, Russia has an aging population. Yes, it has a low birth rate, but there may be other factors that are dominant. The demography is only one of a number of different factors.

- 4.99** Maybe the other factors are all moving in this—the demography might be pushing in a peace direction, but maybe the other factors are pushing in a warfare direction. But all other things being equal, you would say that if Russia had a large and expanding population, you would say their appetite for being in a war for longer would be greater. And so, all other things being equal, them having an aging population is probably going to lead to a more peaceful outcome sooner than if they had a young population. But of course, there's no guarantee that anything will happen in any given year.
- 4.100** What are the providing theories behind why we were maybe seeing, in some of those Scandinavian countries, a bit of a higher fertility rate? More generally, what levers can be pulled politically, societally, culturally, or economically to improve fertility rates in developed countries?
- 4.101** We know more about why there is variation in fertility rate between countries. So there are a couple of factors. One argument is that, in very patriarchal societies like Afghanistan, you have high fertility. That's part of a package including low female education. But in a partially semi-patriarchal society, where there are still relatively well-defined gender roles, so in Japan, for example, or in Southern Europe, Italy, where there's an expectation of, there's more of an expectation of women taking on the carer role, the mother role, when they get married.
- 4.102** Therefore, that makes women more reluctant to get married young and to have children. One of the arguments is, if you take East Asia, Southern Europe, those are the lowest fertility zones. One reason is, they are semi-patriarchal. That's one theory. Whereas Scandinavia and the English-speaking world are less like that. There's less of a trade-off for women. Then you have France and Scandinavia that have subsidized daycare, childcare, better subsidized childcare. That's possibly one factor.
- 4.103** The Anglo-Saxon world has tended to have higher, the English-speaking world has tended to have somewhat higher fertility, along with Scandinavia. One reason is because you can have kids outside of marriage, including single parenthood, or you can remarry somebody else and have kids. So, the more of that fluidity—you can enter and exit the workforce more easily—these are some of the reasons why.
- 4.104** So, actually, even though a single-parent family is not a great thing in many ways, it does lead to somewhat of an increase in the total fertility rate. That greater fluidity of entering and leaving relationships and having kids in and out is associated with somewhat higher fertility rates in the English-speaking world. But what we're talking about really are differences between Italy at 1.3, the Anglo-Saxon world maybe at 1.5, 1.6. We're talking about variation of about 0.3, not massive variation.

- 4.105 But it is worth saying, especially if you go out 100 years, there's a huge difference between being at 1.5, 6, 7 and being at 1.2, 1.3, in terms of what the percentage of your population is that's still going to be around. That is an important difference over a long period. We know that. I'm trying to think if there's anything else from the established demography literature. I think those are the main, the main factors.
- 4.106 Now, sorry, I should mention, in the developed world, religiosity is probably the best predictor of family size. And so, the more religious a society is, in terms of attendance in particular, if you have a society with high religious attendance, you're going to have higher total fertility rates. One part of the world that's seen an increase is the Central Asian republics. After the fall of the Soviet Union, they became less atheist, so less Communist and more Islamic. And there was a religious revival, which has been linked to rising fertility rates.
- 4.107 Whereas, the US has become less religious in the last 20 years, and that's linked to their declining fertility rate. Ireland is another example of that decline in religion.

## Lecture 5: Demography and Democracy

**At a glance** — Demography and democracy — how population composition shapes politics.

- 5.1 Hello, welcome back. So one of the things that we talked about in the previous few lectures was this issue of the broader question of how population shifts affect politics. And in particular, what I'm going to talk about today is democracy and democratization, which is a major topic in comparative politics. There were at one time very few democracies in the world. Then the West democratized to a very large extent.
- 5.2 Then we saw the spread of democracy in Eastern Europe, in Latin America, in Asia, in parts of Africa, particularly from the period from the 1980s to the 2000s. More elections were being held after the fall of communism. Particularly coming into the 1990s, one of the big debates was, how is it that countries democratize? What are the conditions that are needed for countries to become democracies?
- 5.3 You had different debates between more optimistic writers like Francis Fukuyama, who proclaimed the end of history, and others who were a little bit more pessimistic in some regards, like Samuel Huntington. Even though Samuel Huntington had written about this new third wave of democratization, he was also of the view that democracy was more difficult, particularly in Muslim-majority countries, but in countries also that had unfavorable demography, such as a very young age structure.
- 5.4 That gets us to the question that we're going to look at today is, to what extent is the age structure of a society, whether it is a very young population, a very old population, related to whether it can become a democracy? How important is the demographic transition to the democratic transition? Is there a relationship between these two? We'll see there are a number of writers, in particular Richard Cincotta, who argues that this is a very important relationship.
- 5.5 What is then the relationship between demography and democracy? We have a picture here of Tahrir Square, which featured in the Arab Spring. Is there any connection between Egypt's demography and its likelihood of becoming a stable democracy? We'll just revisit the concept of the demographic transition, which I've mentioned a number of times, the transition from high birth rates and death rates to low birth rates and low death rates, one steady-state to another steady-state.
- 5.6 In between that, of course, the death rate falls first, so you get a population explosion until the birth rate comes down. When you get that population explosion, you get the very young population. With a demographic transition, then you get a transition to a more mature age structure, slow down the population growth. What this does is, it puts less pressure on the economy and governing institutions at the elite and mass levels, which, of course, were designed for a pre-existing population, rather than a burgeoning population.

- 5.7 And so what does that mean for the chance of a country becoming democratic? If it goes through a demographic transition and there is less population pressure, to some degree, on the economy and the institutions, does that mean democracy becomes more likely? We'll see that that is the argument of theorists like Cincotta, who make the argument for the relationship between the demographic transition and the democratic transition. However, it is not obvious that having a young population is always bad for democracy.
- 5.8 In fact, it might be necessary because young people can be a revolutionary force that can overthrow autocracy. And so maybe with a young population, you're more likely to get revolution, and you're more likely, therefore, to get democratization. But perhaps, on the other hand, having a lot of young people, particularly young men, who can be recruited into insurgencies, that's great for revolution. But then perhaps when you are a democracy, young men can be a destabilizing force and can make it difficult to consolidate that democracy.
- 5.9 There are also going to be other factors we have to consider, such as the relationship between a young population and an existing older political elite, and what that political elite perceives to be the threat from that younger population, and whether that then affects their willingness to open up the political system. I want to begin with this idea of young people as a revolutionary force.
- 5.10 Jack Goldstone, who has written on revolutions quite a bit, makes the argument that, in fact, demographic forces were important as a conditioning factor, prior to a whole series of important European revolutions and also Asian revolutions. Starting in England with the Glorious Revolution of 1688, moving to the French Revolution of 1789, the 1848 Springtime of Peoples in Europe, the 1848 liberal revolutions, the 1917 October Revolution in the Soviet Union. They all were preceded, argues Goldstone, by demographic rises, which coincided with rising food prices.
- 5.11 We also see similar kinds of population resource pressures in the 19th-century Ottoman Empire, Japan, and China, prior to revolutions there. That you had, because of rising food prices, a certain degree of unrest in the countryside and peasant discontent, fuel which could, in and of itself, not necessarily lead to revolution, but in combination with other factors. This is always, often how demography works. It works in combination with other factors to produce political outcomes.
- 5.12 And so this question of revolution, of popular sovereignty being expressed by rebels against an existing order, seems to be related to population growth. Here's a chart that Goldstone presents in his 1991 book, where he has the population of France, in the period 1650 to 1790, rising pretty steadily from the late 1600s up until the French Revolution in 1789, and tracking food prices, which are also rising.
- 5.13 So that rising population, if you recall the lecture on climate change, population, and resources, with a rising population but a flat food production, you're going to get rising

food prices. Now, of course, the other possibility is you don't have much of a rising population, but you might have a climate event such as a very harsh winter, and that could also raise food prices. But if you look at this chart, what we see is a steady rise in population and a steady rise in wheat prices, preceding the French Revolution.

- 5.14 These staple crops, to the extent those prices are rising due to greater population pressure, that is creating a certain amount of tinder that can be harnessed by other forms of discontent. Again, you need other forms of discontent, so it's not just enough to have these preconditions. You also have to have other factors as well—perhaps ideas such as those of the emergent liberals in the French Revolution.
- 5.15 Goldstone also talks about the concept of elite overproduction. So it's not just that you have population growth in the countryside, given a fixed set of food resources and higher food prices leading to mass discontent. But if you also have an overproduction of elites, and there are only a certain number of positions in the bureaucracy of Louis XIV or whoever the king of France is—I think it was maybe Louis XVI at the time, sorry—if you have only a certain number of positions at court, a certain number of positions in the bureaucracy, in the officer corps, then there are going to be children of the nobility of the elite who don't have a role.
- 5.16 And so that elite overproduction is another dynamic. And when you have elite overproduction, you can have young, relatively elite people with high expectations and no hopes for a status, and they can therefore be important in revolutions as well. Sometimes these revolutions are in the direction of democracy, like the French Revolution, the Glorious Revolution. But at other times, they are in the direction, for example, of fascism or of communism, in the case of the October Revolution.
- 5.17 So there are really two sides to a young population. It can be a force for democratization, as we see in France, 1789, but equally a force for reaction for chaos and illiberalism. And so, in fact, these liberation movements, as Goldstone argues, those who make the revolution don't necessarily take power out of the revolution. With the Arab Spring, perhaps there was one motivation of those who were revolting, but on the other hand, who took power after the Arab Spring? In most cases, democracy was not consolidated thereafter.
- 5.18 There has been a tradition of writing about young populations as a revolutionary force in Europe. Here is Moller, talking about Germany. In the early 19th century, he argues, "The explosion of revolutionary, warlike, nationalist, and anti-French activity towards the end of the 18th and early 19th centuries received social fuel from the large youth cohorts of Europe's initial population explosion." This was the beginning of German nationalism in reaction to Napoleon's invasion of the German states.
- 5.19 Or we have here, Commager writing on the emergence of modern nationalism and imperialistic expansion, which coincided with the first great population increase in modern history. That was in Europe. And then he's saying, well, now in Asia, Africa, and

South America, they are now going through a population explosion, and we're witnessing an upsurge of chauvinism in those societies as well. So is there a relationship here between that youth bulge phenomena and the propensity towards revolutions of various kinds, which could include coups, could include things that are reactionary, illiberal, equally that are liberal?

- 5.20** And so, we can look at the population pyramid of Afghanistan and see an extremely young population there—a lot of potential, possibly, for various uprisings, whether this be the Taliban in Afghanistan, for example, is having a young population. Does it make it more likely that you're going to see these kinds of insurgencies? Whereas a mature society like the United States would not have the same revolutionary potential, the same civil war potential, as a country like Afghanistan.
- 5.21** How does this relate to democracy? On the one hand, we've mentioned that a young population can be conducive to liberal democratic revolutions that can install liberal democracy. So that's a plus. But what if the society is already democratic, as in interwar Germany? Then youth movements may, in fact, be anti-democratic in their results. So youth revolts, especially because younger populations may be impatient with the pace of change, they may be impatient with the messy compromises of pluralistic democracy and rules and norms and procedures that are important for consolidated democracy.
- 5.22** They may not necessarily support independent judiciaries which make rulings which go against what they believe is necessary. And an open society relies upon independent judiciaries and pluralism. It relies upon granting rights to your opponents, not beating them up in the street or banning them. If younger populations are more impatient with these institutions of democracy and procedure, then they actually can be a threat to democracy and can limit the consolidation of liberal democracy.
- 5.23** Perhaps more important than all of this is that, if we're thinking about a society that is autocratic and may transition to a democracy, places like Morocco and Tunisia, if we're going back a number of decades, what is it that tips them over into becoming democracies? If the established elite, the established business community and bureaucracy, look at a society that is very young and chaotic and violent, they are going to be much more reluctant to open up power to the electorate, and especially if the electorate is very young.
- 5.24** If the average age of a citizen is under 20, and you are the established elite, do you want to hand power to that group, to the majority in society? Because you can't trust them, you think that they're going to turn on you and be violent towards you. So this fear of the majority, if it's young and therefore more hot-headed perhaps, more intemperate, less respectful of existing hierarchies and institutions, that can make it more difficult for an established elite to take the risk of sharing power.
- 5.25** Whereas, if you have a more mature age structure, an older population, people are working, perhaps more respect for authority, then perhaps the existing autocratic elite

might say, "Well, maybe the risk isn't that great. Maybe we can find reasonable, moderate people who can ensure an orderly transition." This is one important mechanism for why we might expect younger societies to be less likely to democratize, is because those in power are going to be more reluctant to open up the political system and share power with what they consider to be a threatening majority. Whereas in a more mature society, there may be greater trust that that won't happen.

- 5.26** We can see this instability. If we look at the rise of Nazism, some would argue that the relatively young population in Germany in the interwar period was a factor in the rise of fascism. This is taking place at a time when Germany is a liberal democracy, in the Weimar period. And so anything that involves a rebellion or a reaction is going to move this society in a more illiberal direction.
- 5.27** In 1933, for example, the 20- to 45-age cohort was the largest in German history in both absolute and relative numbers. And those aged 15 to 29 made up a considerably larger share in, say, 1925 than they did in the 1970s. And so the Nazis were able to recruit large numbers of young people into, for example, street movements, as were the communists, by the way.
- 5.28** So you had the far left and the far right, both having street battles and gangs and recruiting large numbers of young people, young men. And if you just look at the age structure of Germany in 1925 versus 2018, you can see how different the population pyramid looks, that bulge of people in the 15-to-30 age ranges in Germany in 1925 versus 2018, much more mature age structure.
- 5.29** How much did that play a role in the rise of Nazism and other fascist movements? We talked about the youth bulge already, the link between young populations and violence, for example, IRA recruitment, the war in the former Yugoslavia, the recruitment of young people into insurgent movements. It is easier in a younger society than it would be in a more mature age structure. We saw that in Iran as well. In both 1979 and 2005, the youth bulge made it favorable to have a rebellion.
- 5.30** Now, we might say, well, in the 2000s, this was a liberal democratic rebellion. In 1979, perhaps it was more of an Islamist, although the actual revolution was a mix of the left and Islamism. That then raises this broader question of, how much is it the case that, as the world goes through the demographic transition, that we're going to see a transition to democracy, and the world will democratize on the back of that demographic transition?
- 5.31** So this question, this theory of the link between these two, as advanced by writers like Hannes Weber and Richard Cincotta, Tim Dyson and others, is a broad phenomenon. On the one hand, young people with the youth bulge are more prone to violence. They're also, with a youth bulge, are more likely to be unemployed, and therefore you have the conditions for instability.

- 5.32 Young people are going to be more attracted to radical change, as opposed to gradual incremental change, so radical transformative ideologies that promise quick changes; and they'll be therefore impatient with existing elites, existing compromises, procedures, and norms. For this reason, that kind of instability that a young population brings can instill fear into established groups, such as the established business community, which fears instability, or even the established, well, as well as the established bureaucracy and ruling class.
- 5.33 And so they might be less willing to share power in a younger society than in an older society. This might be one of the reasons we see that, in older societies, there seems to be a relationship. And so we're going to—and again, demography is always working in an interactive way. It also depends on the economy. Can it attract investment? It depends on the poverty of a society, the ruggedness of the terrain, all kinds of other factors that play into the likelihood of rebellion. Those interactive second-, third-order effects of demography on politics are very important.
- 5.34 I now want to review some of the empirical literature on the connection. What does the data tell us about the connection between the demographic transition and the chance of a country becoming a democracy? We can see on this chart here, starting with the Republic of Korea, beginning in the early 1970s, Korea actually had a large number of young people in that 18-to-30 youth bulge age range as a share of the total adult population. It was around 50%. 50% of adults in Korea were in that youth bulge, 18-to-30 range, really up until the late 1980s.
- 5.35 Then it starts to drop into, below 40%, and in the 1990s, we see Korea become a democracy that's rated free by Freedom House. So the dark black square would indicate a democracy. The white would be a non-democracy, the white squares. That pattern seems to be there in all of the countries listed here, Brazil, Taiwan, Indonesia. Now, how strong this correlation is, is something we're going to look at in a minute. But just to say that, by and large, there seems, perhaps with the odd little reversal, Thailand had a little backslide. But broadly speaking, it seems like there's a relationship.
- 5.36 How strong is that relationship? Richard Cincotta has done a whole series of articles and papers that try to chart this relationship, and it seems to show a pretty strong empirical connection. Here's a map that would say, well, based on a country's stage of demographic transition, we're going to predict the likelihood that they are going to transition to a democracy.
- 5.37 And so you can actually do this on a world map and you can say, well, the regions of the world that seem like they're going to be moving in a democratic direction are in North Africa and South America, for example. These are regions of the world—Turkey, perhaps Iran, Central Asia—places where the demography is favorable for a democratic transition. Whereas if you were to take sub-Saharan Africa, you might say, actually, based on the demography, democratization is going to be more difficult.

- 5.38** Looking at the data, if we just take the former communist states of Eastern Europe, you can see that, for example, in the late 2000s, if we take those that had a share of young adults in the population, that is 15 to 29, divided by the total adult population, the young adult share of less than 35%, those, most of those East European states were rated liberal democracies by Freedom House, whereas those with the population of young adults above 35% were much less likely. In fact, none of them were ranked fully free by Freedom House. That's one piece of evidence.
- 5.39** Let's look globally at a number of other regions, North Africa and South America, particularly the northwestern rim of South America, which is less developed. We can see that the demography is moving in a favorable direction to some degree. The share of young adults, as a share of total adults, going from 45% in, say, Tunisia, down to 36% by 2015. So Tunisia might be a country that you would bet on as having some of the—if you believe in the theory of the demographic transition leading to democratization, you'd imagine that would be the best candidate, compared to—
- 5.40** Although, actually, all of these countries are moving. Egypt has a younger age structure and is not moving in quite as fast a path down its demographic transition. So you might say Egypt would be more difficult than Tunisia. Likewise in various South American countries, Colombia, for example. Here's a chart from Cincotta's work, which plots—we have three years, 1970 in white, 1985 in green, 2004 in black. Then we have five regions of the world, sub-Saharan Africa, Europe, Middle East, North Africa, and so on.
- 5.41** You can see that they are all fitting along that regression line, where we have the proportion of young adults at the bottom, countries where young adults are over half the population of adults. Very few of them are full democracies, whether we're talking about Asia or Africa. We get up to places that have only 30% young adults. Now, that's mainly going to be Europe, but you can see that in those areas with very mature populations, almost every country is a full democracy. This is the pattern that we can find in the data. There seems to be quite a good correlation.
- 5.42** Here's another way of looking at it. The youth ratio, which is again that proportion of the population in the youth bulge age range of, say, 15 to 30, as a percentage of the total adult population—and where that's less than a quarter, so less than 23%, almost every country is rated free as a liberal democracy. And where that is very high, say above 40%, it's mostly unfree. So looking globally, there seems to be this relationship.
- 5.43** As with the relationship, say, between the youth bulge and violent civil war or violent conflict, one of the questions we need to ask is, is this link causal or is it a spurious correlation? Because we know that clearly a country that's poor, that tends to be correlated with having a large proportion of people under the age of 30 and, equally, having a poorly educated population is correlated with poverty and with a young population.

- 5.44 Maybe, or even women not having much education or many rights, that seems to be linked to these factors as well. It's just general low human development. It's not really to do with demography. That would be a counterargument. And of course, the way that political scientists would try and answer this question is, they would try and run a statistical analysis where you control for different factors.
- 5.45 So you might say, OK, within poor countries, we're just going to look at poor countries, and we're going to look at poor countries that have a slightly more mature age structure and a younger age structure. Within poor countries only, to see if, within the poor countries, those that have a mature age structure are more likely to be democracies than those that have a younger age structure. And we've controlled for income because we're only looking at poor countries. That is a way in which—and then we can multiply that.
- 5.46 We can look at within education levels. Within countries where there's very low education, the younger versus the somewhat older countries, is there any difference in their propensity to democratize? If we look at some charts here, for example, we can see that, if you take former East Bloc countries, the proportion of young adults where it's below 40% on the two clusters of bars towards the left, we see a lot of dark colors indicating full democracies or partial democracies; whereas in the countries over 39% young adults, it's all light colors indicating non-democracies.
- 5.47 But the same pattern to some degree holds for income. Over \$11,000 is full democracy, or even over \$3,500 looks to be mostly full democracies, and the poorest countries under \$1,000 at right are mostly non-democracies. So clearly, GDP per capita, the wealth or poverty of a country, is correlated with its likelihood of being a democracy. Secondary school participation, the education level in a country doesn't look to be as correlated, actually, in these charts in the East Bloc. So education level doesn't seem to be correlating as well with likelihood of democratization.
- 5.48 We have these three factors. Again, this is not a statistical analysis; that's the next stage. So what we're going to now do is, we're going to put all of these factors into a model where we can control for everything at once, to see, well, if we can isolate each individual factor and ask the question, does the median age still matter? Do demographic factors still matter in predicting the transition to democracy?
- 5.49 This chart, this is a typical statistical analysis, controls for GDP per capita, controls for the percentage urbanized, the percentage in agriculture, in other words, how urban or how rural is the society, the percent with a secondary education level. So all of those things are being controlled for here; and yet what we see is, the median age is a very strong and significant predictor. In fact, it is the strongest predictor, stronger than GDP, if we look at the 2005 data. So in 2005, in fact, median age is the strongest predictor of democratization.

- 5.50** That would suggest that even controlling for how poor a country is, even controlling for schooling levels, for urbanization, age structure really matters in predicting democratization. Here's another. This is from Hannes Weber's work, a slightly larger model. We now can control for Islam versus Christianity as the dominant religion, colonial experience, population, total population levels, fewer than 5 million, greater than 30 million, unemployment rates, and also ethno-linguistic fractionalization, which I think I mentioned, but that's the degree of ethnic diversity in a particular country.
- 5.51** If we take two random individuals from a country, what's the chance that they will be members of different ethnic groups? The higher that is, the more fractionalized a country is; the more likely, some would argue, they are to have civil war. Here that relationship is close to significance. It's not quite significant.
- 5.52** But the point is here, if we look across these models, proportion of males aged 15 to 29 has various levels of significance. So two star, which means high significance levels in all of these models, in terms of predicting the polity score, which is liberal democratization score. So even controlling for income education and religiosity, we can see in these models that Muslim-majority countries are also significantly less likely to have high liberal democracy scores, compared to Christian countries.
- 5.53** But even controlling for that, even controlling for religious heritage and for population and income and these other factors, which might explain whether a country is a democracy or not, the proportion of males aged 15 to 29 comes out as significant. So that's telling us, actually, this is very important in predicting whether a society is a democracy or not, independently of economic factors, of urbanization, and other human development factors.
- 5.54** The other thing we find is that democracy, even where it exists in young countries, is going to be more unstable. There's a greater chance of backsliding towards non-democracy in a country that's very young. So this particular graph suggests that full or partial democracies with young men exceeding 20% of the adult population have a probability of 23% of becoming a dictatorship within the next five years, whereas that is only 4.6% for democracies with fewer young men.
- 5.55** So 20% versus 5%, much higher share, a much higher likelihood of a young country backsliding out of democracy than a more mature age structure democracy, controlling for a whole series of factors, including religious heritage, GDP per capita.
- 5.56** OK, so we've looked at the evidence then for the demographic transition thesis of democratization. A lot of the analysis is focused on the share of young men, particularly the share of the 15-to-30s, as a proportion of the adult population, that youth bulge measure. We've said that, on the one hand, that share means that there's more likely to be rebellion in the society, which, if it's not a democracy, could mean that you'll have a democratic revolution. But if it is a democracy, could mean actually that you're going to lapse into an illiberal form of government.

- 5.57** But there's another question, which is, should we focus only on young men, or is there also a role for focusing on young women? What does it mean when you have smaller families? Sommer argues that smaller families also affects female participation in society, and that that can also have important effects in improving education in society. Tim Dyson, too, makes this argument, that when people have large families, they're mainly focusing their energies on providing for those families.
- 5.58** They're less likely to be engaged in thinking about politics, in criticizing government, in perhaps devising new solutions for society's problems; and so a smaller, a more mature population also can be more politically engaged, more critical public sphere. That can also be a factor in the push towards democratization.
- 5.59** With women having more time away from child-rearing, they can focus more on educating children, perhaps also lobbying for women's rights, lobbying for shifts in land ownership structure, perhaps a demand for greater property rights in societies where you have large landlords. In all of these many ways, a more mature age structure can be conducive to democratization.
- 5.60** But of course, there is a question which is that, what if you've got a mature age structure, but you have an autocracy? So Putin's Russia, Xi's China. Because these are more mature age structures, more mature societies, maybe we don't have the supply of hot-headed, risk-taking young men that might be needed to have a revolution, and so it's less likely we're going to get revolution. Maybe the same is true in Iran. Maybe if we were in 1980, we would already have had a revolution in Iran by now.
- 5.61** So perhaps, actually, a more mature age structure, especially when you have an autocratic government, might be a negative for a transition to democracy. So that is one thing we do need to bear in mind. While on the one hand, more mature age structures on balance might lead to more consolidated democracy, might lead to political elites being more willing to open up the power structure and transition to democracy, it's also the case in other autocracies that that doesn't provide, perhaps, the supply of people willing to take risks to overthrow the existing order.
- 5.62** In this lecture, we've looked at the relationship between a demographic factor, which is the age structure. And note that that age structure is one step removed from fertility and mortality. We've had a demographic transition. The infant mortality rate has dropped, and so the death rate has gone down. The birth rate has remained high or is perhaps coming down, and that leads to the young age structure.
- 5.63** So the age structure is already the second step on the causal pathway; and then from that age structure, we then look to the impact on democratization. So it's a second-order effect. So fertility, mortality affecting age structure, affecting democratization. So that's the first point. We then ask this question, what is the relationship between a young age structure and democratization? A young age structure can be a spur to

democratization if it involves— simply because young people are more likely to join insurgent movements to overthrow an existing autocratic order.

- 5.64 But equally, we've seen that a large number of young people, because of the instability that's related to, because of anxiety over the threat that that young population may pose to existing hierarchies, procedures, and orders, that could lead the established business and bureaucratic groups to be reluctant to transition to democracy, and that can therefore make democracy less likely to occur with a younger population.
- 5.65 We've seen that there are also other factors involved in a maturing population, including less attention to child-rearing, possibly leading to more people being engaged and thinking about politics, both men and women, and therefore, agitating for reform in a more moderate way. And that might just be a factor in accounting for this relationship, which we've seen in the data fairly clearly between age structure and political freedom.
- 5.66 That relationship appears not to be spurious. It appears to hold, even when we control for GDP per capita, even when we control for urbanization, even when we control for the dominant religion in a society; it still seems to be the case that a younger age structure is less conducive to democracy than an older age structure.
- 5.67 I'm going to leave it there. We've seen that age structure is related to democracy in different ways. One of the questions we might ask is whether, given the continued demographic transition to lower birth rates and older populations, whether we can expect democracy to spread more widely around the world as a result of maturation of age structures. Thank you.
- 5.68 So does a more mature age structure globally also mean more peace?
- 5.69 Yeah. And that is, in fact, the Haas' geriatric peace hypothesis. Yes, the conclusion would be that a more mature age structure, with fewer young men in the 18-to-30 age range, fewer risk-taking people, is one in which you're more likely to have peace. Yeah. And there's also—I mentioned this idea of elite overproduction, both at the mass level and at the elite level. Because if you assume there are only a certain relatively fixed number of positions for elites, the overproduction of elites, to the extent that's not happening.
- 5.70 If communist party members each have one child each, assuming they're grooming their kids for those positions, I'm not saying that that's the only kind of career they can have, but there isn't going to be as much of a problem as if they each had four or five kids, and only one of them could occupy a position high up in the party. So, yeah, I think for a whole series of reasons, that would suggest greater peace, both between countries and within countries.

- 5.71 You explained how poor countries with a high political instability are usually not a democracy. So my question is, is democracy the right tool to fix an unstable country? Or are there better ones which actually can grow the economy and make it stable?
- 5.72 Yeah. I mean, this is a good question, right? Because we do see in some cases, perhaps China and Singapore are examples where, if you have some kind of an enlightened dictator, they can make change more quickly than a messy democracy, like in India, where it's not going to be no—so, it is possible if you have a very enlightened leader, Deng Xiaoping or Lee Kuan Yew, or somebody that can really drive change forward. Yeah, you could probably get things done more quickly in a non-democracy.
- 5.73 However, I guess over the long term, what you would say is that, in the long term, all it takes is one bad leader. It's very hard to be sure that you'll get a good one in a non-democracy, and that can really be a disaster. In the long run, democracy would be a better solution, even if there's a lot of back and forth in terms of how you get there. That would be probably the opinion of people who would be pro-democracy, that democracy is better.
- 5.74 The other thing I should say is, there is a relationship between democratization and violent conflict, which we haven't talked about. Again, it's double-edged. One argument is, well, democracy allows, when people have grievances, they can vote for parties that represent those grievances, instead of fighting each other. So, you would vote for Sinn Féin instead of joining the IRA, for example. So, democratization can actually be an outlet for political passions, and that can make a society work better.
- 5.75 Of course, there is the other view that says, actually, in a democracy, you have to win elections. Especially in ethnically divided societies, parties become ethnic parties, so every election is like a census and becomes, actually accelerates ethnic conflict, accelerates appeals to emotions and to majoritarian sentiments, and therefore leads to more nationalism, more violence. There are both views. Both views exist. There's a debate, so I'm going to leave that there, instead of coming down clearly on one side.

## Lecture 6: Ethnic Conflicts

At a glance — Population shifts and ethnic conflict.

- 6.1 In this lecture, what I'm going to be talking about is the connection between population shifts and ethnic conflict. And so when we're thinking about political demography, I mentioned that human groups, let's say, based on religion or based on ideology or based on ethnicity, human groups are partly culturally created and constructed and reproduced. They're not exactly the same as separate biological species. But that doesn't mean the biology isn't also there.
- 6.2 Especially when boundaries between groups are very rigid, then actually the differences can approximate those of different animal species; and that the biology can actually become very important, especially when one group is growing vis-a-vis another. And so, I want to look in more detail at the way in which differential ethnic population growth functions and how it affects particularly violent conflict. We're going to initially focus more on the developing world, but we'll also bring in discussion of the developed world as well.
- 6.3 First thing to say is, if we look at global trends in armed conflict, since 1945, almost every war has involved combatants within a country engaged in civil war. Of course, Russia-Ukraine is one of the rare exceptions to that, as is Saddam Hussein's invasion of Kuwait. That would be another very rare exception of that. War between India, Pakistan, Bangladesh, that's another example. But there are very few such examples.
- 6.4 Mostly civil wars, we see a trend post-Cold War, after the Cold War, a rise in the number of civil wars, basically—to a peak around 1991, just after the fall of communism and then a decline. There's some instability around the end of communism, wars in the former Yugoslavia and Rwanda, for example, and then a decline, down to about 2007-08. And then we've seen a bit of a rise around the Arab Spring, a few civil wars in places like Ethiopia and Yemen, but at a lower level, perhaps, than at the peak in the '80s and '90s.
- 6.5 That's just some global trends. If we break that down by conflict type, ethnic war is the dominant or is the most common form of civil war. There is still what's called revolutionary war, which might involve a left-right ideological dimension, but primarily ethnic and religious conflict. It's something very important. If we want to understand violent conflict, war and refugees in the world, we need to understand relationships between this ethnic conflict and possibly with demography.
- 6.6 How is demography implicated in ethnic conflict more generally? I mentioned that political demography focuses a lot, not only—it does focus on aggregate population growth and pressure on resources and climate change. We talked about that, but also does focus a lot on the unevenness of the demographic transition. You get this

demographic transition, this explosion in global population, which is now plateaued and will eventually go through a very sharp decline.

- 6.7** But that rise and that decline is happening unevenly, and it's the unevenness that makes it more political, because it's happening at different rates in different places. That affects the balance of political power, cultural power, between groups in a society or between regions in the world. So political demography is very focused on the unevenness of the demographic transition leading to—so you get the demographic transition leading to different growth rates of populations, altering the ethnic religious composition of a society.
- 6.8** So changing ethnic composition of society, what does that mean for political violence in particular? But we also have the balance changing between nations. We saw that France was really worried about the growth of England between 1700 and 1900. England's population very much caught up to France. Then they worried about the German population increase and so on. Nations can worry about each other's population growth.
- 6.9** Civilizations, possibly the West, worrying about the rise of Islam. That could be another civilizational-level concern. But one of the questions always is, well, to what extent is the Muslim world a distinct population, especially when there is conversion that's possible and cultural change that's possible? It might be the case that, in fact, the cultural overwhelms the demographic; but when the cultural is quite fixed, then the demographic becomes more important.
- 6.10** And that's really one of the rules here, is that if the main action is cultural and you're getting large-scale cultural changes, if a lot of Muslims become Christians or Christians become Muslims, that kind of cultural shifting is going to be much more important than the total fertility rate of Muslims versus Christians. But if, in fact, there's very little conversion, and in fact, most people do get religion the old-fashioned way, which is through birth, then demographic dynamics are going to be actually very important in affecting the relative balance between groups.
- 6.11** The first thing to say is that, if we look within a country, the boundary between ethnic groups is almost never completely 100% separate, the way it would be between different species of birds or animals in general. So it's not that complete species separation, but there's a continuum, from more porous boundaries to more inflexible and fixed boundaries. In the more porous situation, people can— identities are fluid and people assimilate easily. In the more fixed situation, identities are, in fact, more fixed and there's less assimilation possible. Identities don't reconfigure easily.
- 6.12** That kind of gets to a debate within theories of nationalism and ethnicity, which is another course that you can take at Peterson Academy that I teach. The question in theories of nationalism and ethnicity is, to what extent are national and ethnic boundaries constructed by elites and flexible and driven by political and economic

interests, or to what extent are they more fixed and inflexible, rooted in people's attachments to symbols and cultural markers and histories?

- 6.13** Therefore, it's harder to reinvent identity. It's harder to have flexible ethnic boundaries. Ethnic boundaries—this question of how rigid or porous ethnic boundaries are is important, and this seems to vary in different parts of the world. Now, different theorists will, if you are a constructivist, you tend to argue that ethnic boundaries are porous and flexible and change over time. If you are more of a primordialist or an ethno-symbolist, you're skeptical that ethnic boundaries can change much over time.
- 6.14** But equally, it seems to be the case that there are different parts of the world where boundaries are more fixed or more flexible. The example of India, the boundary between Hindu and Muslim is quite fixed; that is, intermarriage is extremely rare. There can be a taboo, in fact, on intermarriage. Northern Ireland is another example. Protestant-Catholic intermarriage is extremely rare and very frowned upon.
- 6.15** Then you have a situation that is possibly more fluid, like the Caribbean. Trinidad, marriage between the Indian origin and the African origin population is very common, and there's a large mixed population. So there, in fact, it seems that ethnic boundaries are looser. North America, more towards the loose end; again, it depends, but there's quite a bit of intermarriage. That's increased a lot since the 1960s.
- 6.16** Actually, even in opinion surveys in the US, acceptance of interracial marriage has gone from, I think it was under 10% as recently as the 1980s. It's now 90% acceptance. There's been a big shift in attitudes, and there's been a real shift, but also, I don't want to overstate the amount of interracial marriage, but it's certainly increased. It's to some degree related to the size of different groups.
- 6.17** You can see that, in different parts of the world—so in the British Isles, ethnic boundaries might be looser in England, tighter in Northern Ireland. Certainly, the Catholic-Protestant boundary is much looser, hardly exists in England, but in Northern Ireland, it's a very important boundary. Different parts of the world, different societies have tighter or looser ethnic boundaries.
- 6.18** Some would argue that, where there's a religious boundary that is significant in a society, like India, that makes intermarriage much more difficult than if the boundary is related to language, perhaps, or even skin color; perhaps it's not quite as rigid, you could argue. So ethnic boundary, tightness or porousness varies by the type of group, and it seems to vary by location.
- 6.19** Andreas Wimmer, a theorist of nationalism, writes that this is maybe rooted to traditions of descent lines, in a society with patrilineal descent lines associated with more rigid ethnic boundaries, than societies with more matrilineal descent rules. Equally, histories of violent conflict are often associated with more rigid ethnic boundaries. If there's been a history of conflict, as in Northern Ireland between

Protestant and Catholic, then that will lead to more rigid intermarriage rules, for example.

- 6.20 Let's look historically then at this question of differential ethnic population growth and what that has meant for politics. We can look at any number of countries. The case of Canada, for example, French-Canadians had one of the highest birth rates in the Western world, if we take the 19th century, even though France itself had actually perhaps the lowest birth rate at the time. But French-Canadians remained very Catholic. There was plenty of land for expansion. You had eight children per woman, a lot of whom survived.
- 6.21 Quite rapid population expansion. The French-Canadian settlers of the early 1600s, that was about 10,000 people. And by the time we get to the 20th century, that's 8 million people. So it was a considerable expansion of population. Interestingly, in Canada, you had, in the 19th century, starting in the 1810s and '20s, significant immigration from particularly Scotland and northern parts of Ireland, and to some degree, England. So you had large-scale British immigration, which was seen as a threat by the French-Canadian clergy.
- 6.22 You had two demographic forces. You had British immigration and you had high French-Canadian fertility. They were both leading to different kinds of expansions. There was this battle, arguably, that was somewhat demographic, as to whether—because the question was whether the French-Canadians were going to be assimilated into an English society, which had been mooted by the British government, particularly in a document called the Durham Report.
- 6.23 There was an intention there that, well, eventually, the French-Canadians will assimilate into British identity. That was the belief among some in Canada, but the high French birth rate arguably made that very difficult. In fact, even led the French to expand into some of the border counties of Ontario and into northern Ontario. Without that demographic surge, it's arguable the French-Canadians would have had to retreat to a smaller territorial base.
- 6.24 The French-Canadian Catholic hierarchy used the term "the revenge of the cradle," to refer to this demographic strategy amongst the French as against the British, even though the British immigration did lead the British share to rise. Still, with a high fertility rate, the French were able to fight against that because they didn't have immigration to bolster their numbers. That led to a certain amount of conflict then between these groups, but not necessarily violent conflict. There were some rebellions in the 1830s, but not violent. So this is one situation in which demography, ethnic population, growth rates have played an important part.
- 6.25 Another case, which I'm going to talk about in more detail, is that of Northern Ireland. The first thing to say is that Northern Ireland was created in 1921 when the Republic of Ireland became independent of Britain. So there was a partition of the northern six

counties from the southern 26 counties of the island of Ireland. So the northern six counties became Northern Ireland, which remained with Britain. And there was a whole question as to which counties would be remaining with Britain and which would go into the Republic of Ireland.

- 6.26 And this was based on the calculation of, what is the Protestant share, demographically, of a particular county; and the counties that had a Protestant majority were included in Northern Ireland. There is originally Ulster, which is the northern province, roughly, of Ireland, has nine counties, not six. Three Ulster counties were excluded from Northern Ireland. One of the main reasons is because their Protestant share was too low. It was still significant, those border counties, Cavan and Monaghan.
- 6.27 The northern counties actually had significant Protestant—Donegal is the other one—considerable Protestant population, but not enough to give a secure majority. And so there was a debate as to whether to go for a nine-county or six-county Northern Ireland. Certainly among the Unionists, they decided on a six-county Northern Ireland as something they could control. That gave the Protestants a two-third majority in Northern Ireland. So this was, arguably, a demographic strategy.
- 6.28 The issue, of course, for Northern Ireland is that the Catholic population entered the demographic transition 60 to 80 years after the Protestants. So the Protestants already had a lower birth rate, and there was a lot of Catholic population growth involved. And so, hence the fear of the unionists, as expressed by unionist Prime Minister Terence O'Neill in the Stormont regime.
- 6.29 So the Protestants controlled the government of Northern Ireland. They had a separate assembly, and to some degree had rigged the electoral system, which basically meant that every— and given that people tended to vote for their own ethnic parties, if you were Protestant, you voted for the unionist party, and if you were Catholic, you voted for the Catholic parties, SDLP and later Sinn Féin.
- 6.30 So what that meant was that, because Protestants were a two-thirds majority, they won every election in Northern Ireland. Clearly, you can see how that could become a problem when one party is winning every election because that's expressing the ethnic majority share. Especially if everyone votes for a party based on ethnicity, any shifting in the composition of the population is going to translate very directly into a change in political power.
- 6.31 What happened is, Northern Ireland went from about two-thirds Protestant in 1965 to today, where it's about 50-50. As a result of that, we can see, for example, that you've now had the first Catholic mayors in Belfast, which has traditionally been a Protestant city, the first Catholic-dominated Northern Ireland assembly. That is a shift, and this kind of shift actually did come to pass. It was very much feared by the Protestants, because their view was that, if the Catholics gained control, they would take the

northern counties into a united Ireland, and the Protestants would then be a minority in this sea of Catholics in a united Ireland.

- 6.32** What happens also is that, with shifting ethnic demography with a faster-growing Catholic population, that takes place at different rates in different parts of Northern Ireland. One place where there was a particularly rapid increase in the Catholic share was in the city of Londonderry, which, as recently as the 1920s and '30s, was about 40% Protestant. Between 1961 and 1971, partly as a result of the rise of the civil war there in the late 1960s, a lot of Protestants moved out of the city center. Londonderry goes from 33% Protestant in 1961 to 24% in 1971.
- 6.33** It's worth saying that the unionist government had gerrymandered the constituencies in the city of Londonderry to ensure that, even though Protestants were a minority, they still were able to win a majority of the seats there and still control the city. But with the civil war and afterwards and the reforms, that no longer was possible. Sometimes groups will seek to gerrymander districts to maintain political power if they lose a demographic majority.
- 6.34** One place that occurs is Jordan, where Palestinians are a majority in the Transjordanian group, which is about 40% or so, has gerrymandered the districts to allow itself to remain in power. These are different strategies ethnic groups can use. The bottom line here, however, is, you have certain territory, certain parts of Northern Ireland, really becoming a lot less Protestant. I mentioned in the first lecture, the relationship between demographic or population shifts and politics is often indirect.
- 6.35** One way to test this relationship in Northern Ireland is to say, well, in the counties where Protestants were declining most rapidly, is that where you saw the greatest amount of violence? Possibly the unionists, Protestant paramilitaries engaging in more violence, but actually, as we see in this graph, if you look at the dashed blue line, which is the Protestant growth rate, which is negative in most counties and especially negative in the city of Londonderry, that doesn't seem to be related to the green line, which is the number of Roman Catholics killed.
- 6.36** So where Protestants are falling fastest, we're not seeing Catholics being killed at higher rates. In fact, the relationship sort of shows that the cities, Belfast and Londonderry, are where a lot of the violence was taking place, partly because it's easier to conduct insurgencies in urban areas, rather than in rural areas. So if you looked at that graph, you'd say, well, there's no first-order relationship between demographic change and violence. So we would reject the hypothesis.
- 6.37** However, as I mentioned, often demography works very upstream in the causal chain. So it's maybe two or three steps up the causal chain. Here's a picture of what's known as an interface area, because I want to talk about how demography translates into conflict over territory, even conflict street by street, within urban neighborhoods.

- 6.38** This was an Orange Order parade, which is the Protestant fraternity, Northern Ireland, parading in an area that is an interface area. The public housing behind would be inhabited by Catholic residents. You can see the police vans there. I was actually there at the time and took this photo. You could see the rocks being thrown from the other side. But what you see in Northern Ireland is, in the urban areas, you have segregation by ethno-religious group. Those areas where there's a division, there's often a flash point and tensions.
- 6.39** How then does demographic change play into these territorial dimensions of the conflict in Northern Ireland? Well, one way you could see this is that, in the more Catholic parts of Northern Ireland, to the west, Fermanagh, Omagh, Armagh, you can see there's a higher Orange Order membership amongst Protestants. So where they are more in a tenuous demographic position, they're more mobilized into the Protestant fraternity than in the suburban areas around Belfast, where there is lower membership.
- 6.40** As a share of Protestant population, Orange Order membership is lower. This is one indication of where, this is how demography might be playing a role, is that where you feel more like a minority, you band together more. Now, that doesn't mean that you're going to engage in violence. The Orange Order wasn't involved in violence very much; however, it does suggest a certain degree of mobilization, which can be the prelude to violence.
- 6.41** But perhaps the relationship is a little bit more complicated than that. It seems to be the case that, what occurs is that the Protestants are concerned about losing power, particularly in places like the city of Derry, where they are a minority, and so they've rigged the electoral districts. They don't want to allow for a one-person, one-vote reform. And that unwillingness to reform, because of fear of demographic being outbred, if you like, meant that the Catholic grievances were stoked, and that was fuel for the IRA's campaign.
- 6.42** So the IRA were able to leverage this discontent around the lack of political reform and economic reform. So public housing—there was a degree to which the unionist government tried to deny public housing and other benefits to the Catholic population, in a way to encourage them to emigrate, and they did emigrate. So the Catholic population had a higher emigration rate.
- 6.43** So the strategy, to some degree, of the unionist government was to make conditions less favorable for the Catholics, so that they would emigrate. And therefore, that helped to offset the higher Catholic birth rate. But then that reached ahead when you had the civil rights marches in the late '60s. And so you can see that population is playing a role in the background, even if in the foreground, this looks to be a battle over political rights. The political rights are related to Protestants' anxieties over losing power due to population shifts.

- 6.44 This can take territorial form if we look at a smaller, more fine-grained scale, case of the city of Belfast, which is a segregated city for the most part, not entirely, but for the most part, you have Catholic areas concentrated in west Belfast and in areas known as the—well, yeah, in west Belfast, generally, and in north Belfast. But there are also Protestant enclaves within the west of Belfast.
- 6.45 East Belfast, mainly Protestant, but within west and north Belfast, you have mainly Catholic populations, but you have a neighborhood called the Shankill in west Belfast, which is a Protestant neighborhood within a mainly Catholic area. Where the Catholic street and the Protestant street meet, you've got a peace wall and you have a certain amount of rioting going on in these interface zones.
- 6.46 North Belfast is trickier because the Protestants are—there's a long zone between Protestant and Catholic, and it's harder actually to construct peace walls. And so actually, there's more of, more violent incidents have taken place in north Belfast than in west Belfast.
- 6.47 One question is, what happens when you have a growing Catholic population and a shrinking or stagnant Protestant population? These neighborhoods are going to be expanding, but of course, you have these rigid demarcation zones. So the expansion is going to occur in some areas, but not in others. So one area where the Catholic growth was taking place and resulting in territorial change was in north Belfast. And there is an enclave of Protestants in this mainly Catholic area called Glenbryn.
- 6.48 Here is one of the members of a paramilitary political party, talking about Glenbryn and saying, from the Protestant perspective, saying "it's a small community surrounded by a much larger republican, i.e., Catholic, one. People are hemmed in and they see Ardoyne, the Catholic area, expanding constantly. Continual talk of rising nationalist population makes people feel that even more. It feels if they give any more ground, they will be wiped out as a community."
- 6.49 And this is why they engaged, for example, in a quite-aggressive protest where you saw Catholic schoolchildren walking to school and being yelled at by unionist protesters and being threatened. So that was really all over the media, the Holy Cross protest. Also, Hutchinson's adding that this has resulted, to some degree, in Protestant attacks that, before the ceasefire, the anger of Protestants expressed itself in shootings and bombings, support for paramilitaries like the Ulster Volunteer Force or the Ulster Defense Association.
- 6.50 So maybe at this local small scale, these demographic shifts are playing a role in demographic anxieties, which are playing a role in paramilitary violence, shootings and bombings going on. So what we might say is, maybe even if at the national level, you don't see the counties where Protestants are declining and Catholics are increasing, you don't find more violence at the county level. But when you go down to the level of the street or the block, we do see this relationship between demography and violence.

- 6.51** And a somewhat similar argument, by the way, was made for population pressure on resources, that if you went down to the fine scale of rural communities, you can see local-level violence where there is more population pressure on resources. And so maybe demography is having more of an impact on that very micro, local-level conflict and not necessarily on national level. We'll talk about that relationship in a minute.
- 6.52** But it's not strictly micro-level territorial violence that demography is informing. The big picture of which group is growing and which group is shrinking at the level of Northern Ireland as a whole was a factor in the strategies of both national or, if you like, province-wide political actors. So the IRA invested its hope—and one of the arguments, actually, that Gerry Adams and Sinn Féin made for going into the peace process, which led to the Good Friday Agreement, was that, look, we're growing demographically.
- 6.53** We will eventually win a border poll, a one person, one vote poll, and be able to take Northern Ireland into a united Ireland on the basis of numbers alone. So there's this demographic strategy. You can see that that actually can lead to a shift towards peace; that if the strategy of guns and bombs isn't actually producing dividends, in fact, it's just leading to greater resistance from the Protestants and from the British government, you shift to a more democratic strategy.
- 6.54** If you think the demography is on your side—it's one person, one vote—it makes sense to move to a democratic strategy. So one argument could be that, in fact, favorable Catholic demography might have helped bring peace in Northern Ireland. And there had been conflict for a long time, really since the 1600s in Northern Ireland between Protestant and Catholic. So it's quite an achievement to have made peace in 1998. The first ceasefire was 1994.
- 6.55** It's not to say that, after 1998, there was no more conflict. Conflict often revolved around parading, particularly Orange, that is Protestant, parades. Here again, demography played a role because the Protestant parade routes in some of these towns like Portadown, which was a major unionist parade area, those Protestant parades, because the area had become more Catholic over time, suddenly the parades were more contentious, and you got roadblocks of Catholics who were protesting against these parades.
- 6.56** The Protestants then were rioting with the Catholics, and you got a conflict there as a result of local-level demographic change, which again suggests that the conflicts generated by demographic change can be very local. Now, in the case of Portadown, this then became part of the national news cycle, so that it did have a significance for politics in the province as a whole. That's one of the things that we find, is that the local-level demographic change, in order to matter at a larger scale, has to become used by national-level political figures. I'll talk about that a bit more.
- 6.57** Just to look at a number of other parts of the world where ethnic change seems to be connected to violence or violent civil war, in the former Yugoslavia, in both Kosovo and

Bosnia, the Serbian Orthodox population entered the demographic transition before the Albanian Muslim or Bosnian Muslim populations in Kosovo and Bosnia. Kosovo had at one time been a third Serbian, and that declined substantially throughout migration and lower birth rates. Something similar occurs in Bosnia.

- 6.58** And one study found that areas of Bosnia where the Muslim and Serb fertility gap was greatest were more conflict-prone, and you saw more violence in those particular districts that had a bigger—which is an indicator of more ethnic change being associated with more violence in those districts. Again, that's quite a local-level effect. We also see, for example, Macedonia being very unwilling to take Albanian refugees, because they worried that that would shift the ethnic composition of Macedonia in favor of the Albanian minority and against the Orthodox majority.
- 6.59** Ethnic shifts, ethno-religious shifts in Lebanon have also been identified as being important in sparking the civil war from 1975 until the early 1990s in Lebanon. What occurs is, after the Six-Day War in 1967 and in 1973, you see an outflow of Palestinian refugees into refugee camps in Lebanon, which then become sites of unrest, which shift the composition of Lebanon.
- 6.60** Now, Lebanon, like Northern Ireland, was created to ensure a demographic majority of a dominant group. In the case of Lebanon, it was carved out of Syria under the French and designed to have a Christian majority of around two-thirds. However, low fertility and emigration led to the decline of the Christian population from about two-thirds to perhaps a quarter or a third, maybe a quarter today. But so you had a population that went from majority to minority status. And so, anything that increased the Muslim share was going to be destabilizing in the case of Lebanon.
- 6.61** Now, there's also an additional complication, which is the balance between the Shia and the Sunni. The Palestinians being Sunni are also changing the balance between Sunni and Shia in Lebanon. And then the Palestinians are also a separate group of their own in many ways. The question of whether demographic shifts can destabilize a country. Israel, Palestine, a case where demography has been very much front of mind for both the Jews and the Arabs.
- 6.62** Yasser Arafat of the PLO was already talking about the womb of the Arab woman as being a weapon of warfare. Israel was really worried about the high Arab birth rate vis-a-vis the Jewish birth rate. It's interesting, of course, in the context now of, the Arab birth rates have come down, the Jewish ones have remained reasonably high because of the strongly religious Jewish population, the ultra-Orthodox Jews, and to some degree, the modern Orthodox religious Zionists have relatively high birth rates.
- 6.63** It's quite interesting because clearly, if you were to take the Jewish population of Israel, there is an anxiety of the secular Jews about the rise of the highly religious Jews, but that's masked to some degree by the fact that you have this Arab-Jewish demographic

competition. And so, the highly religious Jews are a threat to secular Jews who are the established elite of Israel.

- 6.64** So they're a threat on one hand, but on the other hand, they're an asset vis-a-vis the Arab-Jewish competition. So it's a complicated picture, with regard to demography in Israel-Palestine. It's just to say that the demographic concern has very much been prominent in that conflict, possibly one reason why you see Arabs in Palestine, in the West Bank and Gaza, and Jews in Israel having higher birth rates than their co-religionists outside of that country.
- 6.65** I want to now look at this question of ethnic conflict. Now, this is primarily in the developing world, but we'll also bring in the developed world. A study from leading political scientists, James Fearon and David Laitin, looked at a data set called the minorities at risk data set, which codes different countries of the world and codes ethnic groups in the world, and looks at the likelihood of civil war in a particular country. It tries to identify which groups get engaged in ethnic conflict.
- 6.66** What they find is, they have two questions. Has a group been in the country since 1800? In other words, is an ethnic group indigenous or is it an immigrant group? If it is indigenous, it has a much higher probability of being involved in ethnic conflict. There's only, I think, one or two immigrant diaspora groups that have been involved in any kind of ethnic conflict, one of which is the Muhajirs of Pakistan who migrated from India. But there are very few such examples.
- 6.67** Secondly, does the group have a territorial base? If they do have a territorial base, they are far more likely to be involved in ethnic rebellion or attempts to separate from the existing country. So essentially the groups that are involved in violent ethnic separatism tend to be groups that see themselves as indigenous to a territory and are territorially compact in a country.
- 6.68** And one question we might ask is, OK, so if that's the case, how does a shift in ethnic composition affect these kinds of groups? And it turns out that, well, where groups consider themselves indigenous to a territory, and they see what they consider to be interlopers or those from other ethnic groups forming a larger composition of their region of a country or perhaps their country as a whole.
- 6.69** So if we take Kenya and Uganda, it would be western regions of the country where you have groups, for example, in the Rift Valley of Kenya who would be resentful of migrants from the coast, similarly in Uganda. Ethnic migration within a country across different territories can lead to conflict. What's known as autochthony, the groups that see themselves as indigenous or autochthonous, resent the outsiders coming into their territory. We can actually scale this up even to the level of a country.
- 6.70** In the case of the Ivory Coast, which had a civil war in 2002, there had been a shift in the composition of the population. The Muslim share had expanded from a quarter to over half the population, as a result of immigration from Burkina Faso and from the

north. This was a factor in the discourse of what was called "ivoirité", that the true Ivorians were being overwhelmed by foreigners. So that was a factor in the civil war there.

- 6.71 There's a concept then of dominant ethnicity, that groups that consider themselves to be the dominant founding ethnic group of a country, that can express a kind of nationalism when they see their share of the population declining. In the developing world, this has been termed "sons of the soil" conflict. There's another dimension, which is often the autochthonous groups tend to be less developed economically, lower education, etc., and the migrants from the possibly more advanced economically or developed area, sometimes with money, often with government support.
- 6.72 So Han Chinese moving to Xinjiang, moving to Tibet, would be an example of that, or Javanese moving to West Papua, seen as interlopers, sometimes seen as being sent by the government, seen as carpetbaggers in a way that need to be resisted. That can become an ingredient. But of course, the question is, when does this become a national-level political issue, or is it just a local-level issue involving local-level land conflicts or conflicts over resources?
- 6.73 Just to say that there is this history then in the world of ethnic change, in some cases being associated with ethnic violence, often within countries, because many countries, particularly in sub-Saharan Africa, are highly ethnically diverse. There are groups which see particular parts of the country as, quote, unquote, "theirs" or their territory, and therefore, even internal migration within a country can be seen as ethnically threatening by particular groups, which see themselves as indigenous to particular regions.
- 6.74 Now, of course, there's a debate in all this between the more constructivist or instrumentalist school of ethnic conflict studies, which would say that, well, boundaries are in some ways constructed by elites. And so you have ethnic entrepreneurs who make the claim that certain groups are indigenous, when either they're not or there's been a flow of different groups over time in the region, and that this debate between the indigenous and the migrant groups is constructed and artificial in some way.
- 6.75 Then the more ethno-symbolist, primordialist school, which would say, well, actually, no, there are myths and symbols and memories of attachment to particular places that are encoded and passed down over time. You do actually have a real phenomenon called indigeneity or autochthony. And so, this is actually a sort of ethno-symbolic response to the loss of congruence between territory and ethnicity. So, as those two come apart, there's a nationalist response to bring them together to produce identity and unity within a territory; and it's that that's leading to these kinds of ethnic conflicts that are connected to ethnic change.
- 6.76 Sometimes democratization also plays a role, particularly if we take the case of sub-Saharan Africa. The conflict in Ivory Coast came about after the country moved from an

autocracy to a democracy. Of course, democracy means one person, one vote. Suddenly demography becomes more important. Sometimes there can be factors that make demography more important. So democratization arguably makes demography more important and may lead to ethnic conflicts.

- 6.77** In this case, an election. If the country has been becoming steadily more Muslim over time, it may not have mattered when you had an autocracy that was in power, it didn't matter for power. Once you have one person, one vote, it really matters for power. Is the Muslim coalition going to get in, or is the Christian coalition going to get in? So democratization in interaction with demography can be important.
- 6.78** Now, you could look at another situation where, if we have high technology making people redundant for warfare, like robots becoming important for warfare, that might make population less important, let's say, for power. Different factors can make demography more or less important for politics.
- 6.79** There are other instances where you may have had—Assam, northeast India, has had significant in-migration of Bengali Muslims into an area that considered itself a Hindu area. This has led to sporadic violence against Bengali immigrants. In some cases, these are well established people who have been there more than one generation. One of the questions is, when do these local-level or provincial-level conflicts become national in scale, leading to national ethnic conflicts?
- 6.80** There is, similarly—and so part of this is to do with migration. It can sometimes be due to differential birth rates within a country from established minority groups. In central Europe, for example, Hungary, Slovakia, Romania, Czech Republic, the Roma population has a higher fertility rate than the majority population. And so there's concern over the expansion of the Roma population. And so the populist right, far right, has certainly used that as well. So minority growth, either through migration or through higher birth rates, can be a factor in conflict.
- 6.81** If we move from the developing world to the developed world, we can see a similar dynamic in the sense that demographic shifting lead to differences in ethnic population growth rate. You have ethnic changes, change in the ethnic composition of the population, playing a role in politics. How does it play a role? This lecture, I'm not going to go in depth to look at the question of the politics of immigration. What I would say, however, is that we can analogize from the developing world to the developed world and see some commonalities.
- 6.82** One is that, where you get demographic shifting, this can express itself in local-level tension and conflict. But unless that local-level conflict or tension becomes a political issue, then the issue is not going to be important in national politics. For example, if we take the case of Switzerland, you did have local-level antagonism to the influx of Italian workers, for example, coming into Switzerland in the early 20th century. But that only mattered for certain regions of Switzerland, not for national politics as a whole.

- 6.83** Likewise, in the United States, the shifting of the ethnic composition of California in the 1980s and '90s was a state-level issue in California. There was a vote in 1994 on a measure called Proposition 187, denying public services to illegal immigrants. That was an issue in California, but it was not an issue of national politics in the United States, the way it did become after 2015 when Donald Trump entered the political scene.
- 6.84** What is the difference there? The difference is that, if you, back in the 1990s, if the Republican leadership, if somebody like Bob Dole, who was running for the Republicans, went out to California and took pictures of himself in the border and hung out with the leaders of the Prop 187, people like Governor Pete Wilson in California, and tried to form a connection between national political issues and what's happening in California—if the Republicans had made that an issue and injected it into national politics, it might have become a national issue.
- 6.85** We see that to some extent in India, where the BJP, the Hindu nationalist governing party, has to some extent made the issue of ethnic change in some of those regions of northeast India, by politicizing those regional tensions, they can make them national issues. But if, in fact, national actors do not politicize, do not care about what's happening in the region, so in the US case, the Republican party did not politicize the issue of ethnic change in California, so it never actually became a dominant issue.
- 6.86** However, it started to become politicized, particularly with Trump, to some degree with Romney, but certainly with Trump in the 2010s, and then it became a national political issue. Now, of course, it's also the case that Hispanic population migrated out of California and Florida to other parts of the US. Partly, the demography became more national, but also the political scene, the political actors, nationalized local developments and turned the issue of immigration into a much more national issue. It had been more of a state-level issue.
- 6.87** The question of, what are the frames through which national politics is conducted, that may differ from the frames in which local or municipal politics is conducted. There is input we can see here from the theory of constructivism or instrumentalism, which would say that it's what politicians decide to politicize and to focus on. That's what really matters, and they can whip up public sentiment or dampen it down. They have a lot of control.
- 6.88** The ethno-symbolist and primordialist schools would say, well, actually, the demography in and of itself leads to discontent in a society. The ethnic change leads to discontent, and it's not really something that is created or whipped up so much by political actors; it almost emerges organically. Political actors have to respond to it. It's a different view. We have these two views, one which gives more power to elites to shape identities and perceptions, and another view which gives more power to ethnic change in populations. Pre-existing attachments interacting with demography is the main engine of this politics.

- 6.89** It's just, again, to reiterate that, when it comes to political demography, especially when we're talking about the differential growth rates of groups, it's important how you see group dynamics. Do you see group boundaries as being relatively rigid and given by people's historic attachments to particular symbols, particular markers, particular myths of ancestry, and attachments to territory and place? If that is strongly path-dependent, then there's going to be little capacity to redefine ethnicity and to have assimilation between groups.
- 6.90** That's the ethnosymbolist perspective. If, however, you're more of a constructivist, you believe that group boundaries are relatively porous, group definitions are fungible, so even if there's demographic change, it doesn't really matter because cultural boundaries are shifting, and group definitions are shifting over time. And of course, we can see examples of both.
- 6.91** In the US case, for example, the Anglo-Protestants ultimately assimilated with Catholic and Jewish immigrants to create a white majority group. So there you see a redefinition of the dominant group from Protestant whites to all whites, whether Catholic, Jewish, or Protestant. That does involve a redefinition of the boundaries of groups. Equally, we can see there's differences—even if the group don't redefine themselves, there's differences between group boundaries.
- 6.92** So Irish Catholics in Northern Ireland, clearly, there's a much stronger group boundary between them and the Protestants in Northern Ireland than in England, for example. And even in Scotland, even though Scotland has a little bit of sectarianism, it's a much more porous boundary between populations there. And so, your view of how porous and how constructed ethnic boundaries are is going to be very important when it comes to assessing the impact of demographic shifts—that is, differences in the growth rates of ethnic groups within a society.
- 6.93** Even if you've got large differences in those growth rates, and composition appears to be changing, if you are a constructivist, your view as well, yes, there's a change in composition, but the groups can redefine themselves. There'll be assimilation. The cleavage lines, the lines of politics that are important, can change rapidly. Therefore, you put less weight on demography as a factor. Whereas if you are an ethno-symbolist or a primordialist, you would think demography is going to matter considerably more and could even lead to violent conflict, not just political conflict.
- 6.94** We've seen some examples. Again, though, instrumentalists and ethno-symbolists would differ in their assessment of how important the shift in the ethnic composition in places like Lebanon and Ivory Coast, and it was for the violence that broke out there. The instrumentalists would argue that it's much more about political entrepreneurs, democratization, perhaps economic factors. The ethno-symbolists would put more weight on the demographic shifts. Thank you very much, and I look forward to your questions.

- 6.95** In the criminal justice system, for example, you have offenders, and you look at population data, static factors, dynamic factors, and you produce predictor scores, so chances of recidivism in the next year or so. If demography is a strong predictor of, say, war or peace, can we use that hard data, give it to computer, give it to AI, analyze it, and say, in the next two years, the chances of civil war in this country is 84%. Is that science developing? I notice you used some data sets here. Is it something that we can expand on?
- 6.96** The answer is yes. So political science, comparative politics, does a lot of quantitative research where—and this is how it would work. We would take all major countries of the world, let's say 150 countries, and we would have a whole series of properties of those countries. What's their population? How much rugged terrain? What's the distance between the highest and lowest point? Do they have oil? What's their oil exports? How recently did they become a democracy?
- 6.97** We'd have all these factors. We could also look at the demographic factors, age structure, for example. We could look at the ethnic composition. How ethnically fractionalized are they, in terms of that ethnic fractionalization measure from zero to one? Then we can put that into, and then we have years from 1945. Now, of course, that relies on people coding country characteristics, often student coders or expert coders. Typically, you'll get two coders, and you'll see, do they agree? If they agree, then that becomes the code.
- 6.98** Then we can run a statistical analysis to see which of these factors predicts civil war. That is a very established literature. What I will say is that the data on ethnicity is not very good, and certainly the data on ethnicity over time is not very good. Because if you imagine what you would need is, you would need census data to be collected on ethnicity over time. But these data sets are year-to-year, because a conflict can break out in any given year.
- 6.99** A census, if it even occurs—Lebanon stopped doing a census, maybe for obvious reasons—but even if it occurs, is only going to happen every 10 years. In many cases, the census did not occur, in particularly the countries that have civil war. We have a rough picture of differences over time, a snapshot. At any one moment in time, we can say, yes, Tanzania is more diverse than Botswana. What we can't easily get is how quickly is ethnicity shifting in particular countries.
- 6.100** We do know across Africa that these countries are becoming a little bit less ethnically fractionalized. As they get urbanized, you get a kind of amalgamation of groups into smaller units, and so you're getting a decline of diversity. We do know—now, where we have good census data in the West, we know, although, again, how many Western countries ask about ethnicity? It's a handful.
- 6.101** In Europe, I think it's only Austria has religion, Ireland, but most of European countries don't have this data. It's very difficult. We have to estimate it. So I would say, this is an

area that needs more work. We can approximate it and get at it, but I wouldn't say we have got good enough data to be able to say, are places that have a faster ethnic shift more likely to have civil war? We don't have good enough data on that.

- 6.102** So, with the data that you have, which seems to be a lot, in your opinion, which countries seem to have got it right, in regards to social harmony? What could a place like the UK learn from places that are doing it right and implement it into the way that we live our lives?
- 6.103** This lecture, I focused a lot on parts of the world that have had violent conflict, to some degree related to demographic shifts. The first question is, if you're a country, how fast do you want to have, how fast a shift do you want, ethnically and demographically? A lot of this depends on what theory you have of the world. If you have a more instrumentalist, constructivist view of the world, you'll think, yeah, we can have rapid shifts, because it's really up to elite manipulation as to whether you'll get a conflict or not.
- 6.104** Whereas if you are more of an ethno-symbolist, you'd say, well, if there's ethno-demographic shifting, that's going to increase the likelihood of conflict. So you want to make sure that shifting is slow. That might be a consideration. Of course, you have to think of other considerations. You maybe want to think of economic considerations, but one obvious policy lever is the immigration level. That could be one lever.
- 6.105** Or within a country, can you in some way incentivize people not to migrate from one region to another, if that is related to tensions? Very often it seems like the central governments are encouraging migration, in China, of Han Chinese into Xinjiang, for example. Sometimes the government has a political assimilationist drive in mind, encouraging this assimilation. I think that's clearly a bad idea for ethnic tensions in a country; even if the government thinks it'll promote nation-building, it clearly seems to be related to ethnic conflict.
- 6.106** So these are just some of the considerations. Then there's a question of how do you manage the ethnic diversity you have? That is a whole big literature in ethnic conflict. Do you recognize groups within your population, so through multiculturalism? Do you only recognize individuals, and so you pursue a more integrationist approach or assimilationist approach? So your ethnic conflict management policy and your immigration policy are two different levers that you can use. And depending on your view of what drives ethnic conflict, that will point you in the direction of what policies to engage in.
- 6.107** As we see a rise in certain ethnic groups in the UK or England, to be specific, is it a possibility in the future in England that we see demographic strategies being applied, let's say, in an election, to gain certain advantages that will affect the policies and the laws being passed in the future?

- 6.108** A demographic strategy might be if a party would benefit from immigration—let's say, if certain kinds of immigrants vote for that party, then they would encourage that kind of immigration into the country. You mean that level of demographic strategy, or it could be appealing to particular ethnic constituencies on the basis of certain issues. In the UK, it might be the Israel-Palestine issue appealing to Muslim voters.
- 6.109** Now, you could imagine, in theory, a political party wanting to encourage immigration that would support it. Of course, there are some who alleged that the Democrats were quite open borders because they wanted immigrants who would vote for them. I don't happen to think that that intentional strategy was going on. I do think, however, that probably Republicans were more concerned about immigration, in part because Hispanic voters were, at least until very recently, were less likely to vote Republican.
- 6.110** It's quite interesting now that that's changed, and Hispanic voters have voted Republican more than Democrat last election to some degree, or at least it's close to 50-50. That could shift the strategies of the parties. I don't necessarily think this has led to differential demographic strategies in terms of who you encourage into the country.
- 6.111** It's worth saying that, in addition to immigration as a policy lever, some countries engage in encouraging fertility, but doing it in such a way that, to some degree, has an ethnic angle to it. So, in the case of Israel, money that is paid to parents to have children, or child benefit, for example, is linked to those who have somebody who served in the military. Of course, Arabs don't serve in the military. Only Jews and some minorities like the Druze do serve in the military. You can see there's an ethnic angle, perhaps, to that population strategy.
- 6.112** These population strategies are used in many countries. Another population strategy is assimilation. If you can assimilate another group, another population strategy might be sometimes to play around with census categories that might, you might say, well, it's alleged that in Sri Lanka there was a division between highland and lowland and that the census categories were changed to make this Sinhalese population into one united chunk, that that wasn't obvious at the time, but that was done deliberately and that then created more of an advantage demographically.
- 6.113** There's a whole number of strategies which different groups and governments engage in to make the demography favorable, perhaps to them, if they think that certain ethnic groups vote certain ways and that they can gain an advantage from that.
- 6.114** Having declining birth rates in mind, especially in Europe, and the lack of workforce, what is research's suggestion on immigration? What is the effective way of selecting immigrants? Is it, like, open borders, first come, first served, or is it like pick and choose?
- 6.115** There obviously are competing demands. There's the short-term economic demand. Let's say you want to post as impressive an aggregate GDP growth rate as possible, then your incentive is to bring in large-scale immigration because, to the extent they

are contributing to some, even to some degree, even if they're a lot poorer than the majority, that will increase the total gross national product, almost certainly.

- 6.116** If your incentive is short-term expansion of GDP, to post a good GDP number, then you want high immigration, regardless of who's coming in. Likewise, if you have a short-term budget problem, you bring in a lot of immigrants who may be of working age, and they're paying in now, even though when they get older, they're going to need the healthcare system. But right now, they're working, and so they're helping to pay the medical costs and the pensions of older people.
- 6.117** In the short- and perhaps even somewhat medium-term, it's a good strategy economically, even though it's increasing the burden for generations down the road. So long-term, it's probably not a good strategy. Now, it depends on a lot of things. What kind of immigrant do you want? Arguably, you want somebody of working age, arguably younger, because they're going to have more years that they're going to be working. This, of course, depends on immigrants who have skills to contribute, who can actually get jobs, not beyond welfare.
- 6.118** But to the extent they are working, and clearly somebody who has more skills in education or is likely to work, is younger, they're going to contribute for longer. But of course, certainly if you have falling birth rates, if you want to maintain the age structure of the population, as I mentioned, you have to have extremely rapid and rising immigration. So in Europe, this study was done, Europe would have to have, instead of 500,000, a million a year, it would have to have tens of millions a year.
- 6.119** One of the results of this is going to be much more rapid ethnic and religious population shifts. I would argue, you could see that this is connected to—now, of course, not everybody would argue this, but there is an argument that probably many people would say, this is contributing in some way to the rise of populism. Now, if you are a pure constructivist, you would say, well, no, it's not the demographic shift. It's about politicians manipulating perceptions and whipping up anxieties that wouldn't exist without the politicians.
- 6.120** Whereas the more ethno-symbolist point of view would be, well, this demographic shifting, almost from a—it's more of a bottom-up perspective that, because of people's existing identities and existing attachments, this is going to lead to tension. And so, depending on which theoretical stance you have, if you're more of an ethno-symbolist, you would say, actually, high immigration is going to lead to tensions almost inevitably.
- 6.121** Therefore, that will be a limit on how much you can pursue a policy of using immigration to counteract the aging problem. As a short- to medium-term strategy, you'll be constrained by populism and by political blowback. And so, as a politician, if you're purely calculating, you would have to say, "I'll have to take into account the political risk of pursuing a high immigration strategy."

## Lecture 7: Fertility Rates

**At a glance** — Fertility rates and their long-run political consequences.

- 7.1 Welcome, everybody. Thus far in this course on political demography or the politics of population change, the main focus has been the effect of population changes on politics, but it's also sometimes worth looking at the other relationship, the effect of politics or cultural attitudes on fertility. This is much more the traditional focus of demographers in institutes of demography. They will look in-depth at the causes of fertility—when women have children, how many, whether at a younger age, an older age.
- 7.2 And so, fertility is the outcome that we'll be looking to explain, in both the developing world, but especially in the developed world. What is it that governs the birth rate? What are the causes of fertility, and what are some of the policy responses that might be adopted in relation to wanting to bring a fertility rate down or increase it?
- 7.3 Of course, we've had a literature that is Malthusian, that is very concerned with overpopulation, that would be concerned with bringing down the population, lowering fertility. But equally now, more and more we're hearing the other side of things, which is, the decline in birth rates is a threat to civilization. We have this headline that Elon Musk says, "Civilization is going to crumble if people don't have more children."
- 7.4 In reality, that's been talked about for a number of years. Going back to Ben Wattenberg's book, *Fewer*, which I believe dates from the early 2000s. More recently, Darrell Bricker and John Ibbitson, a couple of countrymen of mine, have written the book *Empty Planet*, which is also, the book *Empty Cradle*, as well, by Philip Longman, is another title in this genre, looking at global depopulation now as a problem.
- 7.5 And so, trying to understand then fertility decline and perhaps how to raise birth rates, whereas the long-standing concern, going back to the '60s, when, of course, global population was in its most rapid ascent, it was about how to deal with the threat of overpopulation in the world. If we consider the 2020s and we look at total fertility rates, that is the number of children born per woman across her lifetime across the world.
- 7.6 We can see that almost all of the world territorially is now below replacement. In fact, globally, we're at about replacement now. We're going to be below replacement as we move towards the middle of this century, and then we're going to see the global population peak later in this century.
- 7.7 As mentioned already, the Global South is really where you've got above replacement fertility or very high fertility. That is, it's earlier in its demographic transition, and in particular now that birth rates seem to be going down, even in the Andean cone, in the

more tropical parts of Latin America, in a way, the higher birth rates are now more concentrated in sub-Saharan Africa and in west Asian countries, such as Afghanistan.

- 7.8** Southeast Asia has really matured demographically and is now moving towards replacement or sub-replacement fertility in many parts of Southeast Asia. So that's the global map. It's worth saying that almost all the world's population growth is coming in that tropical belt, the Global South. That will be the source. Also, ultimately, if we think about long-distance migration to the developed world, it's going to come out of those high-fertility regions. That also presents particular questions and challenges globally.
- 7.9** I'm looking at fertility. We're looking at the question of what causes birth rates to go up or down. We'll start with the classic literature. There's been the classic literature on the demographic transition from the condition of humanity through time, which has been high birth rates, high death rates. Obviously, not exclusively. We can find people in the late Roman Empire talking about the Roman elite and its low birth rates. So yes, there are particular moments where people have been talking about the elite in the Muslim and the Arab empires having not enough kids, compared to invaders coming in.
- 7.10** So maybe, but aside from a few elites, by and large, the condition of humanity has been, you have as many kids as you can. Half of them die before age 10. Women's education is really low. What happens with the demographic transition as we start in the late 18th century is, we start to see improvements in human development, urbanization, economic growth, industrialization, the emphasis on putting more investment into a smaller number of children.
- 7.11** The infant mortality rate then begins to fall. So high birth rates, high death rates is one steady state. And we're moving then to the low birth rate, low death rate steady state in that first demographic transition. And there's a population explosion as the death rate falls first. Infant mortality comes down, the birth rate still lags, and then eventually the birth rate catches up, and the population stops growing, which is where we are in the West, for example.
- 7.12** Part of what goes on in another demographic process is mass urbanization, the mass movement of people from the countryside to the cities. In Britain, almost everybody is living in urban areas. That's true across Europe. The US case, for example, the US was a largely rural society. It wasn't until 1920 that the urban population began to be a majority. When I say urban, I mean, even in towns that are in the tens of thousands was considered urban. That only really happened starting after 1920, in the US case.
- 7.13** We have large-scale urbanization that is now underway in the developing world. There's a book called *Arrival City*, which I'm trying to remember the author, another Canadian. What is it with Canadians and demography? Anyway, that looked at this phenomenon of mass urbanization in the developing world. We're getting the growth of cities in a very disorganized way, often with large-scale slums, which may not be

hooked up into the city's sewerage and electrical grid systems, but maybe slowly over time gets integrated into those systems.

- 7.14** A lot of these cities, Istanbul and Nairobi and others, started off as small colonial towns almost, of maybe a few thousand or tens of thousands, and they're now multi-million. There's a lot of megacities. There's a big expansion of megacities, Addis Ababa and Lagos, tens of millions of people in these big cities. So, urbanization is happening on a big scale. Most of the world's population, I believe, is now urban. That is a big change from the past.
- 7.15** That's the first demographic transition then, rising industrialization, urbanization, and a shift from more investment, small investment in large numbers of children to more investment in a small number of offspring. In biology, we have the R and K strategies, the strategy where you're focusing on, you have lots of offspring, hope enough of them survive. Other species, like elephants, have smaller numbers of offspring, invest a lot in those.
- 7.16** That's also perhaps in the background here, but that was the shift in a way, as we go from an R to a K strategy. In humanity, the first demographic transition. On this chart, we can look at the human development index across the bottom, which goes from the least developed to the most developed. That index would involve women's education, health, income, and so forth. The higher the human development index in 1990, the lower the total fertility rate.
- 7.17** The highest developed societies, we have Japan, the UK, the US, super high human development, very low fertility. On the other end, we have the Malis and the Nigers and Afghanistans, which have very low human development and very high fertility. You have that line, a correlation line, between the human development index and fertility. That fits very nicely then with that first demographic transition argument, that improvements in human development correlate with declining fertility.
- 7.18** Women's education has been found to be, arguably, the most important factor in lowering fertility, particularly secondary school education, where women then start forming families later, but also have other options besides having lots of children. They can find employment, which also competes to some degree with raising children.
- 7.19** If we consider the age pyramid of a society like Mexico, for example, a classic human development index and declining fertility story, 1950, it has the classic pyramid with a wide base of young people. By 2017, the fertility rate has come down. Mexico is now a much more mature country, demographically; although, interestingly, what you do notice, there's a bit of a youth bulge there. Age 15 to 30 still looks reasonably large in 2017. By 2050, it's projected that the population will be much more mature, with more senior citizens.
- 7.20** So that would be a classic development path where you have economic development correlated perhaps with falling fertility. But then you have a society like Uganda, where

there hasn't been much economic and human development. The age pyramids look relatively similar in 1950, in 2017, and even the 2050 projection. So that's not showing much change in the case of Uganda.

- 7.21 Some have talked about the sub-Saharan Africa fertility stall, where fertility rates come down from the seven and eight children per woman, down to the four, five level, but haven't really gone down to the two level. Others argue, in fact, there are a number of instances of where African fertility has come down to the replacement level, places like South Africa and certain cities like Addis Ababa in Ethiopia. And so, maybe we are going to see that demographic transition in Africa, too.
- 7.22 In terms of those who are worried about overpopulation and youth bulges, very young populations being unstable, leading perhaps to civil war, leading to a lack of democracy, would tend to look for a series of policy remedies to try and address the fertility problem. So, foremost among this would be encouraging women's education, especially secondary school education. A second might be improving access to contraception and family planning. So family planning ministries across a lot of developing countries, access to contraception.
- 7.23 A lot of African countries, for example, women desire smaller families than they have. So, the actual fertility intentions are lower, whereas in developed countries, typically women want to have more kids than they actually do have. So there's actually, perhaps, fertility intentions are for a larger family, rather than the families people actually have. But in the developing world, we often see the reverse, where women want smaller families. They want access to family planning and contraception, and that's not available.
- 7.24 So, improving access to contraception and family planning would be one policy tool. Improving women's education and access to work, so alternatives to having large numbers of children. General economic and human development in terms of the growth of cities and economic opportunities. There is also, it is worth saying, there seem to be examples of where culture sometimes leads the decline in fertility.
- 7.25 There have been academic papers on the rise of what's called the telenovela in Brazil, where TV programs that model two-child families, instead of four- or five-child families, then if those become very popular, those can influence the family size norms in a society. Sometimes culture can also play a role in reducing the family sizes in a society.
- 7.26 Then there's the question of the role of religion, nationalism, and ethnicity in encouraging a pronatal attitude, that is, being in favor of having more children. It's worth saying that the major Abrahamic faiths tend to, for example, enshrine a traditional role for motherhood. They tend to be pronatal. Just, you think of the Bible, "Go forth and multiply." This idea of encouraging women to have more children and larger families, as an example of something that is tied to world religions.

- 7.27 But equally, when you have nationalism or ethnic conflict, and numbers count, if numbers count, whether in war, soldiers, or whether in winning elections in democracies, in ethnically divided societies, could it be the case that, in Africa, with its high level of ethnic fractionalization, where politics runs on ethnic lines, that ethnic competition could be maintaining higher fertility and politicians for ethnic blocs? They want their ethnic bloc to be larger, maybe they are resistant to change.
- 7.28 In other societies, perhaps the role of religious conservatism. So, for example, Islamic fundamentalism in Pakistan might play a role in opposing contraception, opposing women going to work. This could be one reason why Pakistan, with its perhaps somewhat more austere tradition of Islam, has had a slower transition to lower birth rates than Bangladesh, for example. The brand of Islam, whether it's more conservative or not, may be playing a role perhaps in Pakistan, less so in Bangladesh.
- 7.29 It's worth saying, in Pakistan, it's also a somewhat more ethnically diverse society. Pakistan is, compared to Bangladesh. Maybe, again, ethnic competition—is that part of the story? Is it religious fundamentalism, part of the story? I mentioned Israel-Palestine, the high birth rates amongst both Arabs and Jews there, possibly as a result of ethnic competition.
- 7.30 We talked a lot about then the first demographic transition, and the decline from high birth rates and death rates to low birth rates and death rates, as a result of a shift from a largely agricultural economy, where you needed kids to work the land, to a more urban economy, where the value of more children is not as easy to see for economic development. So urbanization, industrialization, smaller families.
- 7.31 We then come to the second phase, which is arguably the world that we live in. And there has, since the late '80s, been the development of what's called second demographic transition theory. So the demographic transition, as I've articulated it about human development and falling fertility, that is the first demographic transition, which is in some ways more related to changes in material life. People are richer, not as many children dying. You don't have to have as many. More urban, more industrial, less agricultural—these are more material factors.
- 7.32 The second demographic transition places more of an accent on culture. That is to say that, in the second demographic transition, once we've gone through the first one, maybe instead of the material factors making such a difference—access to health, enough income, even women's education—maybe now there are going to be some other factors at work. Well, why is that the case?
- 7.33 The first is that the link between sex, access to sex, and procreation has been broken by contraception, effective. Now, there's always been contraception, but it's become a lot more effective since the rise of the birth control pill, for example. Then whether you have a family or not is not about sex, but it's simply about a choice. Once it's a choice,

OK, it could be about economic constraints, but it also could be about values. How pro-natalist are you in your value system?

- 7.34** Values then becomes more important as a correlate of fertility rates. When everybody had to have lots of kids, it didn't matter whether you were super religious or very secular. It didn't matter which ethnic group you belonged to. That was the deal. You needed kids. You needed them to work the land. In an urban, largely industrialized setting, it becomes more a question of values, perhaps, that differentiates those who have more children from those who have less or none.
- 7.35** We see a postponement of childbearing, people having children at later ages, and then a rise of childlessness. And increasingly, the so-called, well, the move to below replacement, or even what's called lowest low fertility, below a total fertility rate of 1.5, spread of below replacement fertility, which is now the majority of the world, and also this idea of lowest low fertility, which we're observing in places like East Asia, Southern Europe.
- 7.36** And there's a chart here from Ron Lesthaeghe, who is one of the key second demographic transition theorists. This is all again in the literature on demography. You can find more of this literature if you focus on demography and what demographers do. The differences between the first demographic transition and the second demographic transition are arrayed on this chart.
- 7.37** First demographic transition is about—it's the rise of a more bourgeois society, industrialized, urbanized bourgeois society, a rise in the proportions marrying. It's the rise in proportions marrying in the first demographic transition, whereas in the second demographic transition, we actually have a fall in the proportion that are getting married. There's an increasing incidence of loose, cohabiting relationships.
- 7.38** People are having children within marriage with the first demographic transition; whereas in the second demographic transition, actually, there's more of a decline of the traditional family, and there are more people having children outside of wedlock. Increasing amount of cohabitation, increasing amount of divorce, so more fluid society. Remarriage rates after widowhood, for example, are lower in the second demographic transition.
- 7.39** If we think about fertility, the first demographic transition is about having fewer kids as a woman, let's say, in older ages. You have your two children and then you stop. With the second demographic transition, in fact, fertility doesn't begin until perhaps age 30. Fertility is delayed. It's not the case where you just have your children and stop at a young age; in fact, fertility is delayed, and you have your kids later. That what's called tempo, the tempo of fertility in demography is later.
- 7.40** Contraceptive technology during that first demographic transition, again, beginning in the late 18th century, moving through to the 20th century, that was not really a contraceptive revolution. Contraception wasn't very effective. It was there, but it was

nowhere near as effective as it becomes in the second demographic transition, much more effective contraception. So this is, again, talking about breaking the link between sex and reproduction. That happens much more in the second demographic transition.

- 7.41 When Malthus was writing in the 19th century, what he was thinking about in terms of people not having children is abstinence, not contraception. It was about controlling fertility through abstinence, possibly through timing of intercourse. But basically, this bourgeois was linked to relatively traditional families, having kids within the framework of marriage. So this more bourgeois pattern, which characterizes the first demographic transition.
- 7.42 Otherwise, what else do we see? The higher-order needs, the more cultural needs, perhaps, are a bigger factor in the second demographic transition; whereas in the first, it was all about, how can we improve our material condition through work, moving to the city, educating our children, investing in them so that they can have a secure lifestyle?
- 7.43 That's then in the second demographic transition, some of these questions around expressive individualism. So individual autonomy, expressive work and socialization values, self-actualization, these things become more important. Some of these cultural value differences are playing a bigger role in the second demographic transition.
- 7.44 We're also getting a change in gender roles. In the first demographic transition, women were still very much in the home. Yes, to some degree working, but it's really in the second demographic transition that we have a real equalizing of sex roles in terms of employment. This is from Ron Lesthaeghe's work on differentiating this second demographic transition from the first one.
- 7.45 One of the features of the second demographic transition is a shift to below replacement fertility, and this chart we've already seen of Western Europe and the decline of fertility rates. So, by the time we get to mothers of the baby boom generation, born after the war, fertility rates in most west European countries are below the 2.1 replacement level. Those mothers would have started giving birth in the late '60s and '70s.
- 7.46 And so, from that period onward, we've had this below replacement fertility in Western Europe, which is now leading, after 40 years of demographic momentum, to population decline, were it not for immigration. Because there's always that momentum, because we've had a previous wave, as you'll recall, of births. That baby boom created a lot of new mothers, even if those mothers have below replacement fertility.
- 7.47 Just the earlier boom of mothers means the population as a whole will go up for a while, and it's going to have momentum, and then it'll start to decline. Once you have now smaller cohorts and fewer mothers, then you're going to start to see the population decline, especially if longevity peaks, and we don't have people living to 200 years. There is a firmer limit at the other end. So yes, we had longevity. We had people

living longer. That was partly behind the expansion of the population. But as that starts to hit a wall, then we're going to see populations start to fall.

- 7.48** This is just another chart of period total fertility rate. I recall the difference between completed total fertility rate, which is where you actually know how many kids women 45 and over had, versus period, where you are using each age group's number of children and creating a synthetic profile of fertility across the reproductive ages, to come up with an expectation of what the total fertility rate will be when it's completed.
- 7.49** Now, that can be confounded, as I mentioned, if the age at which women are giving birth is rising, then the period total fertility rate will be essentially an underestimate of completed fertility. I think a lot of people in East Asia and in Europe are hoping that a lot of women who haven't had kids will have them late, but still have them. We might see the completed total fertility rates be a bit higher than the expected or period TFRs.
- 7.50** But if we look at these period TFRs across different parts of Europe, we can see a consistent variation, where it looks as though the Northern European and Western European total fertility rates in green—if you look at those green and blue lines as we head into the 2000s, they are considerably higher than the total fertility rates of three groups of countries, Southern Europe, Eastern Europe, and German-speaking Europe.
- 7.51** Those all have a very low fertility rate, or a relatively low fertility rate, compared to Northern and Western Europe, which is basically Scandinavia and the English-speaking countries, Ireland, Britain, as well as France. Those countries, and the low countries as well, have a somewhat higher total fertility rate than Southern, Eastern Europe, and the German countries.
- 7.52** It was the case, if we were to go back to 1960 or 1970, the difference between Northern and Southern Europe, or Western and Eastern Europe, was not as great. But it seems like there's been a shift over time, where Northern and Western Europe have around about a half-child advantage over Southern and Eastern Europe and German-speaking Europe.
- 7.53** Even within countries, even within a country like Italy, the north used to have much lower fertility than the south. The south's fertility has really gone down, even collapsed. So now, northern Italy has a higher fertility rate than southern. Maybe that's a microcosm of this pattern of the north seeming to be closer to a higher total fertility rate than the south. Why that is, we can talk about in a minute.
- 7.54** The result, of course, the compositional effect of these shifts in total fertility rate, and particularly the second demographic transition around the world, is that we're going to see a decline in the numbers of people living in regions, of course, that have the lowest fertility rates. So this chart shows a much faster decline of population in Eastern Europe and in Southern Europe than in Northern Europe and Western Europe. But in all cases, because even in Western and Northern Europe, the total fertility rate is now not much

above 1.5 on average. And so we're also going to see significant declines in total population.

- 7.55** But clearly, the numbers here for—if we take Southern Europe, it's a drop from, predicted on this model, which is a UN model, 140 million in Southern Europe in the year 2000, down to, looks to be about 30 million. So 140 down to 30 million by the year 2300. Actually, that actually presumes some rebound in total fertility rate. If TFR remains at a 1.3 level, then in fact, the news could be even worse by 22 or 2300 than that. But what we're seeing is, countries will only be a fraction of their current size.
- 7.56** I mentioned that, at a total fertility rate of 1, which is what you find in cities like Hong Kong and Macau or a country like Korea or coastal China, a country is only one-tenth its size in 100 years. So that's a major change. So we are on the cusp of, potentially, if nothing is changed, a population collapse, which will happen logarithmically. It'll start off as a plateau. It'll pick up speed. And then again, fewer mothers, fewer kids, fewer mothers, fewer kids. It's going to accelerate in that fashion.
- 7.57** Just returning then to—let's look at some other patterns in the data, one of which is the relationship between human development in terms of income, education, and health, along the bottom of this chart, and the fertility rate along the vertical. We can see this classic pattern of a higher human development, lower fertility rate, if we look around the world.
- 7.58** We have Niger at the top, six children on average per woman, which is, if we were to look at this chart in 1990, Niger would be up around 8 or 9. There has been a drop also in the least developed countries in terms of fertility, but they are still much higher than the highly developed societies. If we looked at this chart in 1990, there would be very few countries with below 1.5 or below 2 as the total fertility rate.
- 7.59** Now, there are a lot of countries there, including countries like Chile in South America, and we've got China, and we have Brazil falling. We have a number of countries now really experiencing quite low total fertility rates. It's interesting to break this out by continent. Looking across Africa, we see a traditional pattern of higher human development correlating with lower fertility and vice versa. Very much that sloping line pattern.
- 7.60** We have, for example, South Africa and Cape Verde, more developed, lower fertility rates; Niger, Chad, some of the Sahel countries, with the highest fertility and the lowest human development. But now let's look at Europe here. If we look at Europe and human development across Europe now, most European countries have reached a relatively high human development, but there's still variation between 0.75 and 0.95.
- 7.61** You have Moldova, Ukraine, Bosnia, which are the least developed in terms of income, education, and so on. Then you have the best developed— Netherlands, Germany, Finland, for example, at the other end. What's noticeable here is the line looks flat, or if anything, it's the reverse of what we saw across all of the world or within Africa. So that

it certainly seems to be the case that higher levels of human development are not leading to further declines in fertility to zero. In fact, they may even be associated with somewhat higher total fertility rate.

- 7.62** So it seems as though the pattern is one in which, sorry, one in which you get initially that first demographic transition, higher human development, lower fertility, but then you hit a plateau or perhaps even a hockey stick shape, where higher human development seems to be associated even with somewhat higher fertility. That becomes important when we talk about the question of how to raise fertility rates. It may be the case that higher human development may be part of the answer to raising fertility rates in the lowest low fertility countries.
- 7.63** The consequences of global fertility decline or even collapse are going to be felt in terms of the composition of the world's population. I mentioned, when it comes to political demography, the effect politically comes from the fact that the demographic transition is uneven across the Earth's surface, within countries, between countries, between regions, and here we see this projection showing different parts of the world and how Asia's population will peak around 2050 and start to decline.
- 7.64** Africa's population will continue to grow quite strongly, given current total fertility rates, so that by the time we hit 2100, Africa and Asia are looking similar-sized in their populations. Half of the world's population nearly will be, not quite half but getting on to half of the world's population will be in Africa. And already we're seeing 40% of global births in Africa, and that will move to 50% as we approach mid-century. So a big shift in population power around the world, as a result of global demographic trends.
- 7.65** Europe, North America are a fairly stable chunk of the globe. But of course, the internal ethnic and religious composition of Europe and North America is going to change dramatically. North America, by 2050, that's the US and Canada, are projected to have —only a minority of their population will be of European origin, white, non-Hispanic. Now, of course, we could argue whether the Hispanics are, in fact, white or not. That's another debate. It's an interesting debate.
- 7.66** Actually, a lot of things are implicated in that debate, including this question of whether boundary redefinition and assimilation is more powerful or whether demography is more powerful. It's arguable. What we're seeing, say, in the US case is that there are many people who are part-Hispanic and part non-Hispanic-white who will identify as white, although they will be more likely to identify as white if they're Republican and more likely to identify as Hispanic if they're Democrat voters. So there's an interesting wrinkle there.
- 7.67** But even if we just screen those thorny questions out for the moment, Europe, by the late 21st century, will also be so-called majority-minority. It will follow North America. Australia will be closer, arguably, to the North American pattern, Australia, New Zealand. But Western Europe, the main immigrant-receiving countries of Western

Europe, by the late 21st century, are going to be so-called majority-minority societies as well. So even though their share of the population looks fairly stable, actually, that's due to immigration rather than native fertility.

- 7.68** If we look at age structure, a society like Japan, which has a low birth rate, is elderly. You can see in gray, the increase in the 65-plus age population in Japan from 1950, but particularly taking off in the 1980s and expanding really until the present day. But then, of course, what we see is the population of Japan is slated to decline. It's already declining, so the population of Japan will decline without immigration.
- 7.69** But if you look at the differences between the working-age population in red and the elderly population in gray, that aging is not likely to become a lot worse as we move past 2050. And this is one of the points, is that even with a declining population, that doesn't mean you're just going to get older and older and older. In fact, the age structure will stabilize as long as the total fertility rate remains stable.
- 7.70** Even if that total fertility rate is only one child per couple, as long as it remains one, then you won't get continued aging. If it drops from one to 0.5, then you will have an increase in the elderly population over time. On the other hand, if the fertility rate recovers, then over time, the elderly as a share of the total will decline, and the working age will increase.
- 7.71** You then have in the second chart, Nigeria, where it's really a story of a less mature age structure population and a really small gray elderly segment. It's pretty much all about the increase in the working-age population and a tapering, perhaps if the fertility rates continue to—if they can decline, then maybe Nigeria will have a very large working-age population.
- 7.72** The question will then be, can they create economic opportunities to absorb that population? If so, they're going to boom economically. If they don't create those opportunities, we're likely to see a youth bulge phenomenon, discontent and instability, possibly as a result of that boom in working-age populations. So it could be a blessing, could be a challenge.
- 7.73** This is the United Kingdom, where we see something similar to Japan, a little bit different. The birth rates have not been as low as Japan, up until very recently. Britain's total fertility rate has only declined below replacement, really in the last 10-15 years. It's quite recent. Britain was known as one of those Anglo-Saxon societies like the United States, like Ireland, that had a relatively healthy total fertility rate, but that, really since 2010, has really changed.
- 7.74** What this chart shows is that the proportion of elderly is rising quite rapidly in the 2020s. Really up until about 2060 or thereabouts, you're getting a graying of the population, and by 2100, that's even more pronounced. The working-age share, though, is relatively stable. Britain, in the very long term, if we take beyond 2100, actually, the proportion of elderly and working-age will be relatively stable.

- 7.75** It's just that, it's when you get change, when you go from a young population or a baby boom to a baby bust, the boom is like that pig in the python, and it's working its way through. When it's towards the end of the python, that's when your aging problem is at the worst. Then once the, let's call it the baby boom, when they die off, then that will relieve some of the aging pressure, and you'll have a slightly more balanced age structure in your population.
- 7.76** So, sticking with the developed world, let's look at variations in total fertility rate between countries. I already mentioned that Northern and Western Europe in the last few decades have emerged with higher total fertility than Southern and Eastern Europe, and German-speaking Europe as well; Austria, Germany, Switzerland are in the low fertility zone. High fertility can be found in Israel. In terms of developed countries, Israel really stands out.
- 7.77** One of the questions we might ask is why Israel has a high fertility rate, when it's a developed country with high education, high human development, high excellent health care. And why is fertility lower in the German-speaking world, Southern Europe, Eastern Europe, East Asia?
- 7.78** So, one set of explanations from demography referred to as feminism as the new pro-natalism, that there is a link between gender equality and somewhat higher birth rates. Northern and Western Europe have more gender equality than Southern and Eastern Europe and East Asia, so their fertility rate is higher. Now, of course, that only works if we zoom out to a global perspective. Clearly, Southern, Eastern Europe, East Asia, do not have less gender equality than Pakistan and Afghanistan.
- 7.79** Actually, what we're talking about is a sort of curvilinear shape, where the most patriarchal and gender-unequal societies have really high fertility. The intermediate, let's call it, countries that have intermediate levels of gender equality, that's Eastern Europe, East Asia, Southern Europe, have the lowest fertility; and the most gender-equal societies of Northern and Western Europe have somewhat higher fertility. Although the German-speaking countries have very high gender equality and very low fertility, so they don't fit the model very well, not as well as the other countries.
- 7.80** What are the elements then of this feminism as new pro-natalism model, which is quite popular in contemporary demography? The first is that some countries have a generous subsidized daycare sector, subsidized childcare sector. France and Scandinavia in particular, generous parental leave. Countries that have those generous policies have a somewhat higher fertility rate, is the argument, whereas that is much rarer in Southern and Eastern Europe.
- 7.81** Countries, the Anglo-Saxon world and Scandinavia, where there is somewhat more birth out of wedlock, divorce and remarriage, you don't have to get married to have children. You can cohabit in Scandinavia in particular, so that greater family flexibility. Now, it has downsides, of course, in terms of outcomes, educationally and

economically. But the upside is, it does seem to allow for a somewhat higher fertility rate. That seems to be a pattern. So the Anglo-Saxon, Scandinavian pattern is one of greater flexibility into and out of having children.

**7.82** Similarly, greater flexibility of entering into and out of the workforce, a flexible labor market, more free market capitalist system, certainly in the Anglo-Saxon countries, less so perhaps Scandinavia. But in the Anglo-Saxon countries, a flexible labor force allows perhaps a woman to have children, exit the labor force, have children, reenter the labor force quite easily. That is maybe more difficult in a more rigid labor market, which is not as free. And so that was another explanation that has been used.

**7.83** Now, of course, what we've seen in very, very recently is that the Anglo-Saxon countries and Scandinavia have had substantial fertility decline, just in the last 10 years. So one of the questions is, why this has occurred? Is it to do with something to do with the pandemic? Is it to do with changes in the economy? Could it be to do with cultural shifts like feminism, for example, or the rise of new forms of left-wing identity politics amongst young people?

**7.84** What is the reason for what appears to be a significant decline in the United States, in Britain, in Ireland, and in Scandinavia—countries which were thought of as relatively stable, close to replacement birth rate countries, almost the model, but they've really slipped in the last 10 years or so, 10, 20 years.

**7.85** OK, the explanation around patriarchy is that patriarchal norms, such as this idea that women should be at home to raise the children, or women should support the husband's career rather than advancing their own career, is that will make women more reluctant to have children, because it means they're going to have to give up their own personal ambitions for self-expression or for self-actualization.

**7.86** Therefore, systems such as Southern and Eastern Europe or East Asia, which require that somewhat more traditional gender role, are going to experience a greater decline in fertility than in the more gender-equal countries of Northern and Western Europe. That is how the theory of feminism as the new pro-natalism works.

**7.87** One could argue, one could make the argument, however, that perhaps feminism, like patriarchy, if taken beyond a certain point, may actually lead to declining birth rates as well. If, for example, having children in traditional gender roles are seen as a form of not realizing your potential, or as a form of betraying gender equality, then that is going to have an impact on fertility. A lot of these things seem to involve some kind of a happy medium. But if you seem to go towards an extreme, then it can sometimes lead to declining fertility.

**7.88** The relationship between human development, particularly education of women, and total fertility rate has changed. If we were to look at the year 2000, you could still see that neat downward-sloping pattern between human development index and total fertility rate, where the better educated women had the least kids and the least

educated women had more children. That's really begun to change and shift, particularly in developed societies, where it is no longer the case that the best educated women have the least children.

- 7.89** In fact, it seems more of a hockey stick pattern with the highest human development index societies having somewhat higher birth rates, and in fact, women who have the highest levels of education having somewhat higher birth rates than women with moderately high levels. For example, women with a graduate qualification, a master's or above, have a somewhat higher fertility rate than women with only a bachelor's degree.
- 7.90** Those who don't have any degree have, again, somewhat higher fertility, but it's not dramatically higher. Possibly, the takeaway there is that the old indicators around women's education don't matter really all that much anymore in developed societies for determining or perhaps indicating higher fertility rates.
- 7.91** The cultural argument in second demographic transition theory, that having children becomes a choice as we enter into the second demographic transition—you don't need to have kids to have sex, for example; you don't need to get married—that's a difference. What explains this variation then around the world, where we see some countries like Israel, with a total fertility rate above three? Most of the world, particularly the industrialized world, is well below replacement and is either in the 1.5 to 2 zone, but increasingly in that lowest low zone, below 1.5 children per woman.
- 7.92** We have quite a bit of variation. I want to look at the case of Israel in more detail, to look at subgroups within the population and to ask what it might be that's allowing Israel to have a high birth rate, even though it's an industrialized and modern country. If you break down the Israeli population by religion, we have a number of distinct subgroups that are relatively segregated from each other.
- 7.93** You have the ultra-Orthodox Jews, known as Haredim. So Hasidic Jews would be an example of the ultra-Orthodox. You then have non-ultra-Orthodox Jews in red on this graph, which would be modern Orthodox or reform, or secular. The fertility rates have been relatively stable for the ultra-Orthodox, who have on average about six and a half children per woman, and the non-ultra-Orthodox who actually are above replacement but only barely.
- 7.94** Then we see a pattern of the Druze, which are a minority, which ultimately developed initially out of Islam as a subsect. But basically, they are an ethno-religious minority. Their fertility rate has been going down. The Christian fertility rate has been going down as well. The Arab Muslim fertility rate, the Palestinian fertility rate, stable for a while at an intermediate level of perhaps three; that's been going down.
- 7.95** Looking at more recent data, taking us up to the 2020s, if we look at the ultra-Orthodox, their total fertility rate is fluctuating in the zone of 6 to 7 children per woman. That population, which is only a few percentage of the first-grade class in the

1960s, is now one-third or more of the first-grade class, and is on track, really, to become the majority of Israeli jury as we head into the late 21st century, if nothing changes.

- 7.96** Now, there's a whole debate. Again, I talked about constructivism versus ethno-symbolism. To what extent are the boundaries between groups culturally or politically or economically constructed, and to what extent are they fixed? If we take the ultra-Orthodox, in fact, very few ultra-Orthodox children leave ultra-Orthodoxy, because if you do, you cut yourself off from your family, from your community, from your belief system. It's a very traumatic experience. That is a typical pattern with strict religions. And so, not only is there high fertility, but there's also low assimilation, which gives you very strong population growth potential.
- 7.97** Looking at these other groups, that's the dark blues, the highest line here, Jewish, and that's been pretty stable. Let's look at the Palestinian, the Arab Muslims in Israel. Actually, this is broken down into Muslim religious and Muslim not very religious. But in both cases, that is the light green dotted line and the dark green dotted line. There is a decline. So the pattern amongst the Arabs in Israel is declining fertility, which maps to some degree the first demographic transition dynamics.
- 7.98** So they are declining, although they still, as of the 2020s, have above replacement fertility. Even the less religious Muslims, it's above two, and for the more religious Muslims, it's maybe around three. But on average, the Muslim birth rate is now below the Jewish birth rate. Looking at the Christians, they are below replacement now, as are the Druze, as are those who have no religion, who say they have no religion, so seculars.
- 7.99** So there's a pattern in Israel. If we are then going to explain Israel's high fertility, it's clearly going to be connected to the ultra-Orthodox fast-reproducing population, but also actually the lighter blue line, where it says Jewish, religious Jews, they are around four. And so these are the modern Orthodox or religious Zionists who are not as Orthodox. They are more accepting of modernity than the ultra-Orthodox, but not as much as the seculars.
- 7.100** And their total fertility is around four. Now, they have actually a much higher rate of outflow into the mainstream of the more secular Jewish population. So they don't have quite the same potential to upend the demography of Israel, but they are committing, or contributing to, Israel's high demography. I think if you were to look at Israel, you'd say that religiosity, especially strict religiosity, is the main reason for Israel having a higher total fertility rate and maintaining that high total fertility.
- 7.101** We see echoes of this religious fertility pattern in Western countries like the United States. If we look at this chart, the vote for George W. Bush in 2004 on the horizontal axis and the total fertility rate in the white American population on the vertical axis by state, you can see that the New England states like Vermont and Massachusetts and

the northeastern states, New York, have the lowest fertility, and they are the most democratic.

- 7.102** The highest fertility state is Utah, which is a Mormon-majority state. I think it's still a Mormon-majority state. Had the highest fertility and the highest support for the Republican party. We then get into a number of states in the southeastern US, like Oklahoma, Kansas, and plains states like Nebraska and the Dakotas. They have a higher fertility rate and a higher Republican vote share.
- 7.103** So there seems to be this correlation then in the white population, between total fertility rate and support for the Republican party. And that's actually mediated by religion. Evangelical Protestants have a higher fertility rate than mainline. Protestants have a higher fertility rate than secular individuals. That's correlated with voting for the Republican party. Let's look at more recent data on religion and fertility in the United States.
- 7.104** We see that the people who attend church weekly or more, starting in the 1980s, going through to the 2020s, it looks like weekly attenders, fertility is at the replacement level, now as it was in the 1980s. There hasn't been much change in total fertility rate. And so we have a stable total fertility rate that is at the replacement level for people who attend church weekly or more. It is only particularly those who attend less than weekly, or particularly those who say they have no religion, where we've seen a significant drop-off in fertility, and that's well below the replacement level.
- 7.105** Actually, looking at that yellow line and the blue line for non-religious, there is a gap of almost one whole child in total fertility rate between those who identify as religious and attend weekly or more, and those Americans who say they are not religious. It's worth saying that the non-religious share of the US population has been rising since the late 1980s, from about only around 8 or 9% in the 1980s to now 30% in 2019. So that rise in the non-religious share is a major feature of change in the United States in recent decades.
- 7.106** That seems to correlate with—sorry, I should say, you put these two things together, you've got a rising non-religious share, a falling regular weekly attender share. However, the fertility gap between the regular attenders and the non-religious has widened to a whole-child difference. Now, if we put our demographer's hat on, particularly our projections hat on, we'd say, well, differences in fertility rates are going to matter most if we take a long-term projection.
- 7.107** Like, 2100, that fertility gap between the weekly attenders and the unreligious is really going to tell in the long term. So we would expect a rise in the share of people who are religious in the long term, reading from total fertility rates. Now, of course, we have to factor in how many children of those who attend weekly drop out of the faith, and that's quite significant. But it's really a question of which is moving quicker.

- 7.108** There's a race between, on the one hand, secularization, people raised religiously, moving out of religion, or raised with weekly attendance, moving away from weekly attendance on the one hand, but on the other hand, people who are weekly attenders having a much higher birth rate than those who don't attend weekly. Which is the faster force—is it demography or is it secularization or apostasy?
- 7.109** Any slackening in the rate of apostasy, any decline in the rate at which people move from religion to non-religion, is going to be very important. And what we're seeing just in the last few years is, it seems as though the decline in the proportion of people who are religious has been flattening out and stalling, particularly in the younger age groups. And if that continues, we would then expect there to be, possibly, we would expect that the higher birth rates of the religious will be more important in the long run in shaping the American population.
- 7.110** There are also political implications of this, as we saw with that vote for George W. Bush, this correlation between religion voting Republican. Here is a chart that came out recently in the Financial Times from John Burn-Murdoch, drawing on US data, but also world values survey data. If we take the United States in 1980 and we ask about the average number of children born to conservatives versus progressives, there is no real difference in 1980.
- 7.111** The difference starts to emerge in the 1990s, and the conservative total fertility rate is greater than the progressive rate by about 4%, 0.4 in terms of percentage points. By the time we're into the mid-2020s, the gap has reached, it looks like 0.6 of a child between progressive and conservative. So the gap in terms of family size between conservative and progressive has been expanding, along with the gap between secular and religious expanding.
- 7.112** And the same pattern has been noticed in the developed world as a whole. And I would also say that, in the developed world as a whole, especially if we look at Europe, there's roughly a half-child advantage between those who attend regularly and those who don't attend at all. There is a similar dynamic as what we see in the United States. Religious people, higher fertility than secular people, conservative, higher fertility than progressive, and a widening gap.
- 7.113** A lot depends on what happens with this gap. If this is maintained over a long period of time, the next question will be, how much shifting is there? Do the children of conservative people become progressive? Do the children of religious people become secular? That is true, as long as you're not in a polarized society. The more polarized you get, as in the US case now, where there is increasing unwillingness to date across religious and political lines, the more those lines harden, the less assimilation, the less culture matters, the more demography matters.
- 7.114** So the more polarized the US becomes, the more this difference between religious and secular, conservative and progressive will result in compositional population shifts, with

conservatives and religious forming a larger share of the population. It's worth just saying that in Britain, for example, the most religious part of the UK is actually London because of immigration and ethnic religious retention among non-white British populations. That might be an example.

- 7.115** And even in certain neighborhoods, in the most ethnically diverse part of London, they've actually become more religious over time due to immigration, showing that, in some cases, the demography of religious immigration can overwhelm the cultural process of secularization. And so, it is certainly conceivable that the demography of religion can become more powerful than the cultural process of secularization.
- 7.116** And anything that reduces the rate of religious decline, it doesn't have to be a religious revival. It just could be a slackening or less enthusiasm for moving away from religion. Anything that goes in the direction of weakening secularization is probably going to lead to a more religious society, and certainly in the long term.
- 7.117** There had been some talk in the early 2010s, late 2000s, about a rebound in European fertility as a result, partly, of women having children at later ages, particularly educated women. It doesn't seem as though that rebound has been enduring. We've seen, it looked as though fertility might have been moving towards the replacement level, but that seems to really, since 2010, have moved in the other direction towards lowest low fertility.
- 7.118** Lastly, I just want to go through this debate over pronatalist policies. Over 40 countries now have pronatalist policies, including cash bonuses for children, especially more than two children. I also talked about subsidized childcare, parental leave policies, as well. How effective are these policies? When we talked about the feminism as pronatalism thesis, there was an argument that parental leave, for example, and subsidized daycare might be policies that can increase the fertility rate.
- 7.119** I think the general view is that these policies do not have much of an effect in raising fertility rates. While you could make the argument that policies to reduce fertility, like making contraception and family planning services more widely available, have been effective, the policies to raise fertility rate do not seem to be as effective. We've seen a rise in the number of countries worrying about their fertility levels and engaging in pronatal policies of various types.
- 7.120** There is a bit of a debate. Some would argue that countries that really invest a lot in fertility do sometimes see a benefit. Others that claim, well, actually, they don't really work. You will get a temporary jump in fertility, as in Romania, as in Russia, but that is very rarely sustained. If you consider this chart, looking at Poland and Hungary, and when pronatal policies were implemented in these countries, in the case of Poland and in Hungary, you had a surge in the crude birth rate after the policies were implemented, but then things quickly regressed back to their prior trend level.

- 7.121** Although you can see in Poland, it seems to have been more successful, but still, overall, it doesn't seem to be enduring in many cases. Some would say, well, what you need is, you need to constantly be offering new incentives all the time and drawing media attention to them, in order to keep this front and center in people's minds, to keep the effect of these policies going.
- 7.122** Some would argue, in fact, they can increase fertility by even a few tenths of a percentage point, but that actually really matters. There's a big difference between 1.3 and 1.6, or 1 and 1.3. These little points, they matter in the long term quite a bit, so it's still worth doing these policies. Others would say they aren't really going to make much difference. In any case, we don't seem to have many examples of where countries have been able to restore replacement-level fertility.
- 7.123** There are some countries like Georgia, for example, or Mongolia, which seem to have above-replacement fertility. There has been a bit of a baby boom or a revival of fertility in Central Asia. But in both the Georgia and Central Asian cases, these are relatively religious societies, or they are societies that, in the case of Central Asia, as they left communism, they became less atheistic and more religious. And that, arguably, is why the birth rates increased, not any sort of policy genius in these states.
- 7.124** So, that more or less rounds out our discussion about the drivers of fertility, the policies of pronatalism. One question would be, what if we aren't able to raise the fertility rate and we continue to see these lowest low fertility rates and we do have population collapse? As I mentioned, at a total fertility rate of 1.0, a country in 100 years is only a tenth as large. In 200 years, it's moving towards a hundredth as large. Is that where we're headed?
- 7.125** One argument might be, well, what we're doing is, we're going through a population bottleneck now, an evolutionary process. Now that we have contraception, it's a matter of choice. Therefore, those who possess either the cultural values that prize high fertility, so that could be religious groups like the Amish or the ultra-Orthodox Jews, or even regular religious attenders, they will be the future of the human population.
- 7.126** Or in the case of a purely secular society, it may be that those with the genetics to really like children and to prioritize having families over, perhaps, career advancement or consumer traveling or enjoying consumption, those with the genes that are connected to desiring larger families are also going to eventually win out in this evolutionary race. We're going to get a selection for certain kinds of genetics, certain kinds of cultural patterns.
- 7.127** If we look at populations like the ultra-Orthodox Jews, who are going to be a majority of observant Jews in the United States and Britain in 2050, they are going to be, ultimately, the majority of Jews in Israel— if that is what's happening in the Jewish world, we could imagine that happening in the Christian world as well, possibly through religion.

- 7.128** The US is a case where there are a number of fast-reproducing religious groups. The Amish are one, Mennonites, the Hutterites, to some degree the Mormons, although they seem to be losing their fertility advantage. That will be one to watch. There was a recent paper projecting that the US Mormon population would become much, much larger in the future.
- 7.129** Perhaps in the cases where you do have—the Netherlands is another case, where there's a Calvinist Orthodox Christian population. Finland, there's another such population. Those could become a much larger share of the total. In other countries, maybe it'll be down to those with the right genes to favor larger families that will eventually take over in the population. OK, I'll leave it there. Thanks.

## Lecture 8: Demography and Economics

**At a glance** — Demography and economics — the economic dimension of population change.

- 8.1** OK, so my first degree actually is in economics, believe it or not. And so I felt I had to do at least one quasi-economic lecture today. We're mainly talking about the politics of population change, and we are also going to talk about the politics, but via the economy now. I want to talk about a couple of aspects of how demographics might shape the economy, which in turn might shape politics.
- 8.2** I'm going to talk mainly about the work of the French economist, Thomas Piketty, on inequality, but also on economic growth. With political demography, recall that a lot of this course has been about how population changes affect politics in society indirectly or in a second- or third-order way. You can have birth and death rates affecting age structures, affecting democracy, for example, or violent conflict.
- 8.3** Equally, we can have a fourth-order effect where we have demography shaping economic growth. If you have an aging population or if you have a baby boom, that can affect the economy in different ways. If the economy is affected, that can affect politics in different ways. Particularly, economic growth might have an effect on inequality, might lower it, let's say, which would reduce conflict.
- 8.4** Here in particular, today, I want to look at the question of what a mature age structure does to an economy, and what are the political effects of that. Much of the lecture will focus then on the impact on the economy and on Piketty's argument. Thomas Piketty, he wrote a book on capital, which became a bestseller. It's a 700-page, quite dry tome, and it's very unusual for this kind of a book to do as well as it did. So it is worth paying attention to this particular book.
- 8.5** He's an economist, and again, a lot of this is going to be about economics, but there's quite a bit of demography in the book, which is why I found it quite interesting. Unlike a lot of economics, it's taking a very long historical period, and it's based on historical documents, and it's trying to use this long time horizon, from the late 18th century until the present, to shed light on what's happening in the economy as a result of some of the changes that we've talked about.
- 8.6** He looks a lot at the pre-1914 period, which is unusual in economics. And because he looks at the pre-1914 period, he's able to shed light on, perhaps, some of the unique dynamics, arguing in a way that the pre-1914 period is more similar to our present time period economically, than the period from 1914 until the 1970s or 80s. So really, we're going back to the normal trajectory, and the 20th century was a deviation. That's kind of his argument, 20th century is a deviation.
- 8.7** Using tax records, inheritance records, estate valuation records, rich lists, even literature, going back to the early 19th and late 18th century. So that's his method. The

definition of capital in Piketty's book is anything which generates income. So that is stocks and bonds, bank accounts, physical plant and machinery, but also, critically, your own home as a source of capital, according to Piketty. That's his definition of capital.

- 8.8** You can see that's quite critical in that, because he's including homes in his definition of capital. Home prices have gone up a lot; clearly, that's going to affect the thesis. Because, really, the line between what is capital and what isn't is quite tricky and not well defined. So is a car, an iPad you use for fun, is that really capital or is that really a consumer item? Education, is that a consumer item or is it capital? In any case, we have Piketty's argument, which is really the iron law, that returns to capital tend to exceed economic growth.
- 8.9** The more that the return on capital exceeds the return to working as a laborer, or even as a well paid white-collar worker, the more that returns to owning capital exceed returns to working, that is, to returns to labor, the more inequality there is in society, and that society has moved into a position where returns on capital are exceeding returns to labor, and therefore, the rate of return in the economy exceeds the rate of economic growth, and that's driving inequality, according to Piketty, which is driving political discontent such as populism.
- 8.10** Piketty has also written on the rise of populism as well, and this is part of this more economic argument around discontent, economically. One way to think about Piketty's hypothesis, I've thought of this as a monopoly game. Many of you have played the game of Monopoly, where if you own houses and hotels and someone lands on your square, they have to pay you money. That's an example of returns to owning capital.
- 8.11** Capital accumulation—you make more money from people paying you rents as they land on your square. You can then buy more hotels, accumulate more capital, and they have to pay you even more when they land on your square. That is the process of " $r$ ," or capital accumulation, the rate of return. Then you have income, which is, for example, every time you pass go, you get \$200 or £200. That is your income, but it's relatively stagnant. You just have to keep going around and you only get 200 each time.
- 8.12** That seems to be quite stagnant, but as you accumulate hotels, you can get more and more money through being a rentier. That's the thesis in short, is that the return on capital exceeds the return on income, or so the amount of income you get. And so economic growth, which is tied to labor income, is below the rate of return. The rate of economic growth is below, and that leads to inequality.
- 8.13** The solution would be a card that you might pick out of the pack in a Monopoly game, which is, you are assessed for street repairs. In other words, every hotel or house you own, you have to pay a tax. And that's Piketty's solution, is to tax capital in order to reduce inequality. It's clearly coming from the economic left. But it's an interesting concept as to why we are in the position where returns to capital are exceeding labor income. Why are we in that position now?

- 8.14** Because he's making the case not that this is for all time, but he's making the case that certainly this was what obtained prior to 1914 and that we are back in that kind of an economy with high inequality. What does this have to do with demography? What it has to do is that demography is very much interwoven into his argument. That is, that when you have faster population growth, you tend to have a faster GDP growth, income growth, which means less inequality. When you have slow or declining population, you would tend to have slower economic growth and more inequality.
- 8.15** Because inequality is about the difference between rate of return on capital and labor income or growth rate. So aging slows down the rate of economic growth or the rate of growth of incomes, and that increases his formula,  $r > g$ . And that means that more money is made by those, if you like, owning houses and hotels on the Monopoly board than those who are making their 200 for passing Go.
- 8.16** And so accumulation of capital leads to accumulation of rents to capital owners, and that dwarfs the money paid to workers of all kinds, including people working in very high-wage jobs. There's a difference here between rentier versus earners, those making money from rents on capital ownership, including stocks and bonds, and those making money through labor income. Rents versus income, that's the basic concept, and that rents is becoming more important than income in this more mature demography that we are living through.
- 8.17** Thinking about the general theory of economic growth, in economics, you have resources, you have capital, you have labor as inputs into the economic growth process, along with technology. The Piketty argument is in effect that labor in combination with technology is what gives you economic growth. But what's happened is, in fact, that we're getting a decrease in the labor supply due to an aging population. So there's actually less economic growth coming through a rise in the labor supply.
- 8.18** Capital, thinking again of those hotels and houses on the Monopoly board, is less equally distributed than income through labor. And therefore, the more capital is as a share of the economy, the more unequal the economy. When you have population stagnation, you have economic stagnation and lower economic growth. That makes capital a bigger share of the economy. Of course, labor is shrinking as fewer people are there. You get labor as a smaller share of the economy.
- 8.19** As a result, you're getting more inequality. More inequality then leads to more political tension. Populism rises, polarization rises. That's a very multi-step argument, from population changes through to slower economic growth, through to inequality, through to political tension, is kind of the argument. It's like a fourth-order effect of demography on politics, a kind of indirect multi-step process, but still matters up the causal chain that we have this.
- 8.20** The model that Piketty develops is that the rate of capital accumulation and the rate of return to capital minus the economic growth rate gives you your level of wealth

inequality. A big factor in the growth rate is the number of children people have. So he goes back to— he thinks we're really in a new normal, which is really the old normal, just with, we had the hiatus of the 20th century.

- 8.21** He wants to argue that the 20th century was a bit of a blip, and it's more normal to have capital-dominated societies, where inheritance and not income is what really determines one's position in society. In the literature of Jane Eyre and Charlotte Brontë and others, and Balzac, they were talking about aristocratic society. The aristocracy, really, it was your inheritance that counted; you needed to marry into wealth in order to stably remain in the upper class. If you fell out of the upper class, you really fell far. It was a very unequal society, based on returns to estates where inflation was very low.
- 8.22** So Piketty argues that inflation actually helps to make things more equal because capital loses value when there's inflation. In the long 19th century in Europe, very few major wars. Wars tend to destroy capital. So very few wars, relatively low inflation, stable returns to landed estates. And he has a quote from *Sense and Sensibility*. He says, well, in this novel, "John has just inherited the vast Norland estate, which brings in £4,000 a year."
- 8.23** Now, that income was 100 times the average British income. So you had extreme wealth inequality in societies that had aristocracies. And that is compared with his half sister who only had—his half sisters Elinor, Marianne, and Margaret, who only had four times the average British income, versus they had 100 times the average income. There's just this vast difference in wealth between those who had capital and those who didn't have capital. Capital really was the secret to being wealthy in Europe in that long 19th century, from the late 18th century to 1914.
- 8.24** He notes that there was a considerable difference between the United States and Europe, that the United States had immigration. It was a new society. It was expanding. There were more opportunities for people to make money through labor. And so you had fast economic growth through fast population growth. So there's a tie with demography, is immigration leading to population growth, leading to economic growth in the American case. In that kind of a society, income and economic growth are more important; ownership of capital is less important.
- 8.25** You get less inequality than in Europe, with a more stagnant population capital, the larger share of the national economy, and therefore more inequality. Prior to 1914, he cites, just for illustrative purposes, he points to the novels of Jane Austen, Henry James, and Honoré de Balzac, talking about main characters and how they really understood that, in that kind of an unequal capital-dominated economy, you needed to have capital so you could charge rents, and that was your way into the upper echelons of society.
- 8.26** If you weren't there, then you were really in the poor classes. So, Rastignac in the Balzac novel, he has the option to become a lawyer, but even in a well-paid position

like lawyer, you can't really make enough to be sustainably in the upper class. You have to marry into somebody who is part of a family that owns landed estates, and then you would be able to be secure. You need to be in that 1%, if you like, that is 20 or 50 or 100 times richer than the average. That 1% was really important in the period from the late 18th century to 1914. So the 1% takes on more importance then.

- 8.27** We then have World War I, where all of a sudden you need mass mobilization, taxation. There's a lot of destruction of buildings and capital stock. You then have the Great Depression, where the value of capital really declines, and there is massive inflation. Here's Saul Bellow talking about suicides after the stock market crash of 1929 and people driving around and they own a lot of land that is worth nothing. You have a lot of capitalists who've lost their shirts. The value of capital due to war, due to Depression, shrinks in the society.
- 8.28** Wars and Depression destroy the value of capital. Invaders sometimes take capital away from former owners. Wealth is taxed to support the war effort. The Depression wipes out the value of stocks and shares. Inflation devalues capital. If you owe money to a moneylender, you want inflation to be high. If you're lending money, you want inflation to be low. If you are a capital owner, you want low inflation, you want stability. Where you have high inflation and instability, that devalues capital.
- 8.29** So the ratio between capital and labor favors labor income in these periods because the capital stock is wiped out. Of course, it reduces the size of the economy, but in terms of inequality, it actually improves inequality. Even during the economic financial crash of 2007-08, there was a decline in inequality due to that event. So war and depression then reduce inequality.
- 8.30** Of course, after the war, you had an economic boom, argues Piketty. In that economic boom, 1945 to '73, there was also a baby boom expanding the population. You also had population become better educated. There was the end to the gold standard in 1971, leading to more inflation. There was freer trade. There were all these mechanisms of economic growth. GDP per head, which is economic growth rising rapidly. Labor income is rising, and that reduces this  $r > g$  formula that he has. Rate of return to capital minus income gets narrower. That reduces inequality.
- 8.31** We've now had a sustained period in the 20th century, from 1914 to 1973, where inequality has been reduced due to economic crashes and war destroying capital. Then after the war, there's a baby boom, more population, and there's economic growth. But since 1973, for Piketty, we have gradually been returning to the period prior to 1914 of greater income inequality—back to the normal, in a way. Income becomes more unequal.
- 8.32** When capital accumulates in a period of stability, the capital share of the national economy rises back to previous levels. If you look at charts that Piketty has in his book of the share of the economy that is made up of capital and labor, you can see—charts

here in Britain and in France show that the capital share drops dramatically in 1914 and really doesn't recover until we get into the late 1990s. And so we're really just coming back.

- 8.33** If you look at this graph, it looks like we go off this cliff in 1914 and we only gradually return back in the late 1990s towards something like the way society was. The US doesn't have that same cliff drop in 1914, so it never really had the big change, the big drop in inequality, that Europe had. But still, it now seems to have rising capital as a share of the economy.
- 8.34** So this story that Piketty tells is one of a return to normal, a return to capital-dominated societies. That's as a result of weaker economic growth, stability, less inflation, but also slower-growing populations. So the mechanism is maybe a little different. Slow growth might have been caused by other factors prior to 1914, but in the post-1975 period, demography is playing a role.
- 8.35** One way it's playing a role is through—if you think about inheritance, if you have a lot of children, if you're a rich individual, then your money is divided in many ways. So that actually spreads the wealth a certain amount. And that can be happening across the economy. But if you have only one child, then all the wealth from two individuals and their families funnels down into one individual. And so, falling birth rates mean fewer people to inherit wealth, and that concentrates wealth.
- 8.36** Inheritance, which was very important prior to 1914, is becoming important again as a source of wealth. It is very important now, as it was prior to 1914 to inherit. And inheritance or marrying into money is becoming more important, the way it was, according to Piketty, the way it was prior to 1914 in the aristocratic world. And he looks at patterns of inheritance, again looking at probates and wills and gifting, showing that there has been an increase in the importance of inheritance since the middle of the 20th century.
- 8.37** So that's risen and that that's been a factor. That is, differences in terms of inheritance are very important. And if you think about even an economy with very costly housing, those who can inherit housing are going to be much better off. When houses don't cost very much, it's not going to matter as much whether you inherit a house or not. But when houses do cost a lot, it will matter more. So that will be a factor in inequality.
- 8.38** In this simulation in France, if you change the inheritance flow based on different rates of economic growth, inheritance is much more important at a low rate of economic growth than at a high rate of economic growth. If we have got lower rates of economic growth, as it seems to be the case, which is partly connected to lower rates of population growth, then inheritance is going to be more important.
- 8.39** Inheritance is more important, that's going to fuel inequality and the lucky ones who inherit or marry into capital. So it's all about capital, the owners of capital, inheriting capital, marrying into capital. That becomes the nature of how you wind up in the more

advantaged part, as opposed to the disadvantaged part. And the top 1% is extremely important as a share of total national wealth, as it was prior to 1914.

- 8.40** The risk is of a spiral of capital accumulation generating rents, which leads to more capital accumulation. The example of that Monopoly board, the owner of the house or hotel, getting more rents, buying more hotels, and that spiral of inequality. That is really kind of where this goes. Now, what does this mean politically, if we have a stagnating population, which leads to a stagnating economy, and more inequality; does that then pose political risk? The argument from Piketty is that, if you have economic inequality, you will have political instability.
- 8.41** So, commenting in 2014, argues that in France, it's striking to see that in 1920, political majorities adopted steeply progressive taxation, even though in 1914, they balked at even a 2% tax rate. Once they saw the instability of the Bolshevik Revolution, they became convinced they needed to adopt a progressive taxation rate. Well, so the argument there is that political instability might lead to more progressive taxation solutions.
- 8.42** This gets to Piketty's argument for a wealth tax, rather than an income tax, and to make the tax a global capital tax, rather than a national tax. This is his progressive argument, as we've got to tax wealth, not income, to lower the "r" and not the "g," to reduce the inequality. He does, however, accept that this is extremely difficult politically, to have property taxes and wealth taxes. That's politically unpopular, because even those who don't have a lot of property or a lot to give to the next generation, for them that's still very important that that not be taxed.
- 8.43** They're less likely to see these indirect effects of tax, essentially capital accumulation. They are unlikely to see the effects on economic growth, for example. He also argues that, in fact, an inflationary monetary policy may be good for combating inequality. We talked through that Piketty's thesis about demographic maturity leading to economic stagnation, leading to political tension. Now, against this, we have a number of critiques.
- 8.44** The most common critique from economists is that you can't separate the rate of return to capital and the rate of income or the rate of economic growth, that they are connected, and that it's a bit nonsensical to separate these and assume that they operate completely independently. Thomas Piketty, his argument was, in fact, we do see a lot of independence of economic growth and returns to capital.
- 8.45** In particular, there's a difference, let's say, between Karl Marx's view and Thomas Piketty. Karl Marx believed that capital accumulation would drive down the rate of return. Piketty believes capital accumulation does not affect the rate of return. Those are very separate worldviews. Economists, not just Marx, but Larry Summers, for example, would say more capital accumulation will lower the rate of return to capital,

and that that will naturally shrink the difference between the capital rate of return and economic growth.

- 8.46** Also, other economists have criticized Piketty, saying that he ignores depreciation of capital costs, which are significant. So it's not costless to own capital the way it would be on a Monopoly board. You're actually losing money all the time through depreciation. So we have a debate amongst economists over the Piketty hypothesis. There are other challenges to Piketty's thesis, and one of which revolves around the question of—that a lot of what Piketty is arguing is an increase in the capital stock, is in fact just rising house prices.
- 8.47** There has been an increase in land prices in urban areas. Cities have become more desirable to live in through the 20th and 21st centuries. That's one of the reasons for the growth. There are more single people. So cities have become more attractive and land prices in cities have gone up. So capital accumulation is one reason why, due to higher land prices is what we're talking about, not necessarily an increase in capital and the rates of return to capital. And those land prices could fall and rapidly change the picture.
- 8.48** Returning to this question of population. So part of this theory is a stagnation of population leading to economic stagnation. Charles Goodhart argues actually that what occurs in the period of globalization of the economy in the '90s and thereafter is that, because of globalization, capital owners were able to use labor all over the world. They were able to offshore production and use the labor supply from cheaper regions, places where people were paid less.
- 8.49** That created a golden age of capital and a golden age of rents. So lowering the cost of labor and boosting rents and profits. But he argues that with the demographic transition spreading beyond the West to the developed world, workers will start to become more scarce, and therefore, they will be able to command higher wages. So if you imagine, if the demographic transition really spreads around the world and the supply of labor really starts to shrink, then it won't be possible, really, to leverage cheaper wages in the developing world and offshore production as easily.
- 8.50** Goodhart's argument is that we are on the cusp of a reversal, that labor will be increasingly fought over and be in short supply, and capital will have to pay more for its labor. That is going to change the equation as well. That inflation may also be on the rise—again, that's a long-term debate. Others say it will be declining. This is just to say that, economically, clearly there's a connection between what's happening in terms of global population trends and what's happening in the global economy, and that can have political effects, too.
- 8.51** That's really the point of this lecture. Partly the point of looking at Piketty's argument is to say that, in addition to talking about perhaps the more obvious political demographic phenomena such as north-south migration, the youth bulge, and the

impact of population aging on pensions, for example, and on budgets and military spending, it's also important to look at the impact indirectly of demographic shifts on economies, and what that is going to do to politics. It's something else we need to think about within this big picture of political demography. Thank you.

- 8.52** How important do you think inequality is in driving political instability? There's been people who have argued, there's a book called *The Spirit Level*, where they argued that inequality, income inequality, is a driver of all types of instability in society and negative outcomes. And there's others that have argued that, actually, income inequality isn't that relevant if you look globally. It's like, *The Spirit Level*, they compared more developed countries. But if you looked at the globe as a whole, that you don't really find an effect of income inequality. So what is your view on that?
- 8.53** Income inequality, in my view, is certainly not so closely tied to right-wing populism, for example. Now, it is more closely tied to left-wing populism, but even there, a lot of the evidence seems to suggest that people's actual incomes are not a huge driver of their politics. One way of seeing that is to take Britain as an example. Really, from 1945 up until New Labour in the election in 1997 of Tony Blair, if you were working-class, you were more likely to vote Labour. If you were poorer, you were more likely to vote Labour. Wealthier, more likely to vote Conservative.
- 8.54** The link between income levels, between class and voting, is now basically gone in Britain. The class makeup of the—if we take the 2019 election, the class makeup of Labour and Tory voters was identical, basically. Certainly, support for something like Brexit was higher amongst people with lower incomes, but that effect is much less noticeable than the difference between those who favor lower immigration versus higher immigration. The immigration attitudes are much more closely connected to, let's say, populist voting.
- 8.55** Certainly across the West, I would say that the old left-right economic question, redistribution, free market, the link to class, the link to income is much weaker now than it would have been in the, say, '60s and '70s. It's cultural attitudes that are much more closely tied to voting behavior. That's what's called the realignment.
- 8.56** So, yeah, it seems as though people don't compare themselves to the 1% much. The 1%, Jeff Bezos and Bill Gates having billions and billions doesn't seem to drive the perceptions of a lot of voters. It's very important for people, particularly in the elites, who are concerned with these questions. But for most voters, a lot of the literature on class and inequality in voting, it seems to me, doesn't show that this is a big driver of voting in the West.
- 8.57** Now, maybe in the developing world, where you have, particularly, either crony capitalism or expropriation, you have people who simply own all of the resources through inheritance or through military force or through some kind of extraction. There, you might actually see more of this. I think that is one question for Piketty.

- 8.58** To be fair to him, his book is not really big on the political science and the link between inequality and politics. I think, yeah, I guess I would say, I'm not sure the evidence is that strong, that inequality per se, people on lower incomes, that that is really a big driver of how voting works, at least in developed countries today.
- 8.59** You mentioned that when we see an aging population, there's an impact on, let's say, military spending. So my question is, in the near future or in the long term, is this going to be something that may affect the way that China or the US shape geopolitically, and also in terms of how they spend militarily, when you look at their population age in the near future and in the long term?
- 8.60** I think this gets back a little bit to the geriatric peace hypothesis of Mark Haas, that when you are facing pressure to spend more on the elderly and you have fewer young people, either young people willing to risk their lives or parents not wanting their young people's lives to be risked, that will induce more caution and less risky geopolitical behavior. But it also means, of course, that the US might be more reluctant to get involved in foreign conflicts.
- 8.61** I mean, that's clearly a hallmark of the Trump administration, is not wanting to get involved in, quote, unquote, "forever wars." I think this will simply make international interventions in, like, Iraq or Afghanistan—that kind of intervention may become more rare when populations are older. Maybe there's a recruitment crisis. A lot of militaries are finding it difficult to recruit enough soldiers in Europe or the US, so there's a recruitment crisis.
- 8.62** Now, that could have many sources, including a decline in patriotism in the young population. So there are other factors that are playing a role. But if you can't recruit a military, if you can't fund that military, then, yeah, I would expect these countries would be more cautious in getting involved in military adventurism. The flip side of that is, countries might also be more cautious in— they may not wish to intervene in Ukraine, in Israel-Palestine, in Taiwan. Countries might be more reluctant.
- 8.63** So that, maybe China will be more reluctant to invade Taiwan, but then the US might be more reluctant to defend Taiwan. So it's not exactly clear what this all means, although Haas would argue that the net effect of an older population will be fewer wars and more peace in general, both within countries and even between countries.